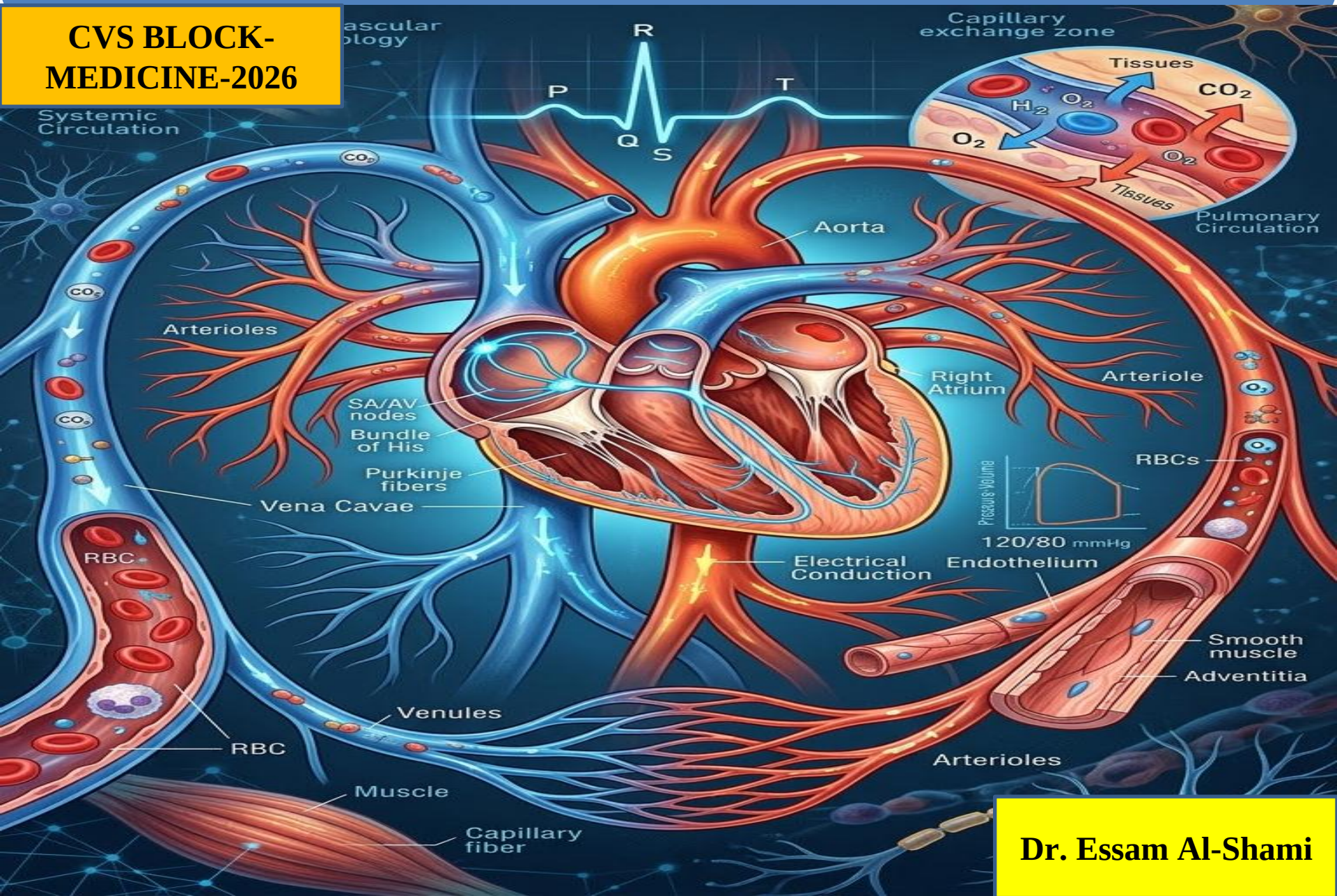
Cardiovascular system

CVS BLOCK-MEDICINE-2026



Dr. Essam Al-Shami

Cardiovascular system

Function of Cardiovascular system

- Deliver oxygen and nutrients to body cells
- Remove carbon dioxide and other waste products

Major Components Cardiovascular system

- **The heart:**
  - Pumps blood
- **Blood vessels:**
  - Allows blood to circulate to all parts of the body

The heart



The heart

Component

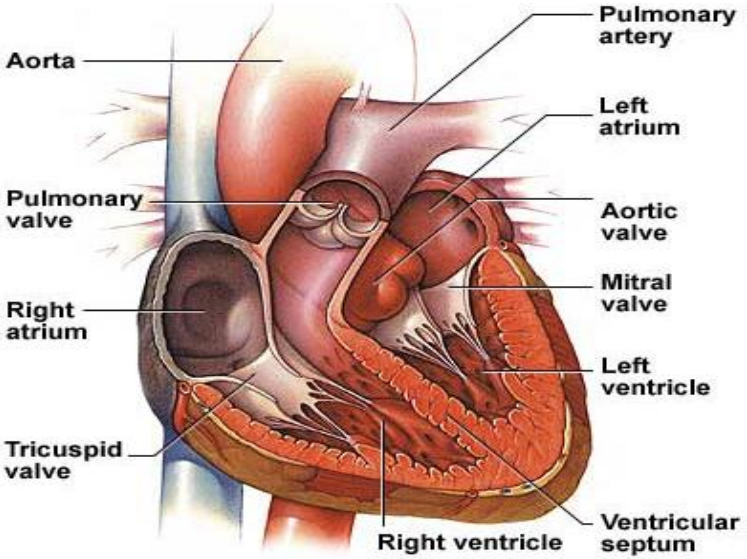
1-Cardiac Tissues

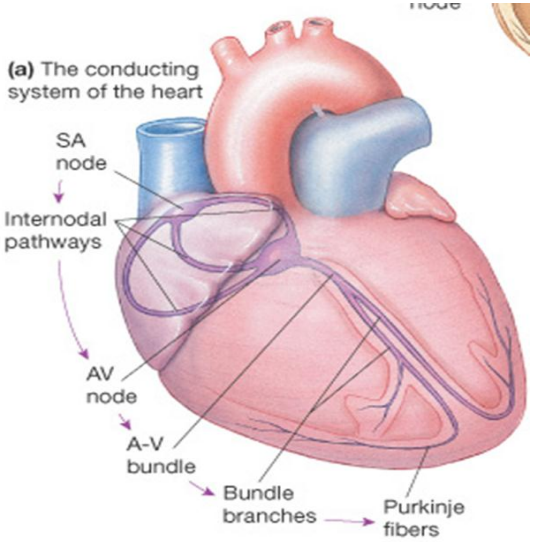
2-Cardiac Chambers

3-Cardiac Valves

4-Associated Great Vessels

5-Conducting system

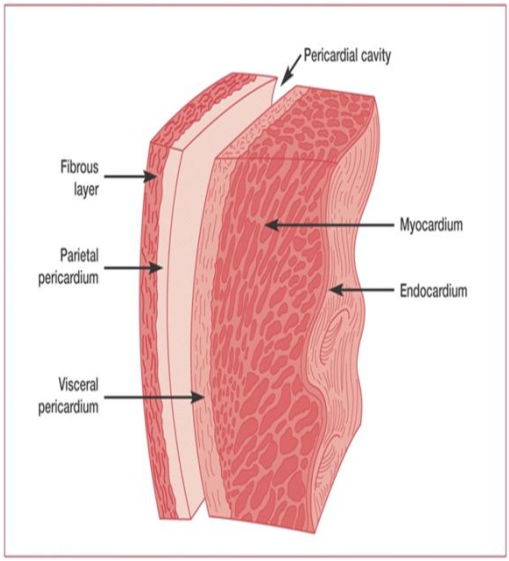




1-Cardiac Tissues

A-Pericardium

- ❖ TWO PARTS: the fibrous and serous
- ❖ Pericardium secretes lubricant (serous fluid)
- ❑ Pericardium is a membrane protecting and anchoring heart to diaphragm and sternum
- ❑ Allows sufficient freedom movement
- ❑ Prevent heart friction with surround parts
- ❑ Prevent over dilation of heart



B-Myocardium

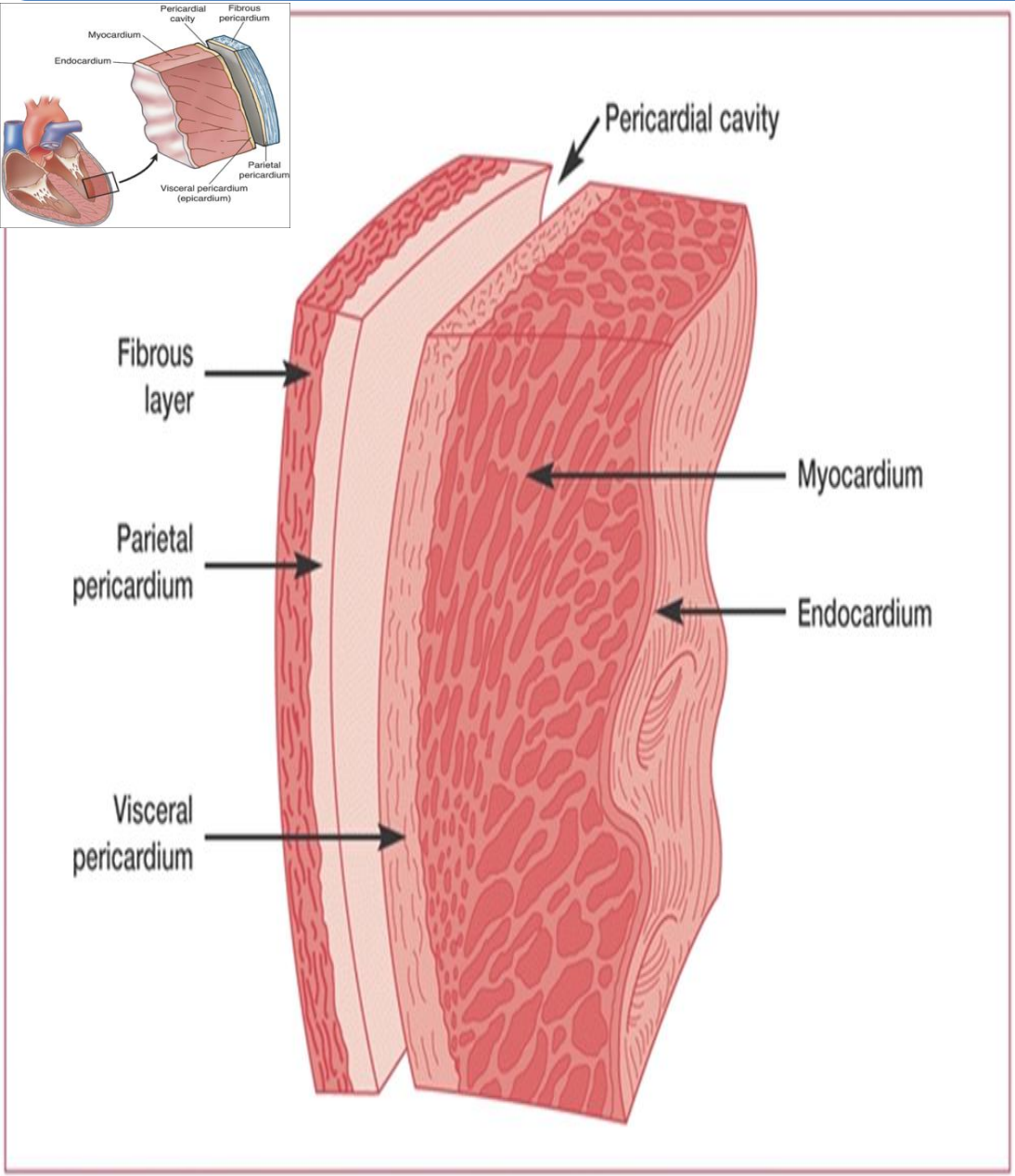
- ❖ Contains contractile muscle fibers
- ❑ Responsible for pumping

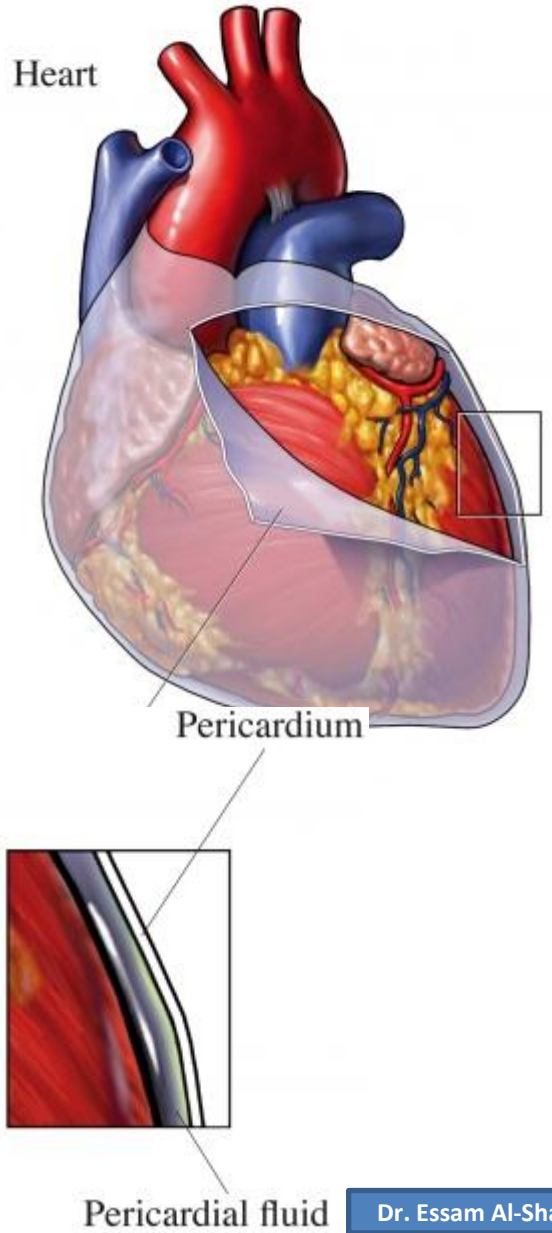
C-Endocardium

- ❖ Thin layer of endothelium & lines cardiac chambers
- ❑ Protect heart chambers and valves
- ❑ Controlling the ionic composition of the ECF

Abnormalities

Cardiac Tissues

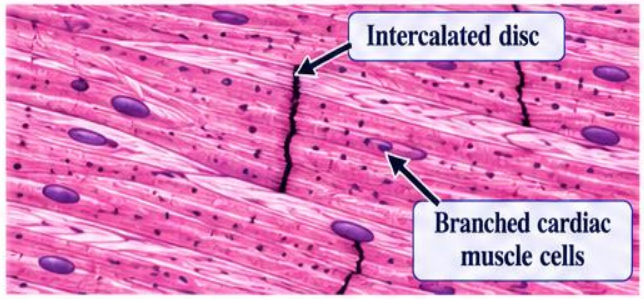




Cardiac Muscle vs Skeletal Muscle: Key Structural and Functional Differences

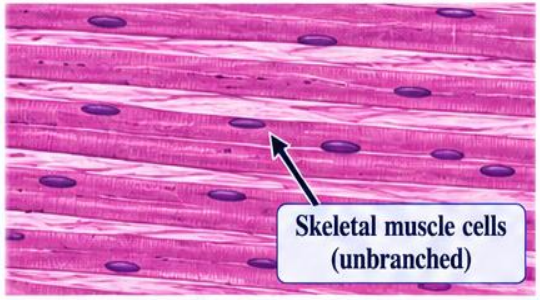
Cardiac Muscle (Heart)

- Involuntary
- Striated
- Branched cells
- Intercalated discs
- Single central nucleus

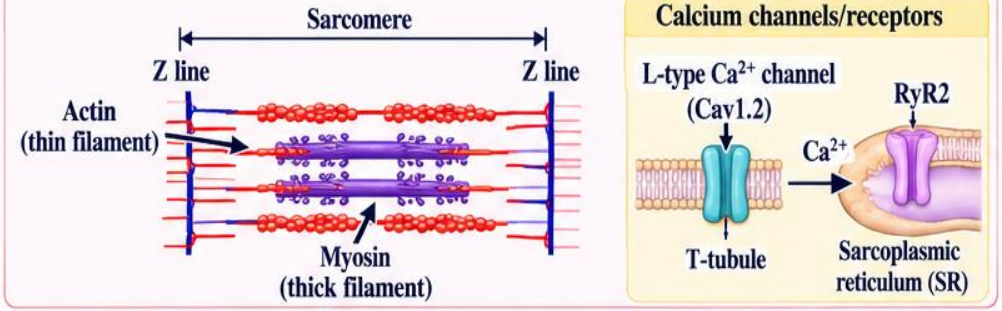


Skeletal Muscle (e.g. biceps)

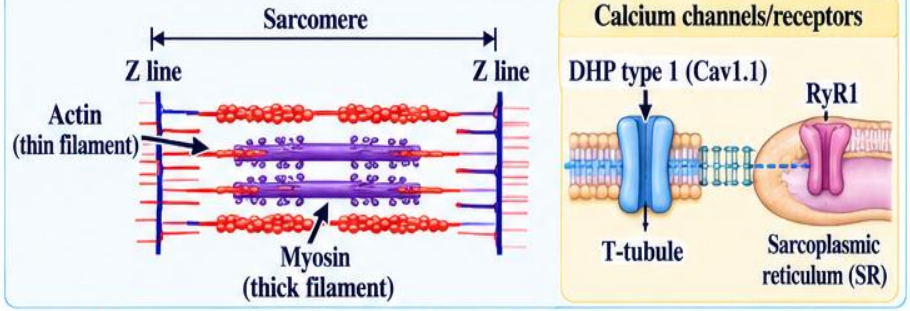
- Voluntary
- Striated
- Long cylindrical cells
- Multinucleated
- No intercalated discs
- Peripheral nuclei



Sarcomere and main proteins (Cardiac muscle)



Sarcomere and main proteins (Skeletal muscle)



Feature	Cardiac Muscle	Skeletal Muscle
Location	Heart (myocardium)	Skeletal muscles (e.g. biceps, quadriceps)
Control	Involuntary (autonomic nervous system, pacemaker)	Voluntary (somatic motor nerves)
Cell shape	Short, branched cells	Long, cylindrical cells
Nuclei	Single, central nucleus	Multiple, peripheral nuclei
Striations	Present	Present
Intercalated discs	Present (gap junctions and desmosomes)	Absent
Sarcomere	Z line to Z line; same basic organization	Z line to Z line; same basic organization
Myofilaments	Thick (myosin) and thin (actin) filaments with troponin–tropomyosin	Thick (myosin) and thin (actin) filaments with troponin–tropomyosin
Calcium channel/receptor	L-type Ca^{2+} channel (Cav1.2) and RyR2	DHP type 1 (Cav1.1) and RyR1
Coupling mechanism	Calcium-induced calcium release (CICR)	Mechanical coupling (DHP–RyR1)
Number of mitochondria	High (very abundant)	Low (relatively fewer)
Special features	Rich mitochondria, extensive capillary supply, T-tubules at Z line, continuous rhythmic contraction	T-tubules at A–I junction, large SR for Ca^{2+} storage, larger fiber diameter, can develop high force and tetanize
Main function	Rhythmic, continuous contraction to pump blood	Movement, posture, heat production

Functional properties of cardiac muscles

Property	Definition and physiological significance
Automaticity	**Ability of specialized cardiac cells to generate impulses spontaneously.**
Rhythmicity	**Ability to generate impulses at regular intervals.**
Excitability	**Ability to respond to an adequate stimulus by generating an action potential.**
Conductivity	**Ability to propagate electrical impulses from cell to cell.**
Contractility	**Ability of myocardial cells to develop force and shorten.**
Refractoriness	**Period during which cardiac muscle cannot be re-excited, or requires a stronger stimulus to respond.**

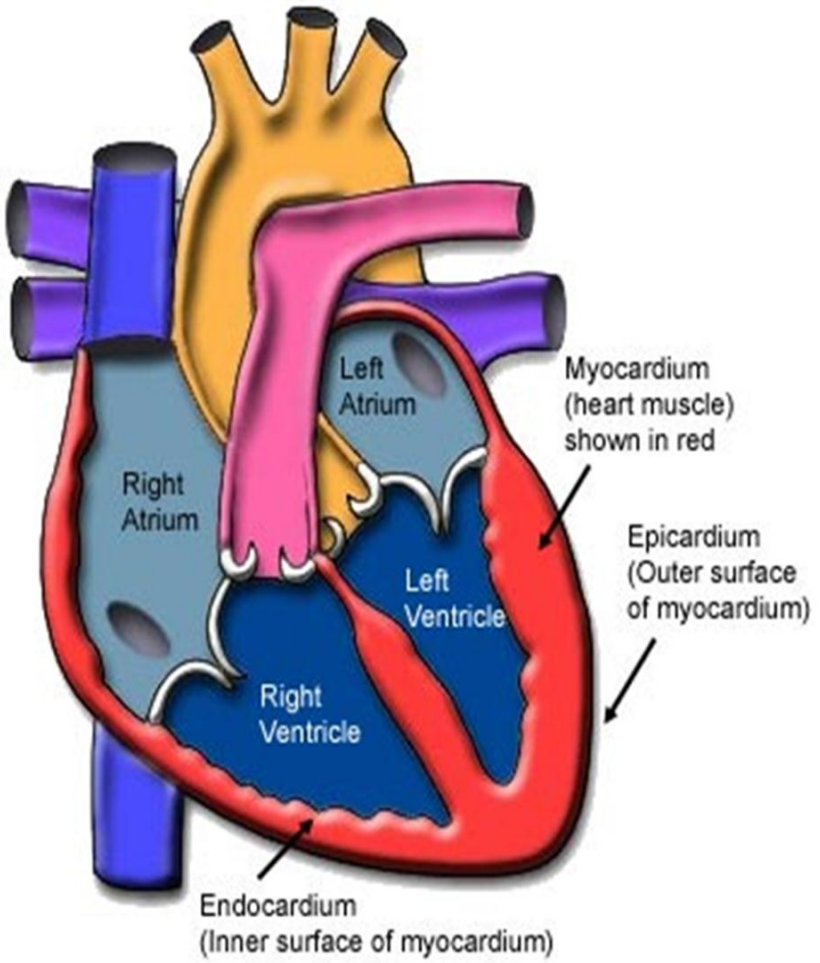
2-Cardiac Chambers

Atria

- ❖ Thin walled
- ❖ Superior
- ❖ Receiving chambers
- ❖ Right atrium (BP: <10)
- ❖ Left atrium (BP: <10)
- ❑ Pump blood to ventricles

Ventricles

- ❖ Thick walled
- ❖ Inferior
- ❖ Discharging chambers
- ❖ Right ventricle (BP: < 25)
- ❖ Left ventricle (BP: < 120)
- ❖ Thickness of LV 3 times of RV
- ❑ Pump blood to lung & whole body

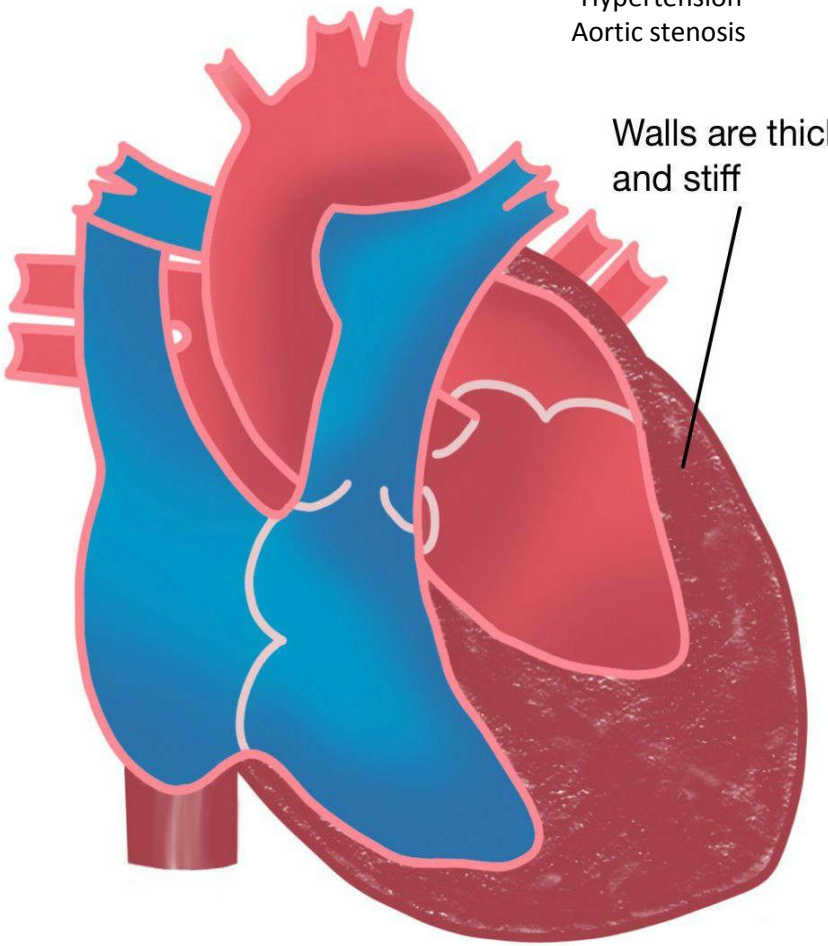


Abnormalities

2-Cardiac Chambers

Heart failure with preserved ejection fraction
(Diastolic heart failure)

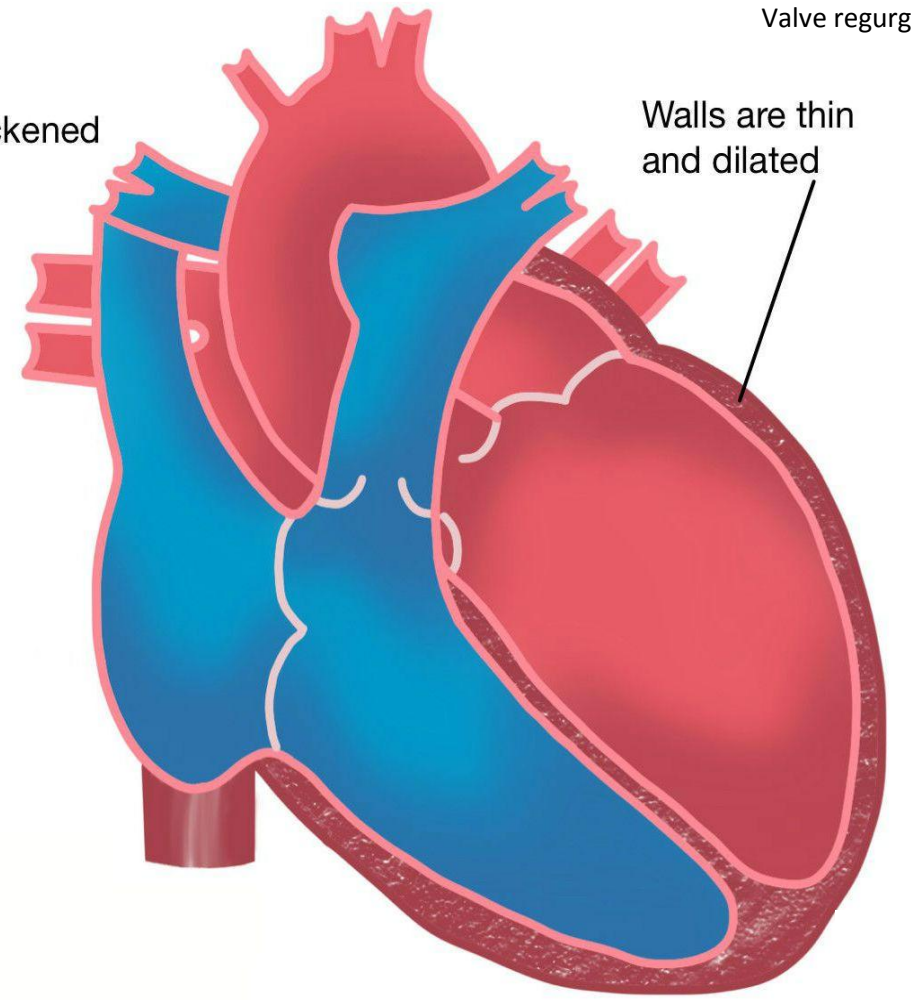
Hypertension
Aortic stenosis



Heart struggles to fill

Heart failure with reduced ejection fraction
(Systolic heart failure)

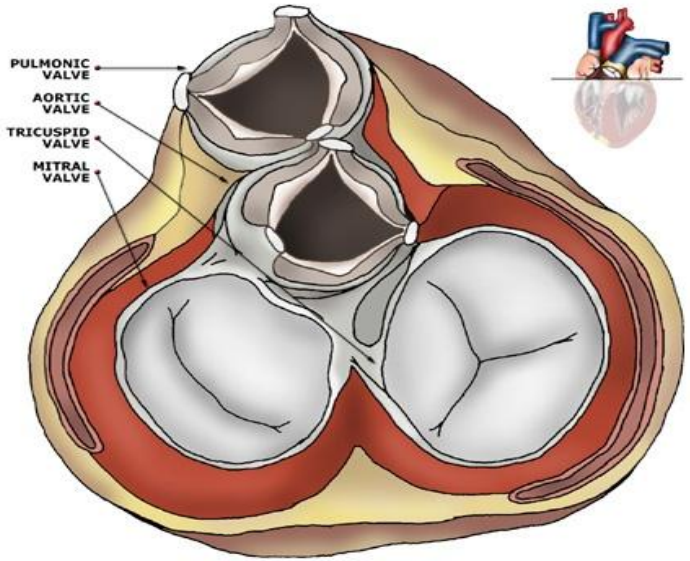
Myocarditis
Valve regurg



Heart struggles to pump

3-Cardiac Valves

- ❖ Composed of connective tissue & endocardium
- ❖ Composed of 2-3 flaps
- ❑ Allow blood to flow in only one direction
- ❑ Valves open as blood is pumped through
- ❑ Close to prevent blood backflow
- ❖ Regulated by pressure gradient

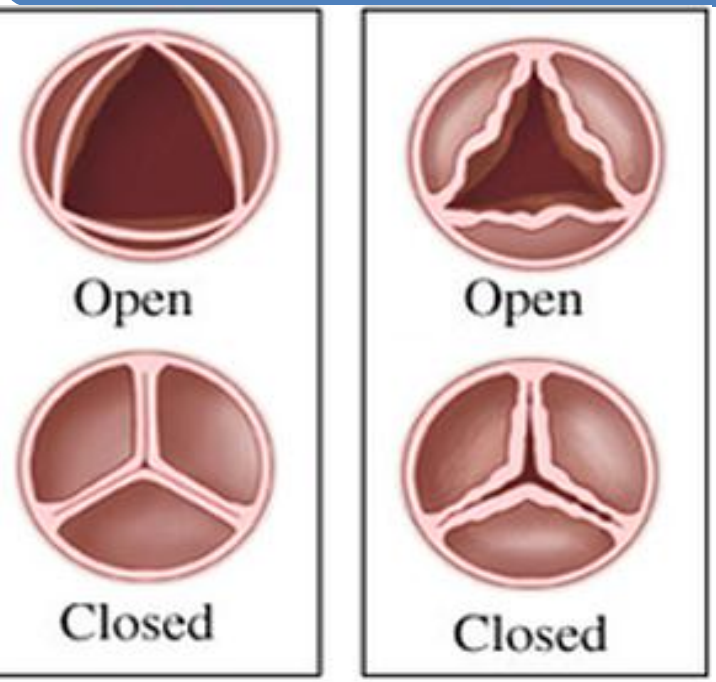


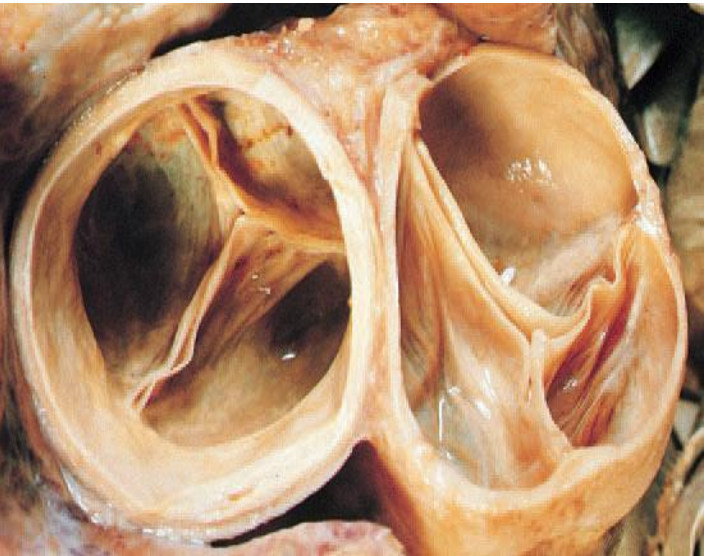
Four valves

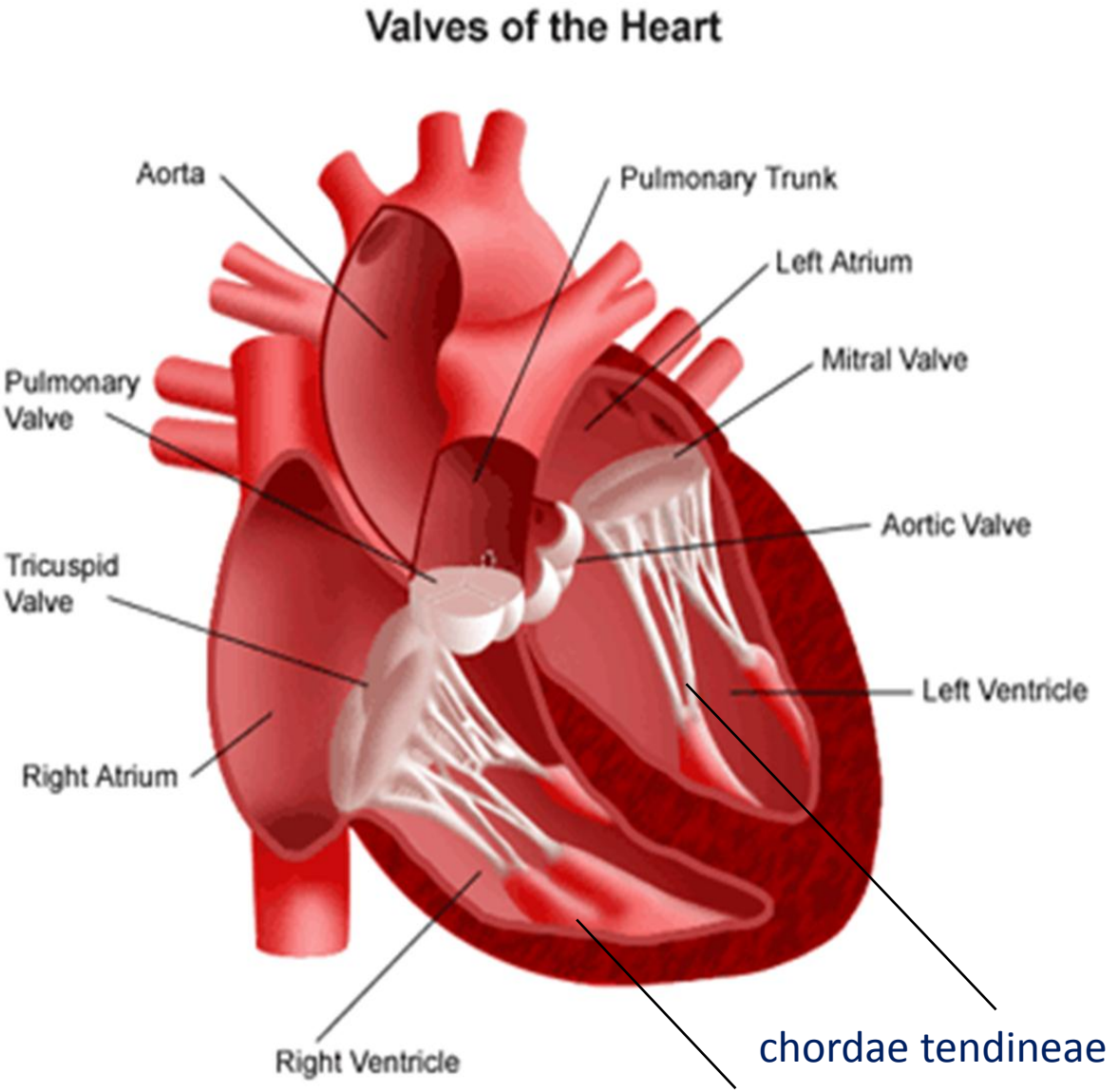
- ❖ **Atrioventricular valves:**
  - ❖ between atria and ventricles
    - ➢ Bicuspid valve “Mitral” (left)
    - ➢ Tricuspid valve (right)
- ❖ **Semilunar valves:**
  - ❖ between ventricles and arteries
    - ➢ Pulmonary semilunar valve
    - ➢ Aortic semilunar valve

Abnormalities

Cardiac Valves







4-Associated Great Vessels

Aorta

- ❖ Leaves left ventricle to systemic circulation
- ❑ Carry oxygenated blood from heart to body organs
- ❖ Blood pressure 80-120 mmHg

Vena cava

- ❖ Carries blood from body to right atrium
- ❑ Carry deoxygenated blood from body organs to heart
- ❖ Blood pressure < 10 mmHg

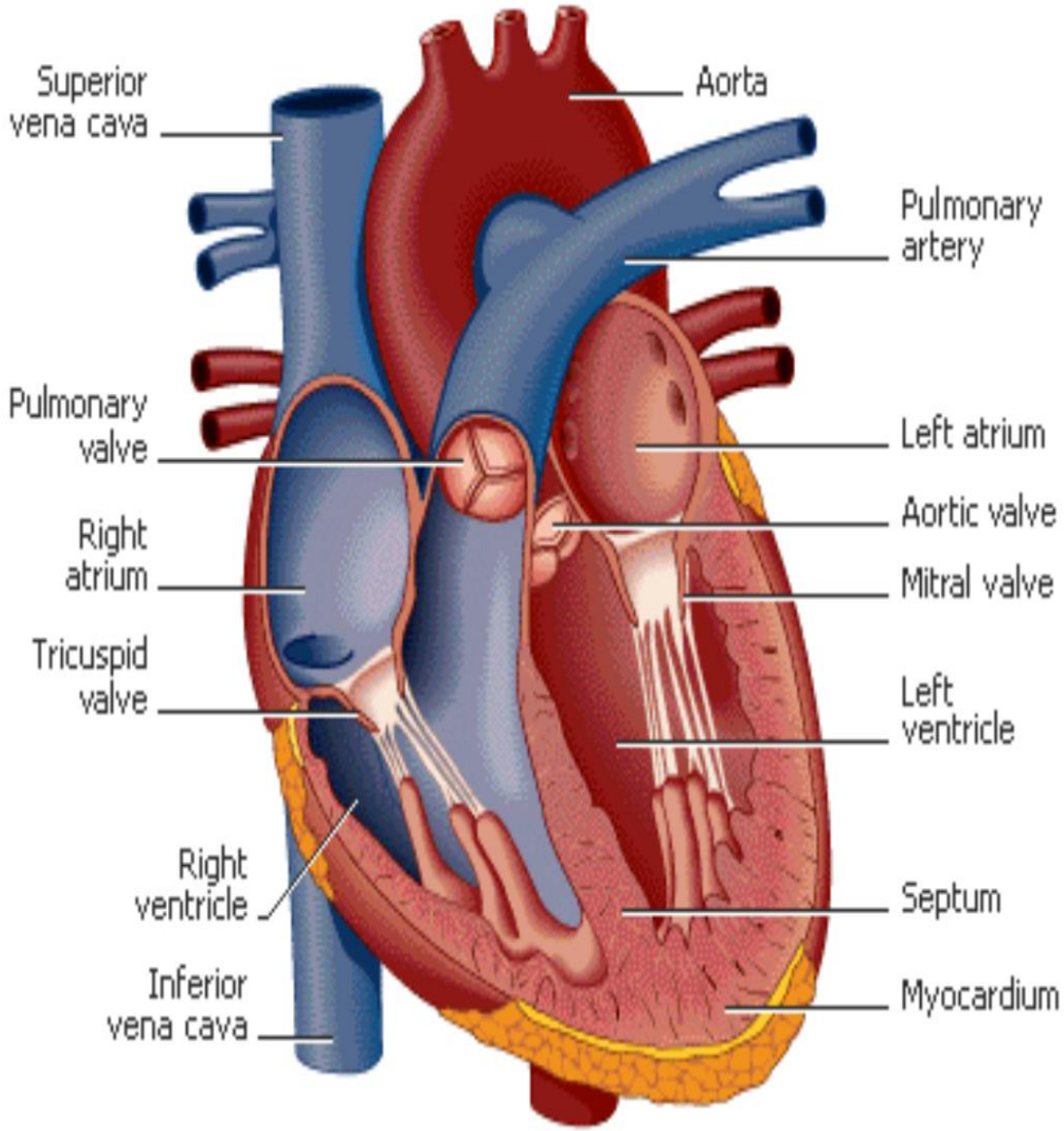
Pulmonary artery

- ❖ Leave right ventricle to pulmonary circulation
- ❑ Carry deoxygenated blood from RV to lung
- ❖ Blood pressure < 25 mmHg

Pulmonary veins

- ❖ Carries blood from lung to left atrium
- ❑ Carry oxygenated blood from lung to LA
- ❖ Blood pressure < 20 mmHg

4-Associated Great Vessels



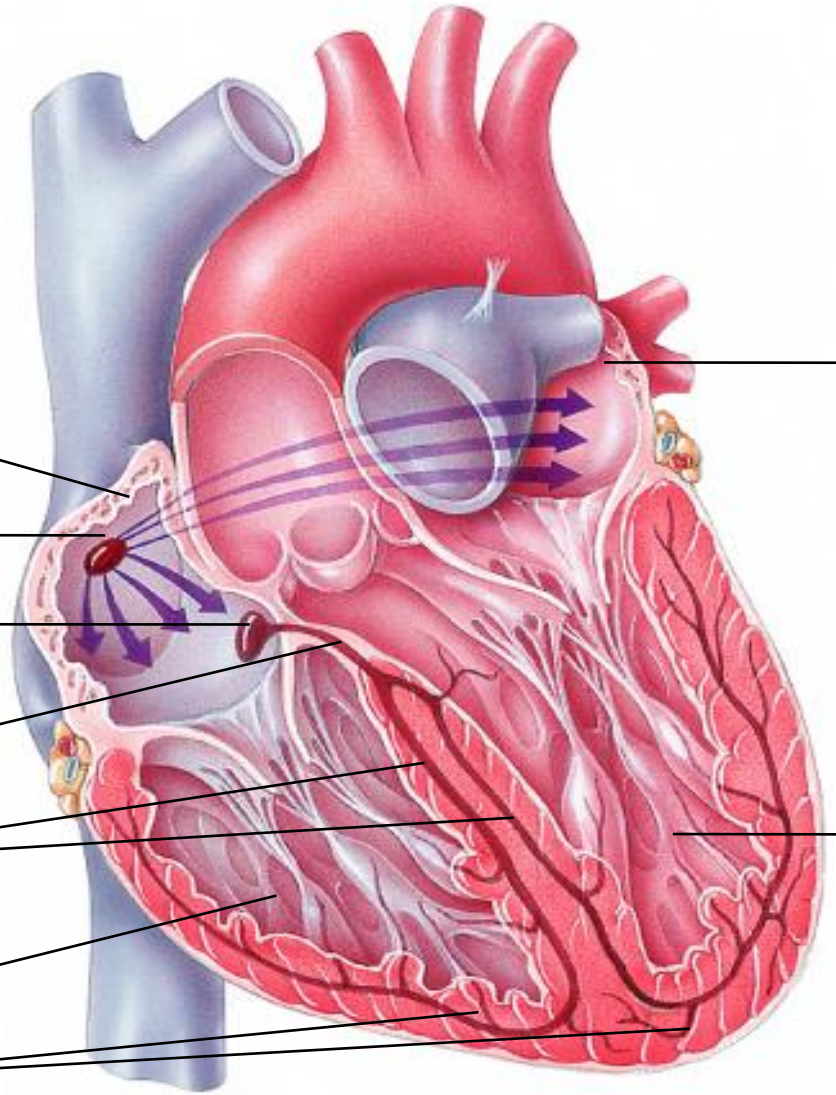
5-Conducting system

- Intrinsic conduction system (nodal system)
- Specialized cardiac muscle fibers
- Self-excitable
- Act as pacemaker
- Form conduction system
- Repeatedly generate action potentials that trigger heart contractions
- Pacemaker cells include:
  - Sinoatrial node (SA)
  - Atrioventricular node (AV)
  - Atrioventricular bundle (bundle of His)
  - Bundle branches
  - Purkinje fibers

Conducting system



Frontal plane



Left atrium

Right atrium

1 SINOATRIAL (SA) NODE

2 ATRIOVENTRICULAR (AV) NODE

3 ATRIOVENTRICULAR (AV) BUNDLE (BUNDLE OF HIS)

4 RIGHT AND LEFT BUNDLE BRANCHES

Left ventricle

Right ventricle

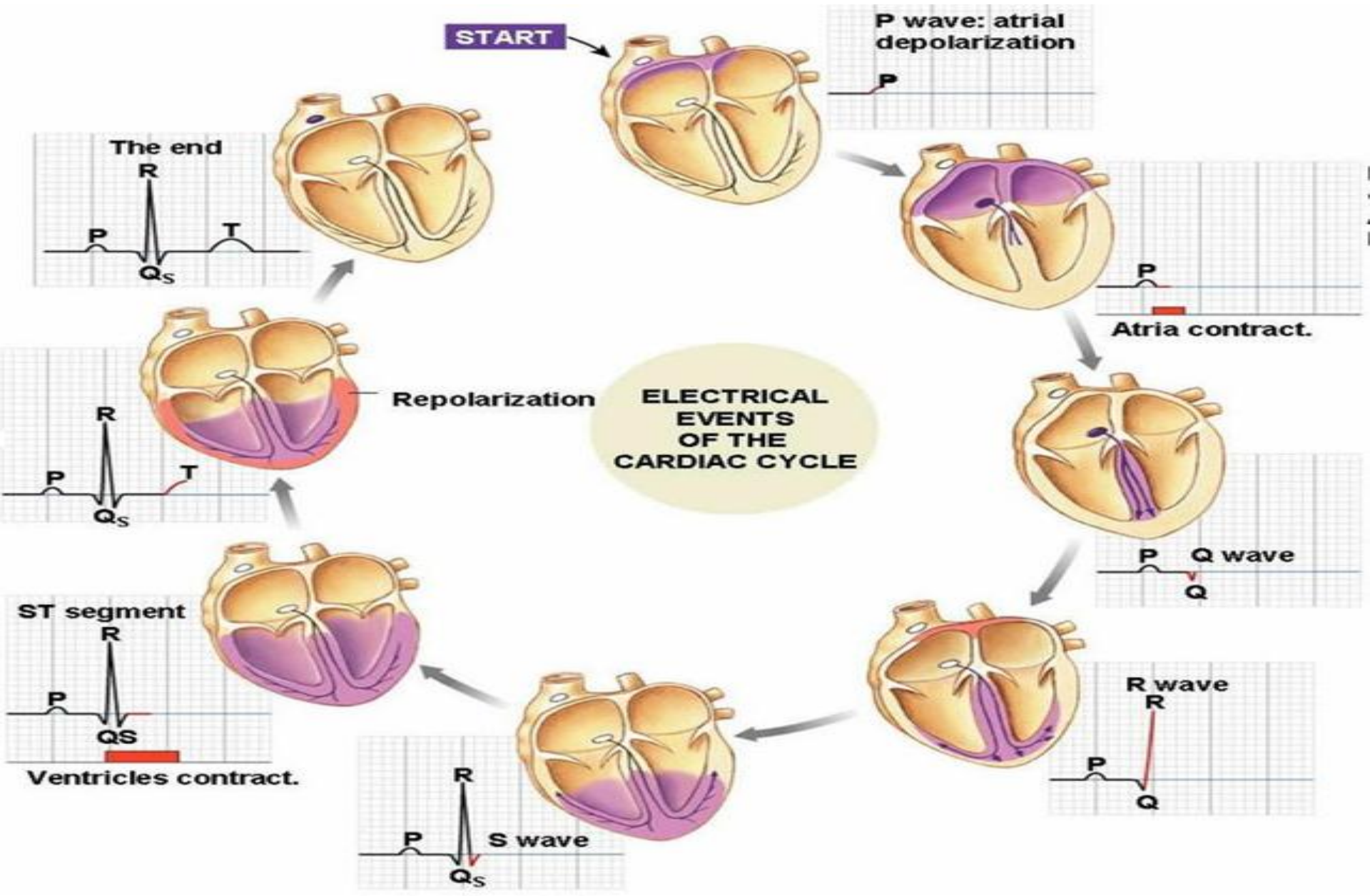
5 PURKINJE FIBERS

Anterior view of frontal section

Conducting system

1) Begins in sinoatrial (SA) node in right atrial wall generate the action potential (AP)
   - AP spreads through atria via internodal pathways
   - Speed 60-100 ppm
   - Atria contract
2) AP reaches atrioventricular (AV) node in interatrial septum
   - Speed 40-60 ppm
3) AP enters atrioventricular bundle (Bundle of His)
4) AP enters right and left bundle branches which extends through interventricular septum toward apex
5) AP finally reaches Purkinje fibers & conduct AP to remainder of ventricular myocardium
   - Ventricles contract

Conducting system

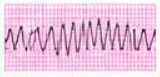
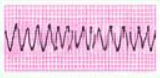
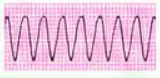


Bradyarrhythmias and Tachyarrhythmias: Types, Features, Causes and Treatment

1. Bradyarrhythmias (HR < 60 bpm)

Type	ECG example	HR (bpm)	Rhythm	Origin	Common causes	Main treatment (examples)
1. Sinus bradycardia		< 60	Regular	Atria (sinus node)	• High vagal tone • Athletes • Hypothyroidism • Drugs (β-blockers)	• Treat cause • If symptomatic: Atropine • Consider pacemaker
2. Sinoatrial (SA) block		< 60	Irregular (with pauses)	Atria (SA node)	• Increased vagal tone • Ischemia • Drugs (β-blockers, digoxin)	• Treat cause • Atropine if severe • Pacemaker if symptomatic
3. Atrial arrest		< 60	Irregular	Atria	• SA node failure • Hyperkalemia • Drug toxicity	• Treat cause • Pacemaker
4. AV block (1st degree)		Usually > 60	Regular	Both (atrium and ventricle)	• Increased vagal tone • Drugs (β-blockers, CCB, digoxin) • Ischemia	• Often no treatment • Treat cause
5. AV block (2nd degree Mobitz I)		Usually < 60	Irregular (with progressive PR ↑)	Both	• High vagal tone • Inferior MI • Drugs (β-blockers)	• Atropine if symptomatic • Usually no pacemaker (except persistent)
6. AV block (2nd degree Mobitz II)		Usually < 60	Irregular (dropped QRS)	Both	• His-Purkinje disease • Anterior MI • Degenerative fibrosis	• Pacemaker (usually required)
7. Complete heart block (3rd degree)		< 60 (escape rhythm)	Regular or irregular	Both (AV node or below)	• Ischemia/MI • Degenerative conduction disease • Drugs, myocarditis	• Atropine (temporary) • Permanent pacemaker (definitive)

2. Tachyarrhythmias (HR > 100 bpm)

Type	ECG example	HR (bpm)	Rhythm	Origin	Common causes	Main treatment (examples)
A. Supraventricular tachyarrhythmias (origin in atria or AV node, usually narrow QRS)						
1. Sinus tachycardia		> 100	Regular	Atria (sinus node)	• Exercise, fever • Pain, anxiety • Anemia, hypovolemia • Hyperthyroidism	• Treat cause • β-blocker (e.g. bisoprolol)
2. Atrial flutter		250–350 (atrial)	Regular (2:1, 3:1 etc.)	Atria	• Structural heart disease • COPD, pulmonary disease • Hyperthyroidism	• Rate control: β-blocker or CCB • Anticoagulation • Cardioversion
3. Atrial fibrillation		> 100 (variable)	Irregular (irregularly irregular)	Atria	• Hypertension, IHD • Valvular heart disease • Hyperthyroidism • Alcohol ("holiday heartt")	• Rate control: β-blocker (bisoprolol) • Anticoagulation • Cardioversion (if unstable or symptomatic)
4. AV nodal re-entrant tachycardia (AVNRT)		150–250	Regular	AV node (junction)	• Often no structural disease • Triggered activity	• Vagal maneuvers • Adenosine (1st line) • β-blocker or CCB • Catheter ablation
5. AV re-entrant tachycardia (AVRT, WPW)		150–250	Regular	AV node/ accessory pathway	• Accessory pathway (WPW syndrome)	• Adenosine (if no AF) • β-blocker or CCB • Catheter ablation
6. Multifocal atrial tachycardia (MAT)		> 100	Irregular	Atria (multiple foci)	• COPD, hypoxemia • Electrolyte disturbance • Theophylline, digoxin	• Treat underlying cause • β-blocker or CCB
B. Ventricular tachyarrhythmias (origin in ventricles, wide QRS)						
1. Ventricular tachycardia (VT)		> 100	Regular (monomorphic) or irregular (polymorphic)	Ventricles	• Myocardial infarction • Cardiomyopathy • Electrolyte imbalance • Drug toxicity	• If unstable: Synchronized cardioversion • Antiarrhythmic (e.g. amiodarone) • Treat cause, ICD (long term)
2. Torsades de pointes (polymorphic VT)		> 100	Irregular	Ventricles	• Prolonged QT (drugs, electrolyte imbalance) • Congenital long QT	• Magnesium sulfate • Correct K^+ / Mg^{2+} • Remove QT-prolonging drugs
3. Ventricular fibrillation (VF)		> 200	Irregular (chaotic)	Ventricles	• Acute MI • Severe heart disease • Electrocution, trauma	• Immediate defibrillation • CPR • Epinephrine, amiodarone
4. Accelerated idioventricular rhythm (AIVR)		50–110	Regular	Ventricles	• Reperfusion (after MI) • Drug toxicity	• Usually no treatment • Treat underlying cause

Key differences

- **Bradyarrhythmias:** HR < 60 bpm (risk: low cardiac output, syncope).
- **Tachyarrhythmias:** HR > 100 bpm (risk: reduced filling, ischemia, HF).
- **Narrow QRS:** supraventricular origin. **Wide QRS:** ventricular origin (or aberrancy).

Emergency treatment

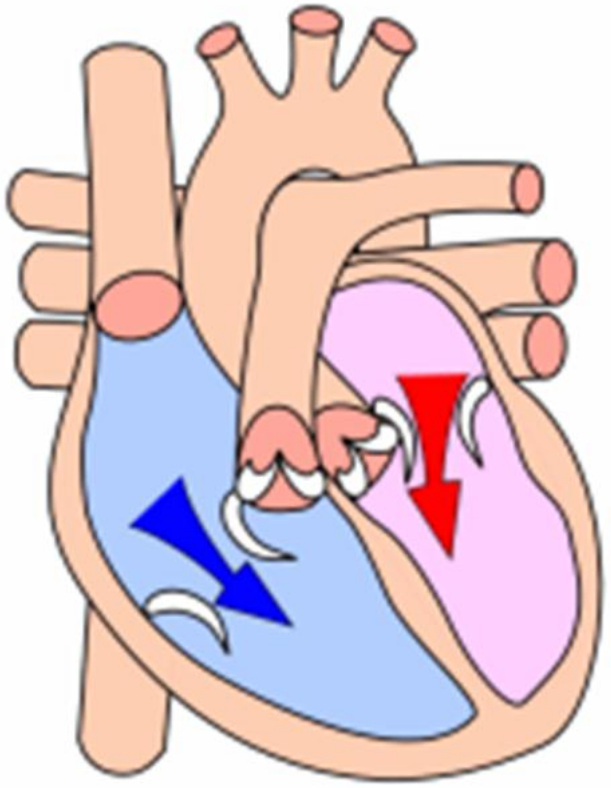
- **Unstable bradycardia:** Atropine → temporary pacing → permanent pacemaker.
- **Unstable tachyarrhythmia (narrow QRS):** Synchronized cardioversion.
- **Unstable tachyarrhythmia (wide QRS/VT or VF):** Defibrillation + CPR.

Abbreviations

HR: heart rate **AV:** atrioventricular **CCB:** calcium channel blocker
IHD: ischemic heart disease **MI:** myocardial infarction **WPW:** Wolff-Parkinson-White
VT: ventricular tachycardia **VF:** ventricular fibrillation **ICD:** implantable cardioverter-defibrillator

Cardiac cycle

All events associated with one heart beat

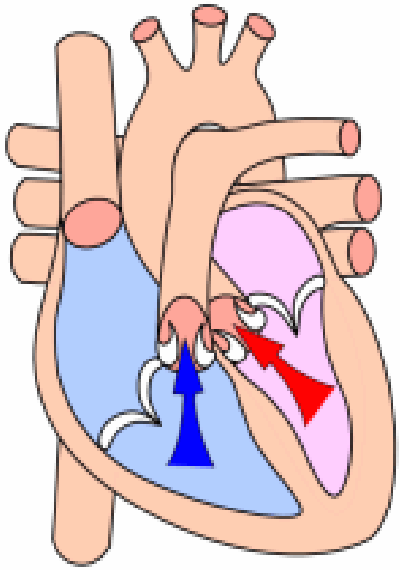


ATRIAL SYSTOLE

- LASTS FOR 0.1 SEC
- ATRIAL DEPOLARIZATION
- CAUSES ATRIAL SYSTOLE
- A FINAL 25mL OF BLOOD TO EACH VENTRICLE
- END OF ATRIAL SYSTOLE IS ALSO END OF VENTRICULAR DIASTOLE
- END DIASTOLIC VOLUME (EDV) IS 130 mL

Cardiac cycle

VENTRICULAR SYSTOLE

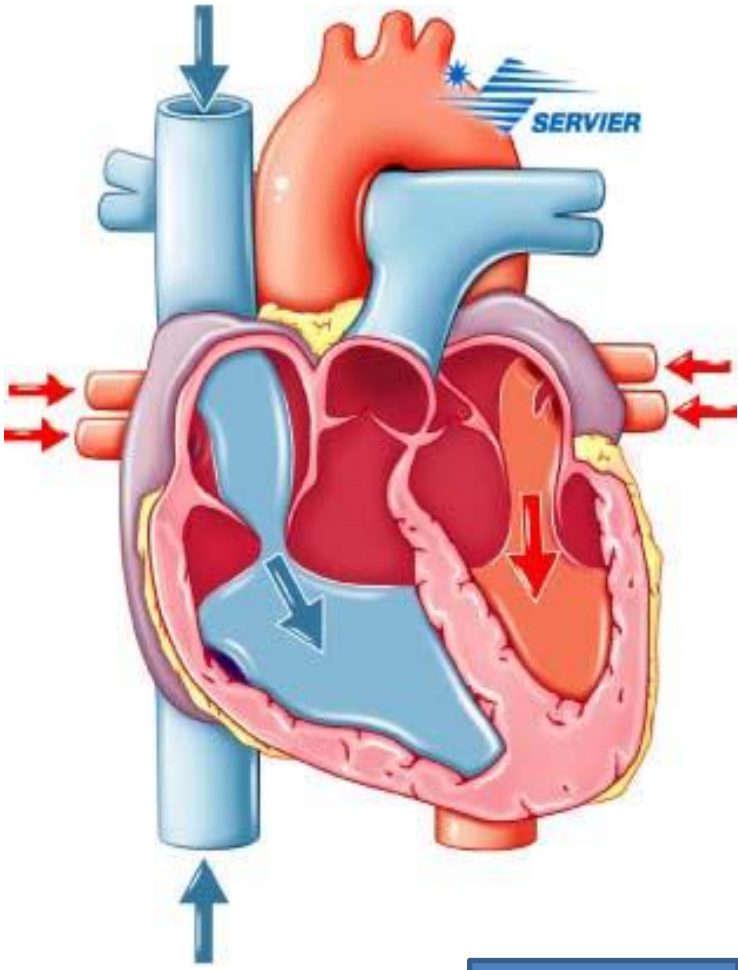


- LASTS FOR 0.3 SEC
- IT IS CAUSED BY VENTRICULAR DEPOLARIZATION
- THE SL VALVES OPEN WHEN THE LEFT VENTRICULAR PRESSURES ABOVE AORTIC PRESSURE(80 mmHg)
- THE RIGHT VENTRICULAR PRESSURE RISES ABOVE PULMONARY PRESSURE (20 mmHg)
- THE LEFT VENTRICLE EJECTS ABOUT 80 mL (STROKE VOLUM) INTO THE AORTA
- THE RIGHT VENTRICLE EJECTS THE SAME VOLUME INTO THE PULMONARY TRUNK
- END SYSTOLIC VOLUME (ESV) IS 50mL IN EACH VENTRICLE
- ↑Contractility (force of contraction) of ventricles leads to ↓ ESV

Cardiac cycle

RELAXATION PERIOD

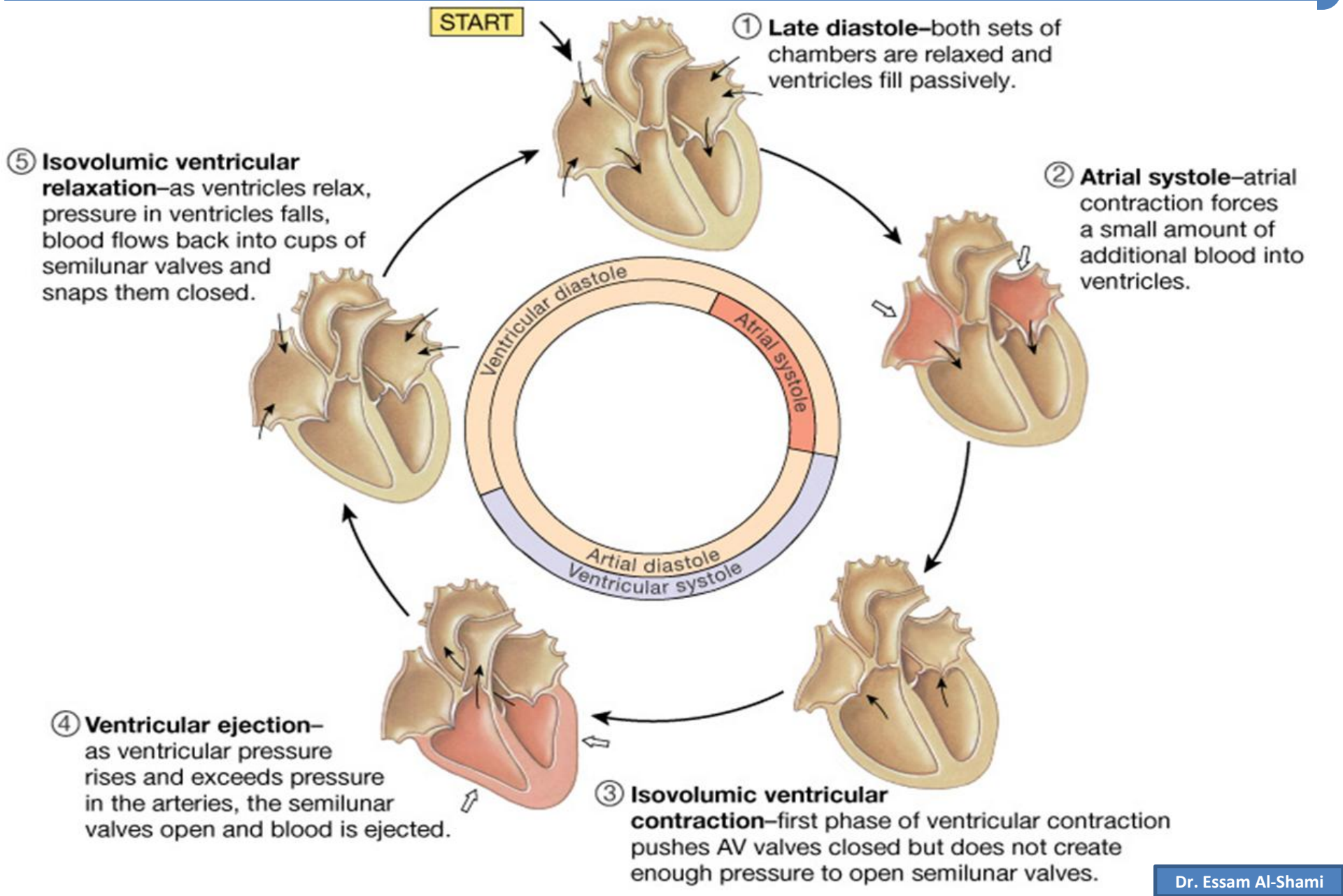
- BOTH ATRIA AND VENTRICLES ARE RELAXED .
- IT LASTS FOR 0.4 SEC.
- VENTRICULAR REPOLARIZATION CAUSES VENTRICULAR DAISTOLE.
- BOTH VENTRICLES FILLED PASSIVELY OF 55ml OF BLOOD
- NORMAL HEART RATE IS 60-100 BEAT/MINUTE
- WHEN HEART BEATS FASTER THE RELAXATION TIME SHORTENS.

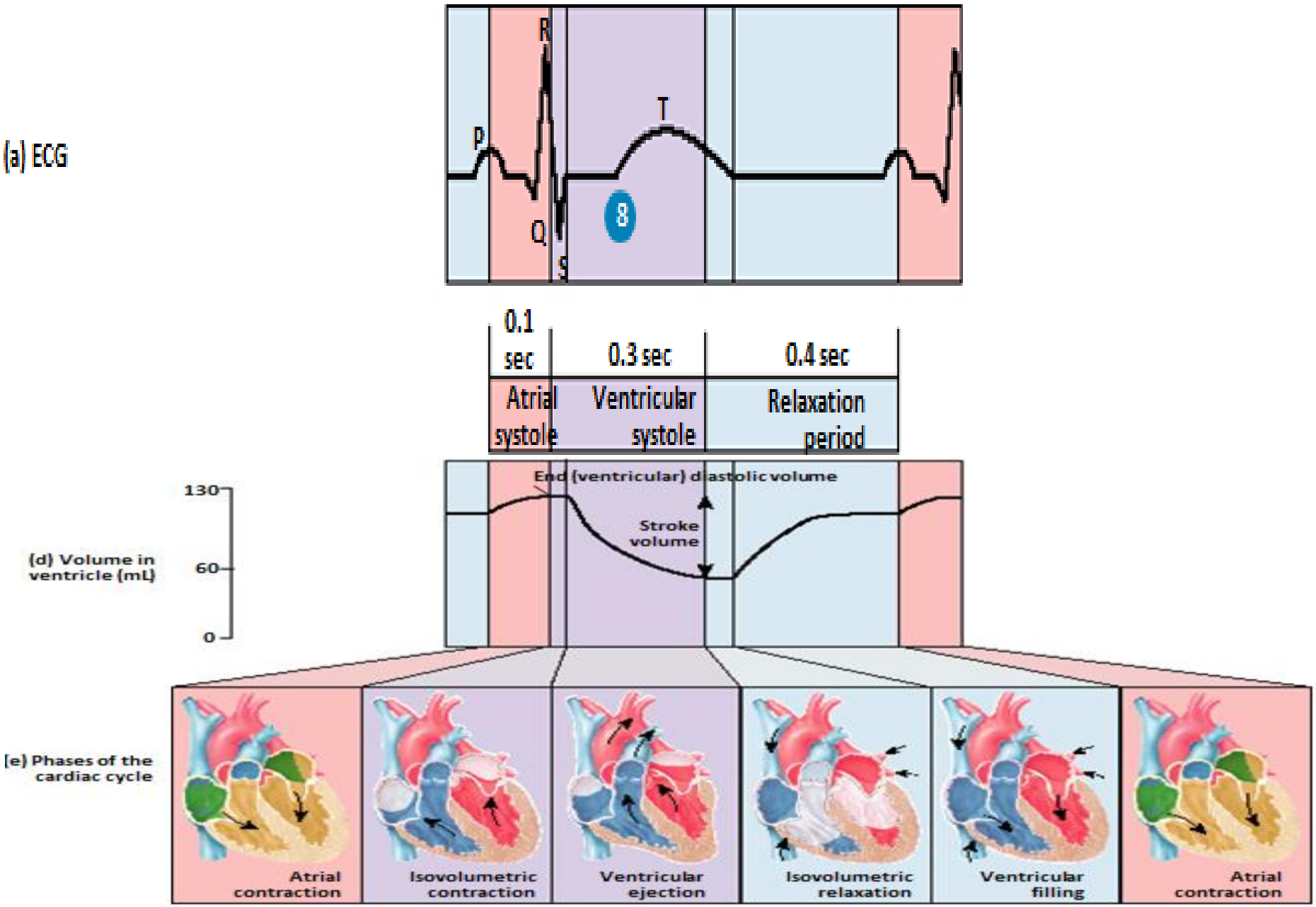


EJECTION FRACTION (EF)

- EF=SV/EDV=80/130 *100=61%
- Normal range of EF is 50-70%
- Measured by Echocardiograph
- Diagnosis of heart function
- EF decrease in:
  - Cardiomyopathy
  - Mitral stenosis or mitral regurgitation
  - Low blood pressure
  - Atrial fibrillation

Cardiac cycle




(a) ECG
R
P
T
Q
S
0.1 sec
0.3 sec
0.4 sec
Atrial systole
Ventricular systole
Relaxation period
End (ventricular) diastolic volume
Stroke volume
130
60
0
(d) Volume in ventricle (mL)
(e) Phases of the cardiac cycle
Atrial contraction
Isovolumetric contraction
Ventricular ejection
Isovolumetric relaxation
Ventricular filling
Atrial contraction

Cardiac Action Potentials

Phase 0

- ❑ Stable resting membrane is -90 mv
- ➢ Rapid ↑ Na+ conductance → ↑ Na+ inward → depolarization

Phase 1

- ➢ Brief period of repolarization (initial repolarization)
- ➢ Brief ↑ K+ outward

Phase 2

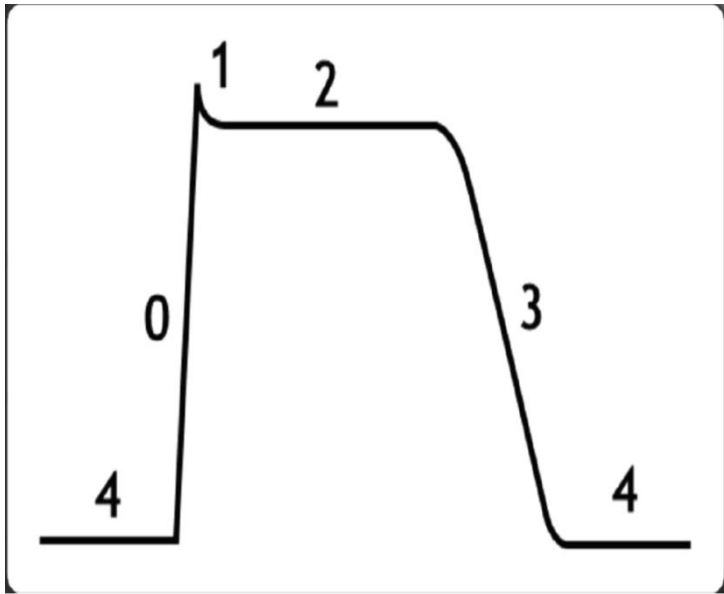
- ❑ Plateau of action potential
- ➢ Transient ↑ Ca^{2+} conductance → Ca^{2+} inward
- ➢ Transient ↑ K^{+} conductance → K^{+} outward

Phase 3

- ❑ Repolarization
- ➢ High ↑ K^{+} conductance → K+ outward

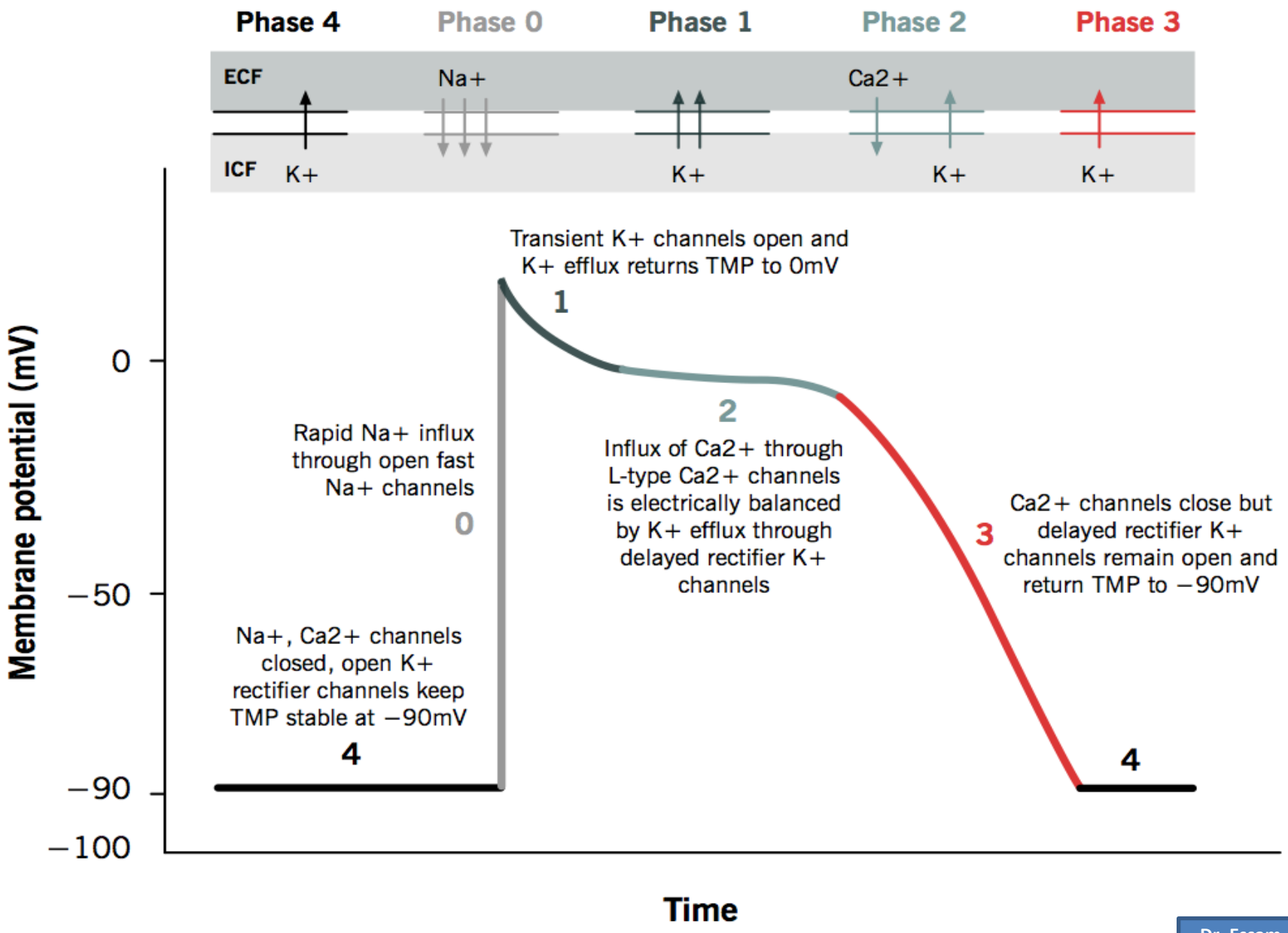
Phase 4

- ❑ The resting membrane potential
- ➢ Na^{+}/K^{+} ATPase pump
- ➢ Na^{+}/Ca^{2+} Exchanger



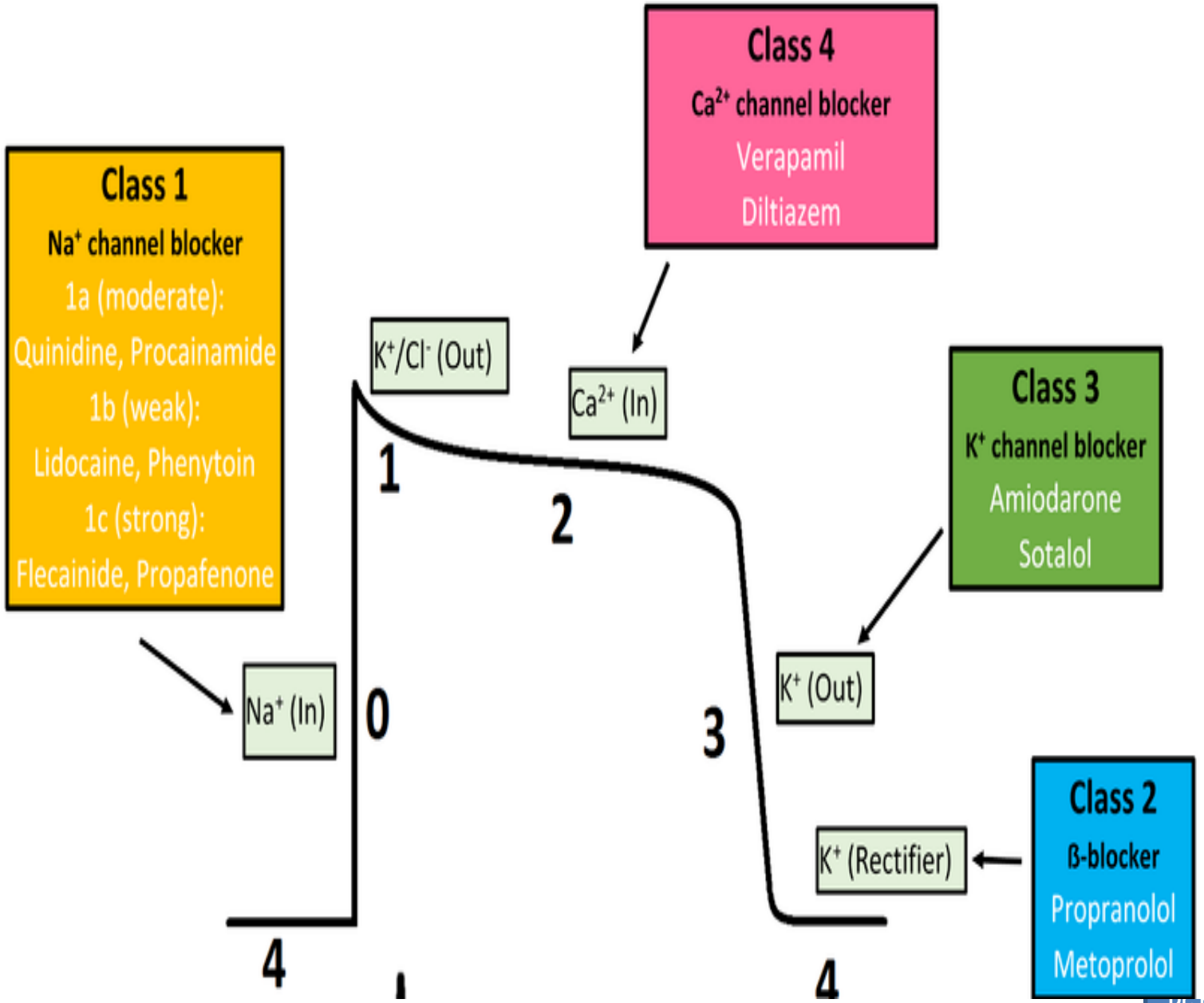
Action potential of cardiac muscles

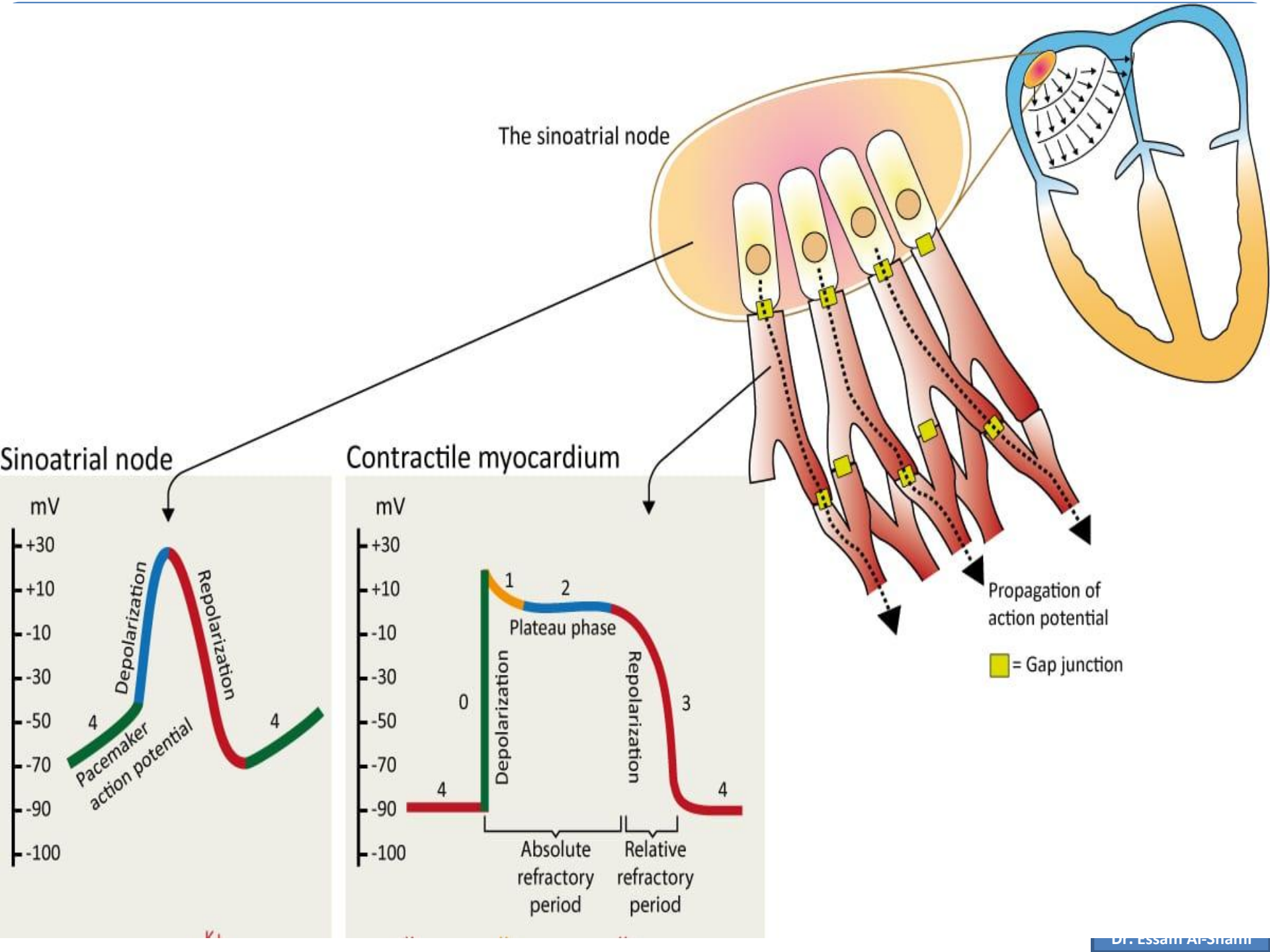
Grigoriy Ikonnikov and Eric Wong



Dr. Essam Al-Shami

Antiarrhythmic drugs

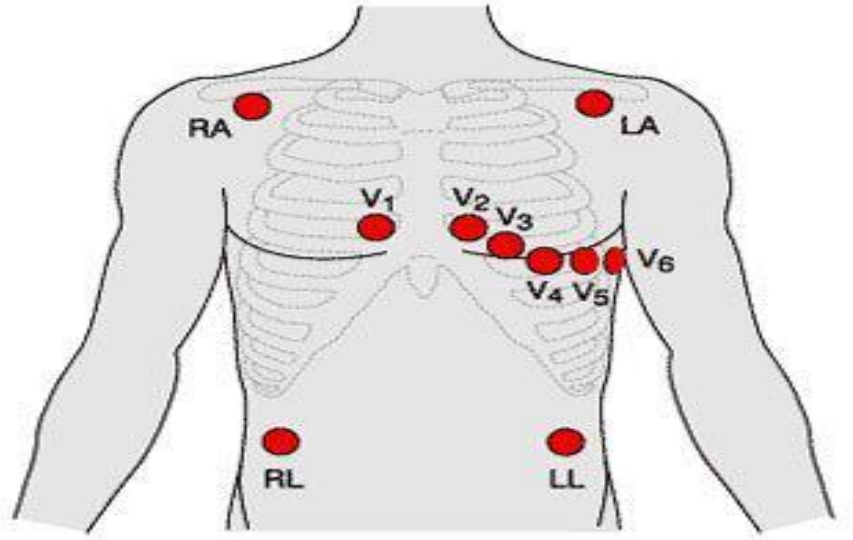



The sinoatrial node
Sinoatrial node
mV
+30
+10
-10
-30
-50
-70
-90
-100
Depolarization
Repolarization
4
4
Pacemaker action potential
Contractile myocardium
mV
+30
+10
-10
-30
-50
-70
-90
-100
1
2
Plateau phase
0
Depolarization
Repolarization
3
4
4
Absolute refractory period
Relative refractory period
Propagation of action potential
= Gap junction

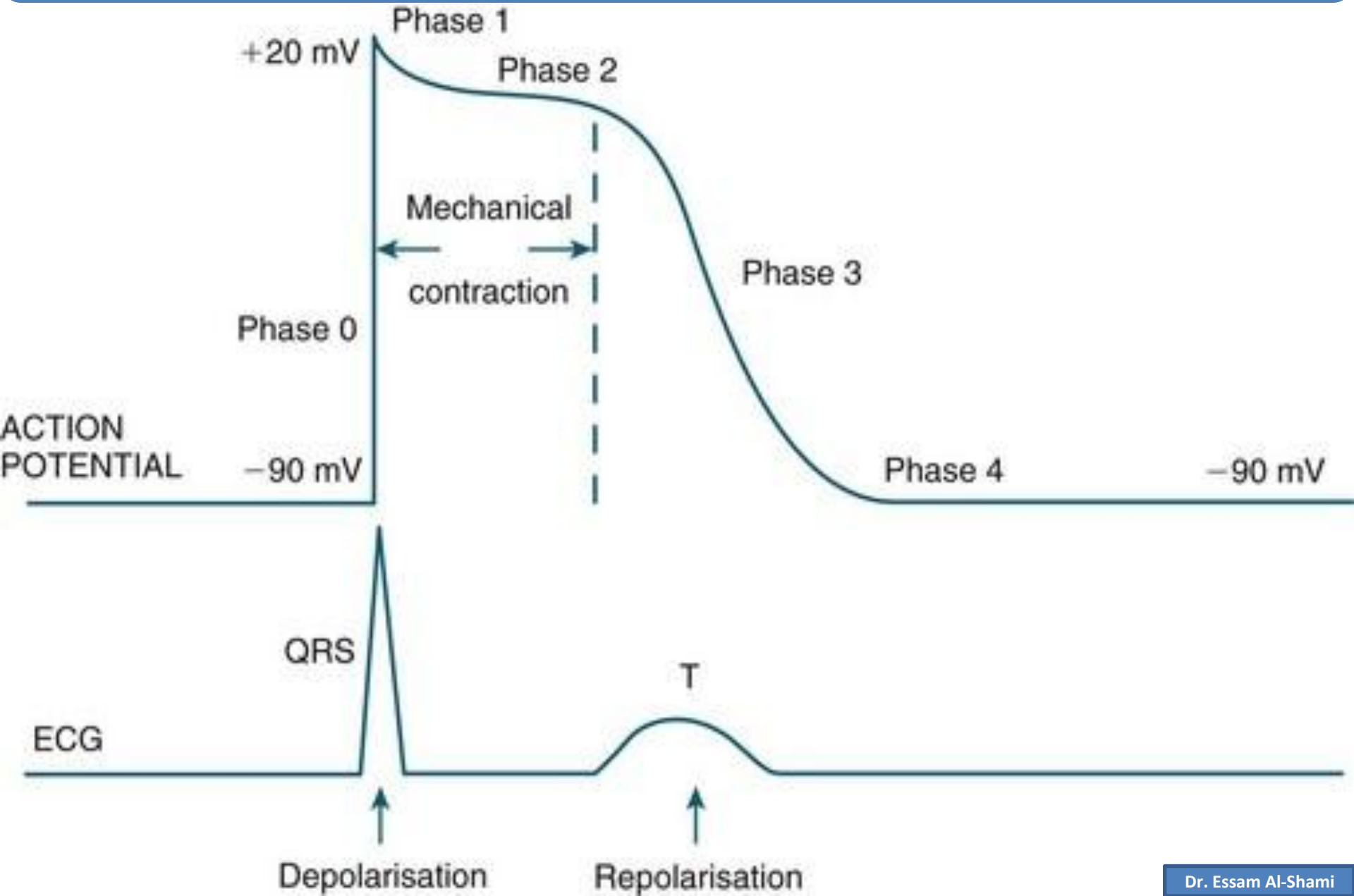
Electrocardiogram (ECG)

- is the process of recording the electrical activity of the heart over a period of time using electrodes placed on the skin.
- These electrodes detect the tiny electrical changes on the skin that arise from the heart muscle's electrical pattern of depolarizing and repolarizing during each heart beat.
- 12 leads & 10 electrodes





Electrocardiogram



Electrocardiogram (ECG)

1. Bipolar limb leads:

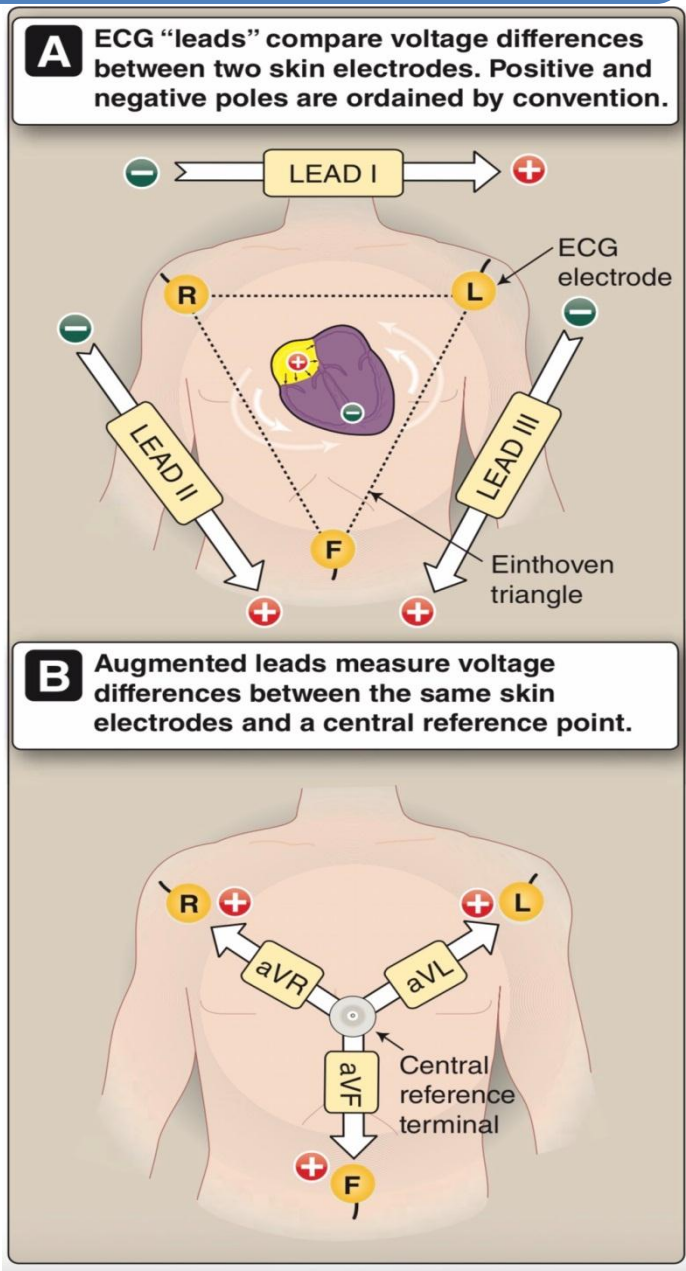
- ❑ There are three bipolar limb leads compare voltage differences between two skin electrodes
- ❑ I, II, III

2. Augmented limb leads:

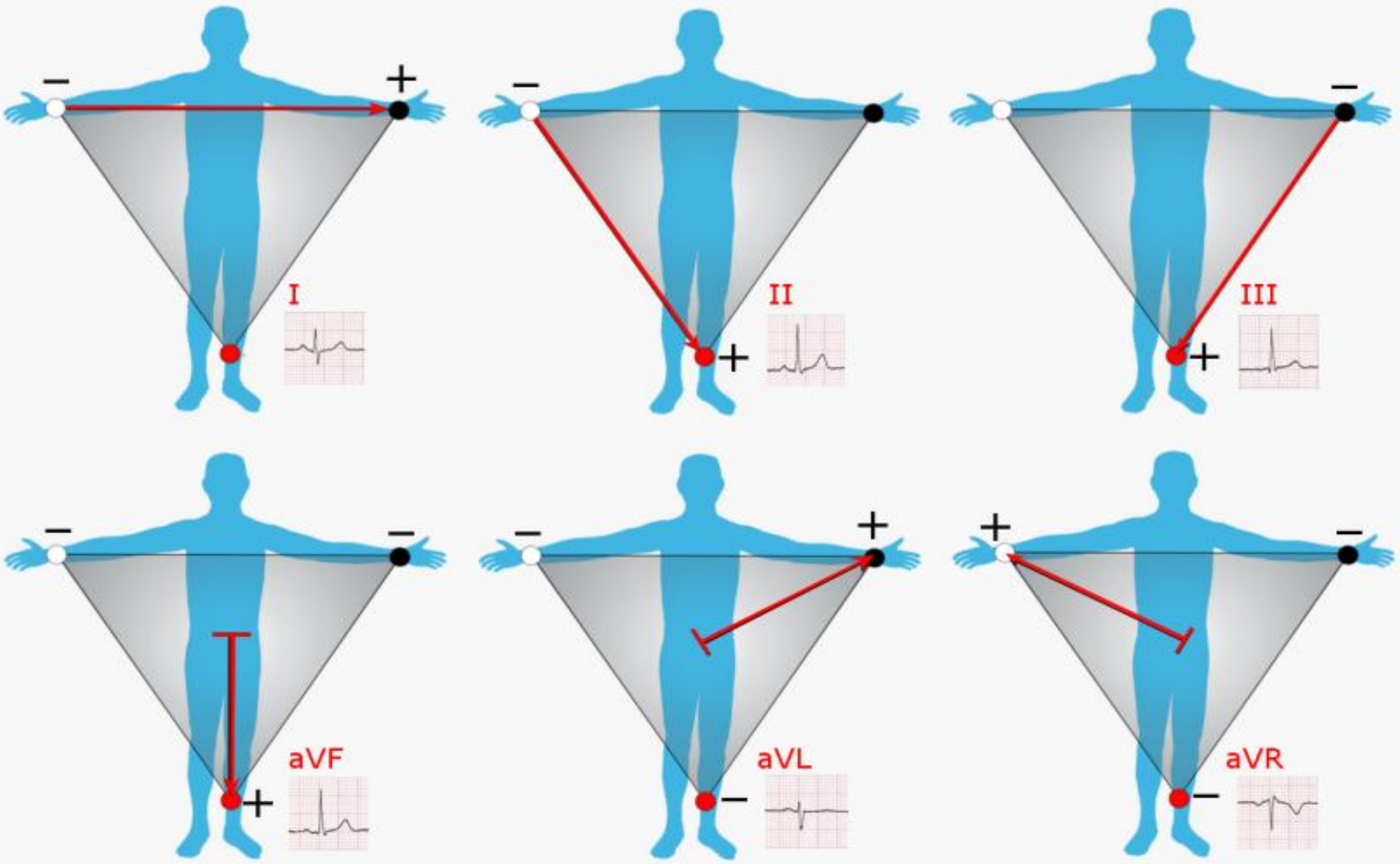
- ❑ Three unipolar leads compare voltage differences between common reference point (central terminal that is held close to zero potential on the center of heart)) and one limb
  - ➢ aVR , aVL , aVF

3. Precordial leads:

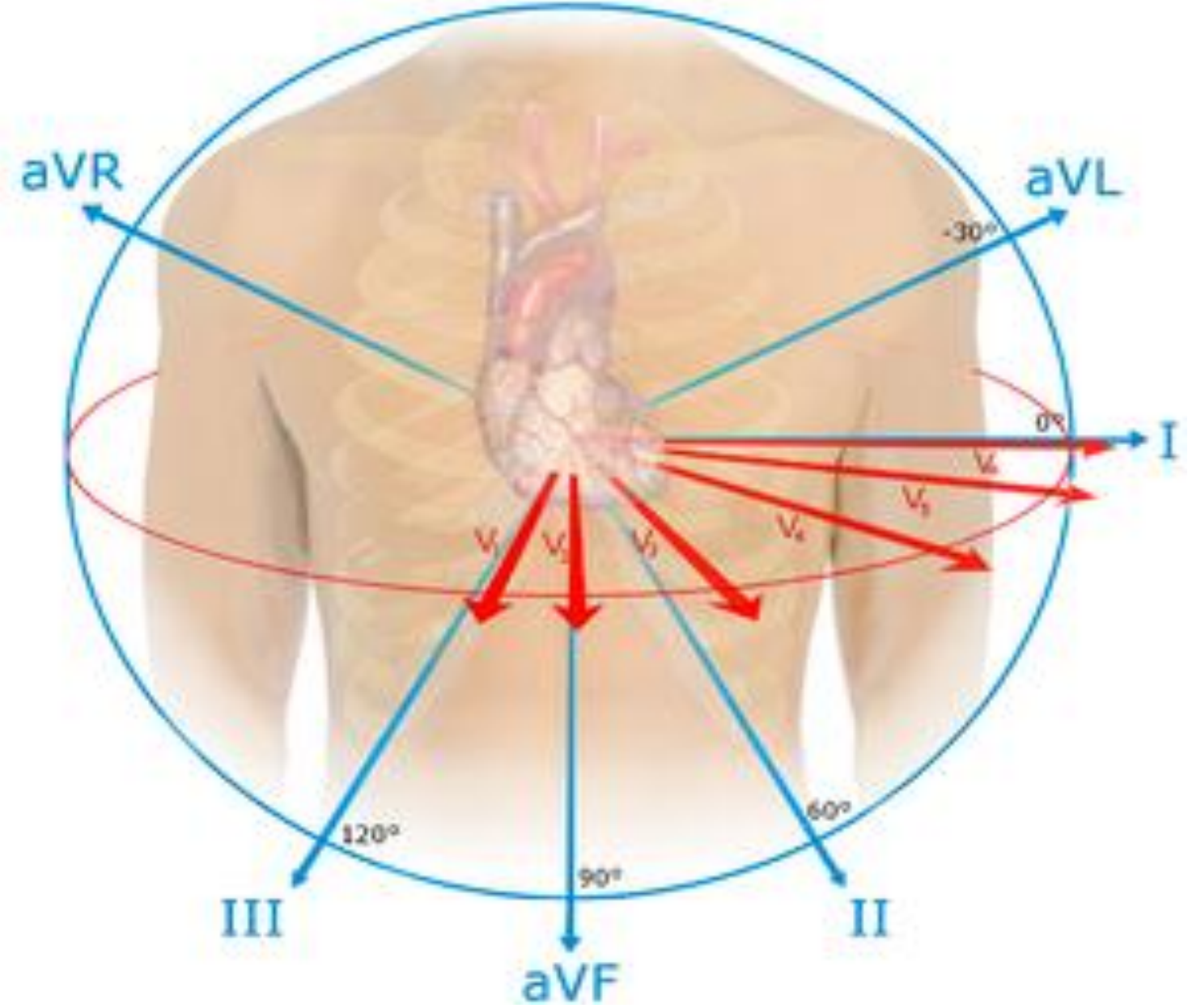
- ❑ Precordial or chest leads compare voltage differences between the common reference point (central terminal that is held close to zero potential on the center of heart) and six additional skin electrodes placed in a line directly above the heart
  - ➢ V 1, V2, V3, V4, V5, V6



Electrocardiogram (ECG)

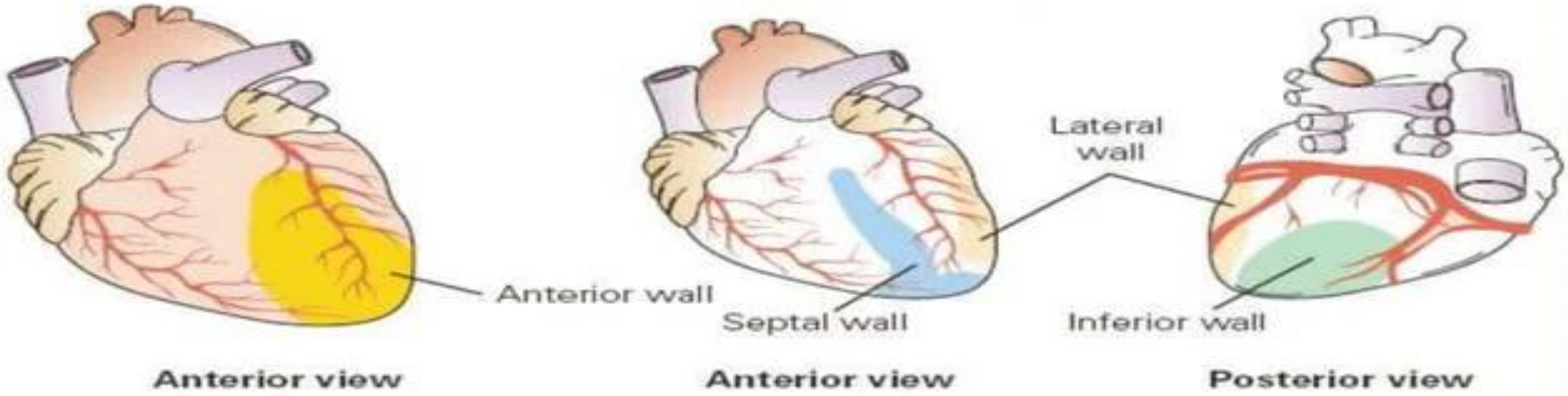


Electrocardiogram (ECG)



Electrocardiogram

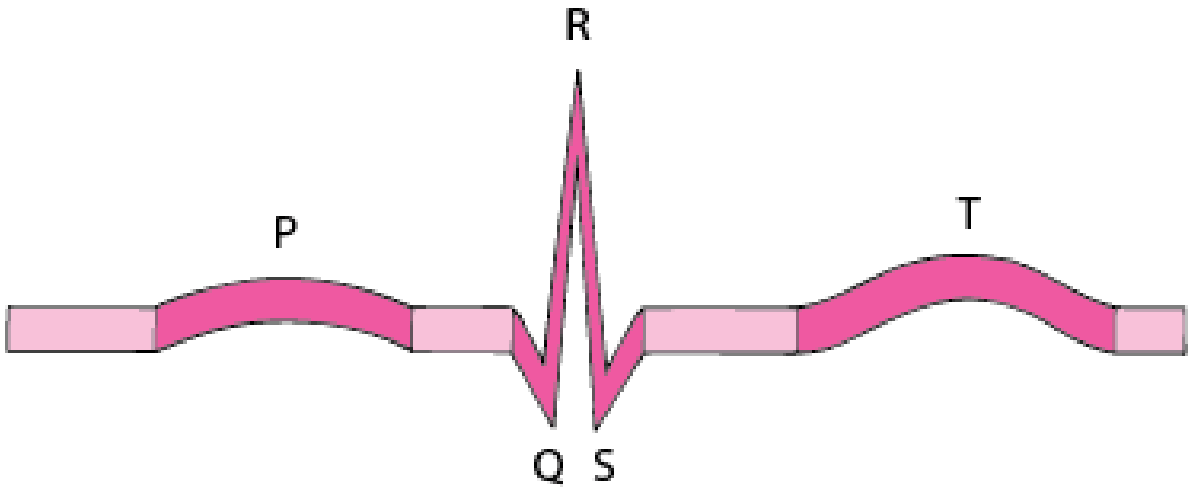
Ischemia, Injury, or Infarction in Relation to the Heart



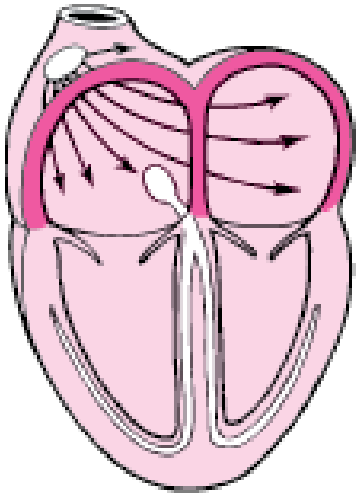
Location of MI by ECG Leads

I lateral	aVR	V_1 septal	V_4 anterior
II inferior	aVL lateral	V_2 septal	V_5 lateral
III inferior	aVF inferior	V_3 anterior	V_6 lateral

Electrocardiogram (ECG)

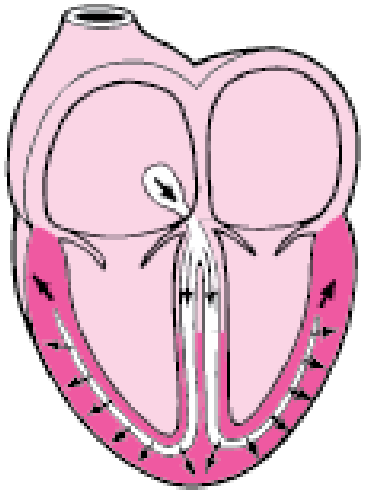


P Wave



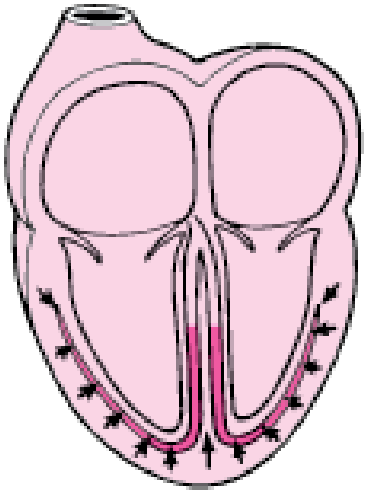
Activation of the atria

QRS Complex



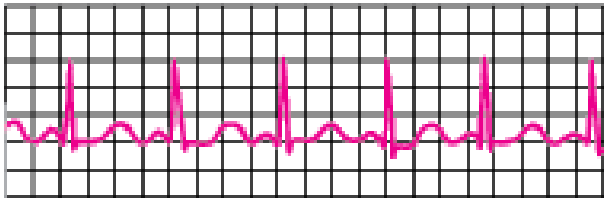
Activation of the ventricles

T Wave

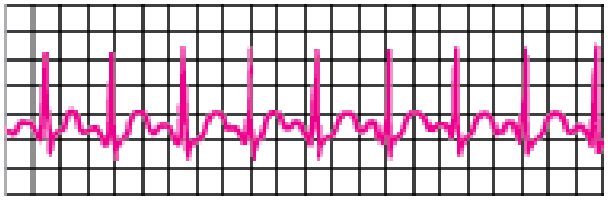


Recovery wave

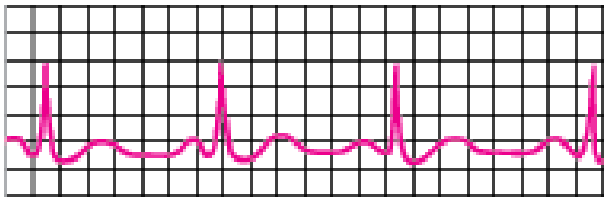
Normal Heartbeat



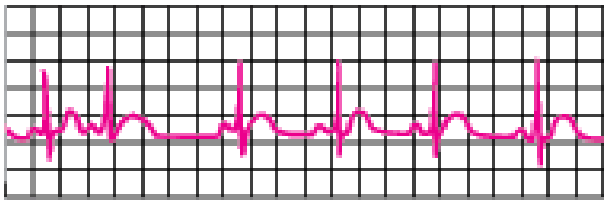
Fast Heartbeat



Slow Heartbeat

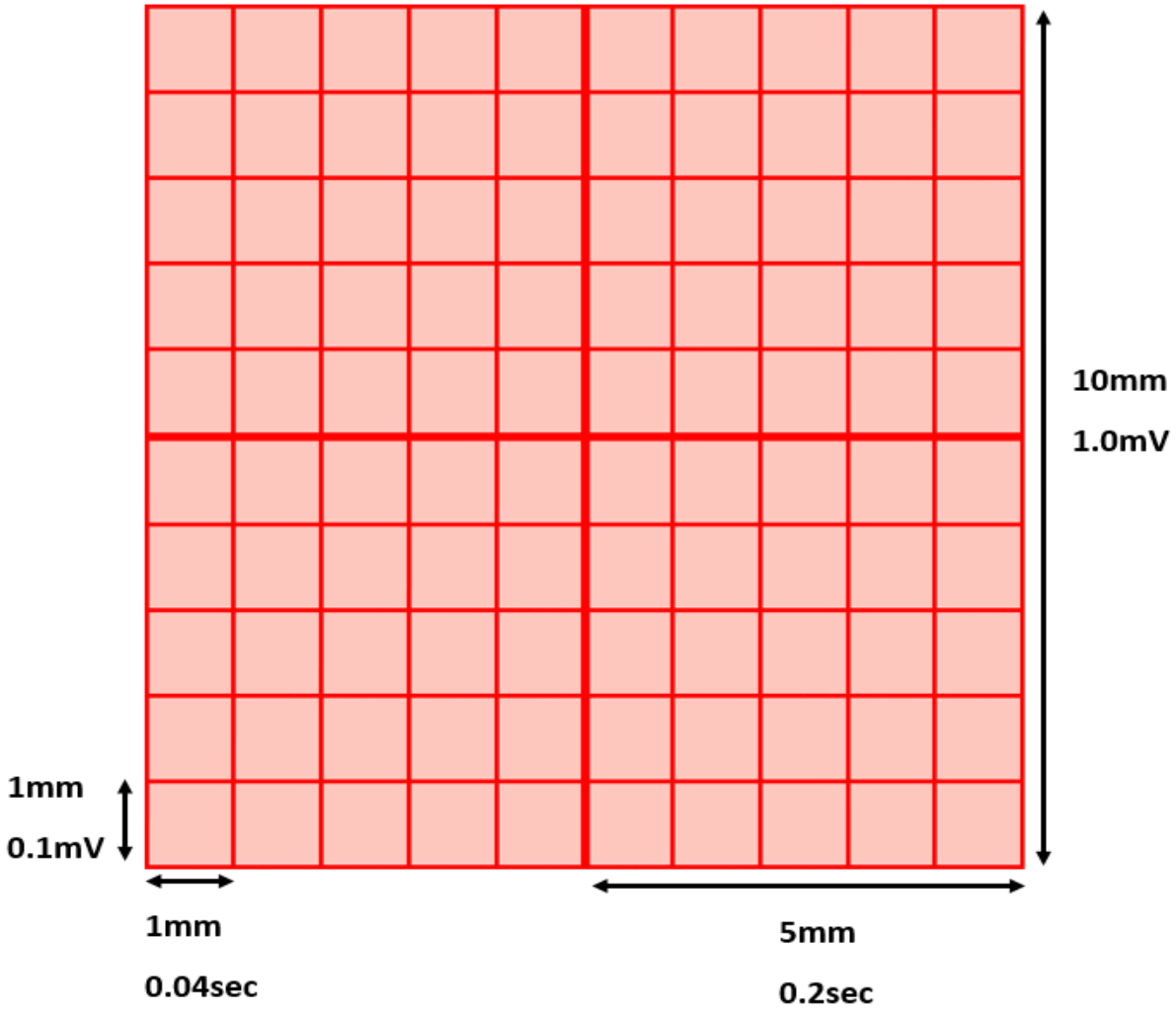


Irregular Heartbeat



Electrocardiogram

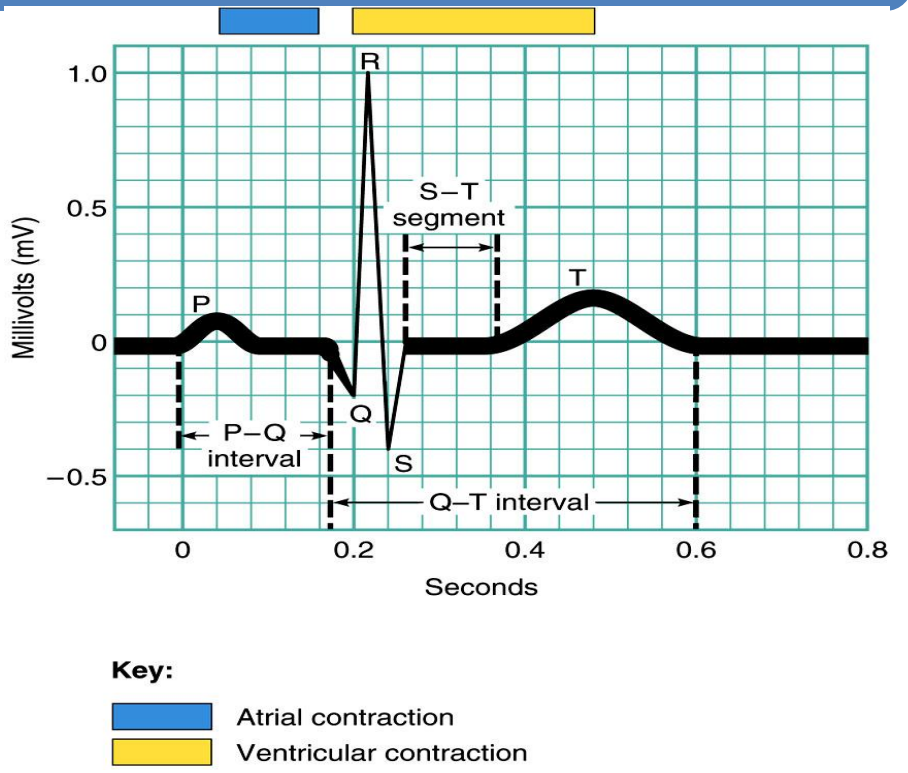
Content of ECG



Electrocardiogram

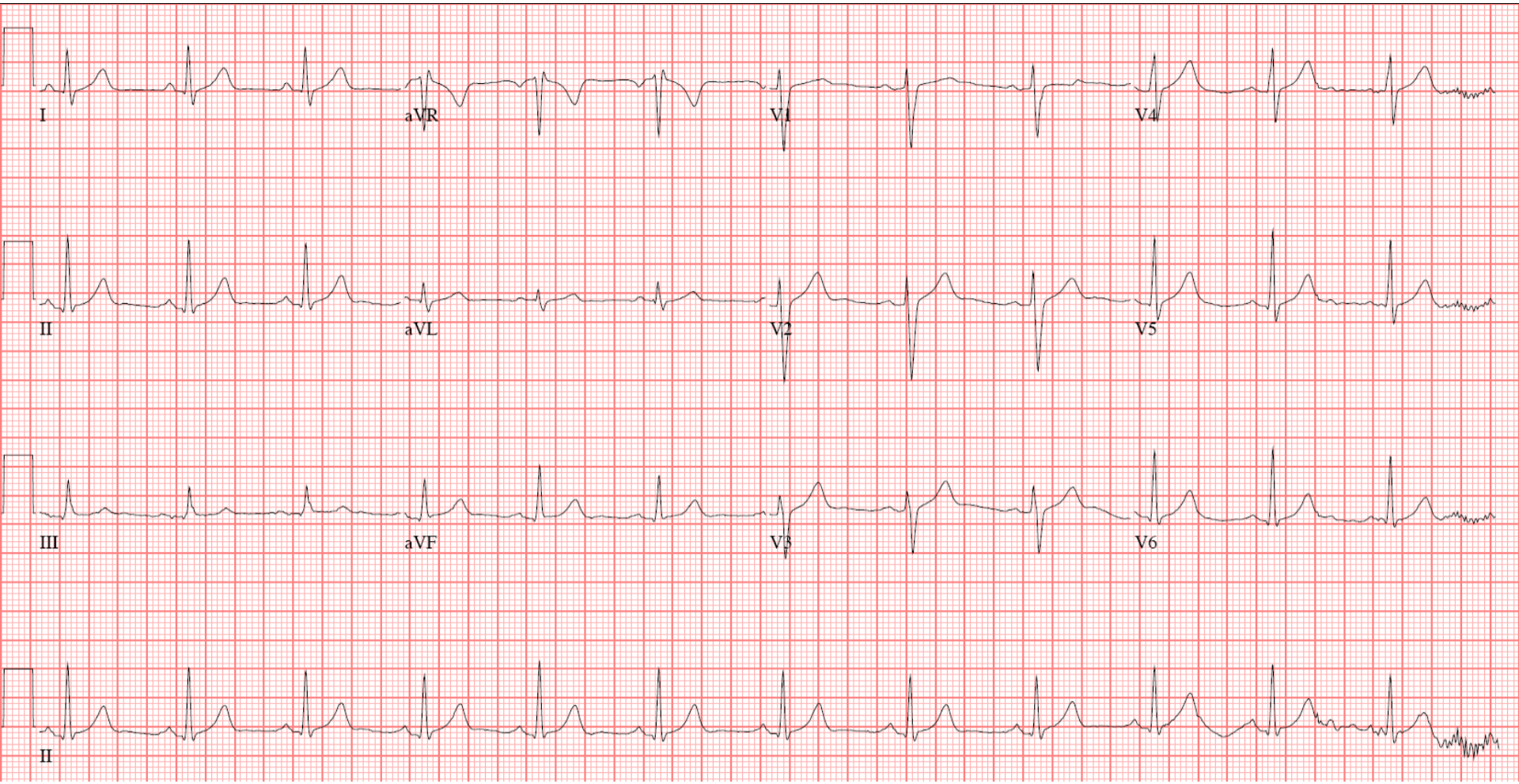
Content of ECG

- **P wave**
  - It is +ve wave
  - Duration 100-120 ms
  - Atrial depolarization (left & right)
- **PR interval**
  - AP travel from SA to AV
  - Duration 100-120 ms
- **QRS complex**
  - It is +ve or -ve wave
  - Duration less than 120 ms
  - Masks atrial repolarization
  - Ventricles depolarization
  - Beginning contraction of ventricles
- **S-T segment**
  - Continued contraction of ventricles
  - Isoelectric
- **T wave**
  - It is +ve wave
  - Repolarization of ventricles
- **Q-T Interval**
  - Depolarization and repolarization of ventricles
  - Duration less than 440 ms



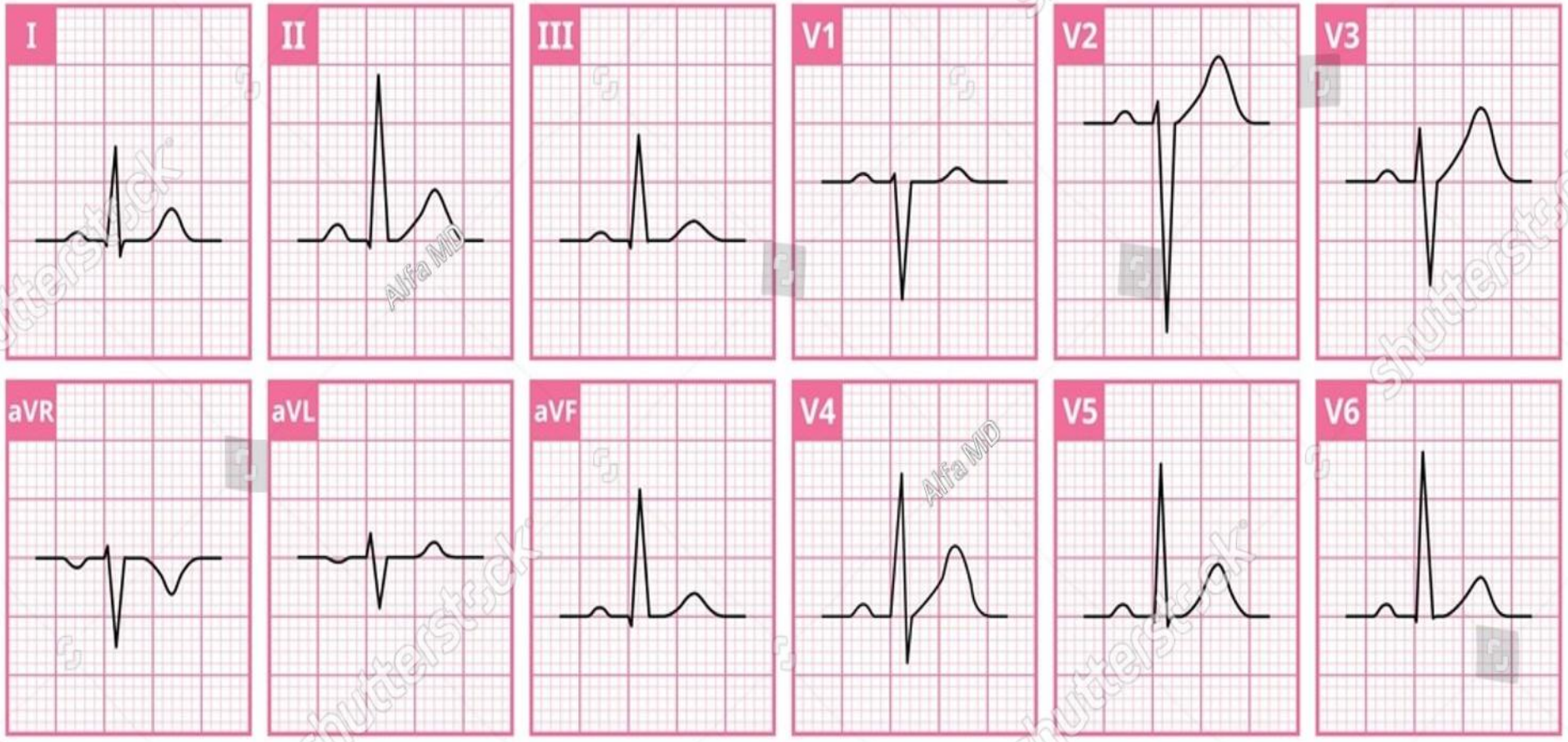
Electrocardiogram

Normal ECG



Electrocardiogram

Normal ECG



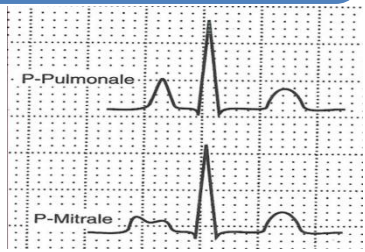
Electrocardiogram (ECG)

Importance

- ❖ **Assessment & diagnosis of certain diseases:**
  - ➢ **Arrhythmia**
  - ➢ **Myocardia infarction**
  - ➢ **Cardiac ischemia**
  - ➢ **Cardiac changes**
  - ➢ **Pericarditis**
  - ➢ **Drugs toxicity**
  - ➢ **Electrolyte disturbance**

Electrocardiogram

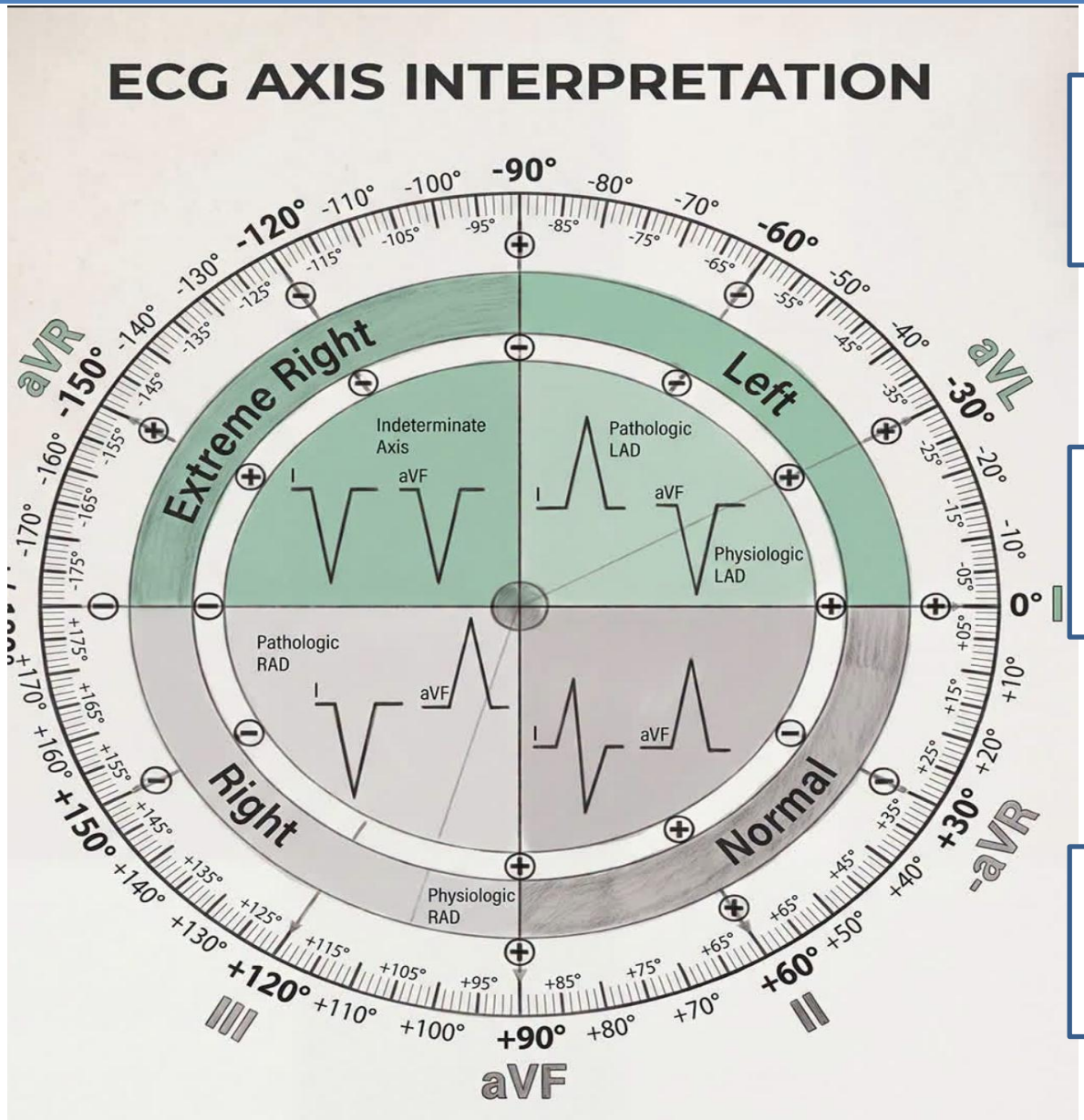
Abnormalities



- **P wave**
  - P Mitrale → Left atrial hypertrophy
  - P pulmonale → Right atrial hypertrophy
  - P inverted → Dextrocardia / AV nodal rhythm (AV node abnormality)
  - Disappearance → Atrial arrhythmia/ Hyperkalemia
- **PR Interval**
  - Long duration → First degree of HB/Parasympathetic activity
  - Short duration → AV nodal rhythm/Sympathetic activity
- **QRS complex**
  - Long duration → Hyperkalemia
  - Increased amplitude → cardiac hypertrophy
- **T wave**
  - Inverted → Severe V. hypertrophy/Ischemia/Digitalis toxicity
  - Elevated → Hyperkalemia/ Exercise/ Sympathetic activity
  - Flattened → Hypokalemia
- **S-T segment**
  - Depressed → Ischemia/Hypokalemia /Digoxin/ RBBB/ LBBB
  - Elevated → MI/ Pericarditis/ Left ventricular hypertrophy
- **Q-T Interval**
  - Long duration → Genetic abnormality /Some drugs



Electrocardiogram



Left axis deviation:
LVH
LAFB
Inferior MI

Right axis deviation:
RVH
LPFB
Lateral MI

Left axis deviation:
VT
Emphysema
Arm electrode reversal

(a) Normal ECG

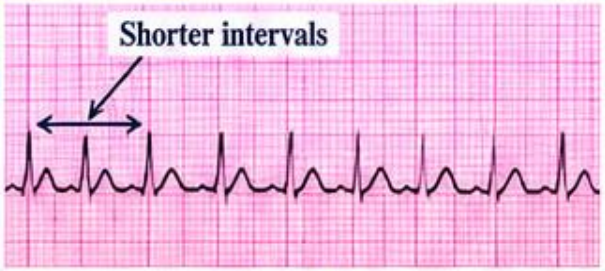


Rhythm: Regular; HR: 60–100 bpm; **P waves:** 1:1
PR: 120–200 ms; QRS: <120 ms

Normal

- Normal sinus rhythm
- Healthy heart (no arrhythmia)

(b) Sinus Tachycardia

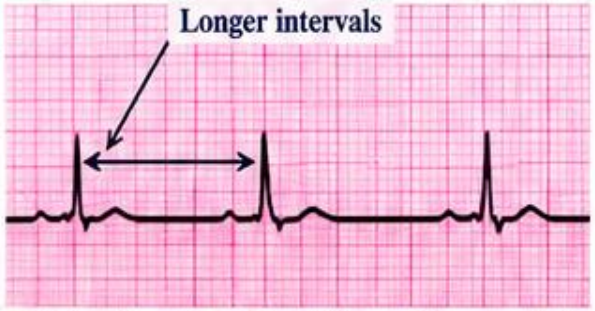


Rhythm: Regular; HR: >100 bpm; **P waves:** 1:1
PR: 120–200 ms; QRS: <120 ms

Related to

- **Sinus tachycardia**
- **Causes:**
  - Exercise
  - Fever
  - Pain, anxiety
  - Anemia, hypovolemia
  - Hyperthyroidism

(c) Sinus Bradycardia

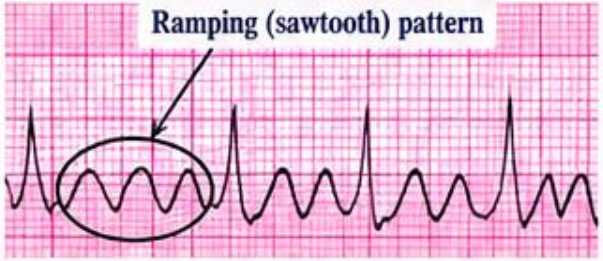


Rhythm: Regular; HR: <60 bpm; **P waves:** 1:1
PR: 120–200 ms; QRS: <120 ms

Related to

- **Sinus bradycardia**
- **Causes:**
  - High vagal tone
  - Athletes
  - Sleep
  - Hypothyroidism
  - SA node dysfunction
  - Drugs (β-blockers, digoxin)

(g) Atrial Flutter

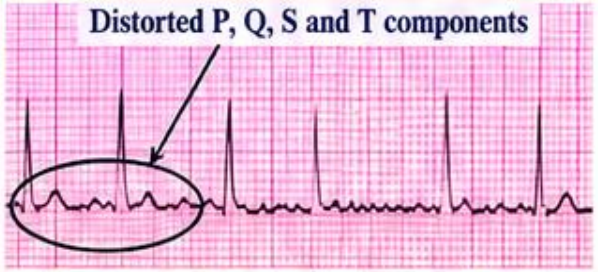


Rhythm: Regular; HR: Any; **P waves:** >1 (flutter waves)
PR: No; QRS: <120 ms

Related to

- **Atrial flutter**
- **Common causes:**
  - Structural heart disease
  - COPD, pulmonary embolism
  - Hyperthyroidism

(h) Atrial Fibrillation

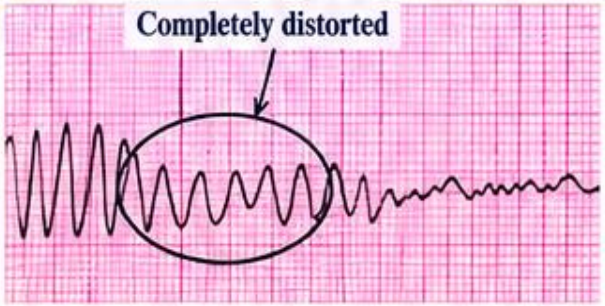


Rhythm: Irregular; HR: Any; **P waves:** >1 (absent)
PR: No; QRS: <120 ms

Related to

- **Atrial fibrillation**
- **Causes:**
  - Hypertension
  - Ischemic heart disease
  - Valvular heart disease
  - Hyperthyroidism
  - Alcohol ("holiday heart")

(i) Ventricular Fibrillation



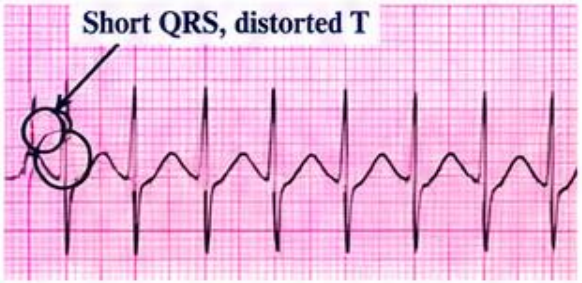
Rhythm: Irregular; HR: >250 bpm; P waves: No
PR: No; QRS: >120 ms (chaotic)

Related to

- **Ventricular fibrillation**
- **Causes:**
  - Acute myocardial infarction
  - Severe heart disease
  - Electrolyte imbalance
  - Electrocution, trauma

Dr. Essam Al-Shami

(d) Supraventricular Tachycardia (SVT)

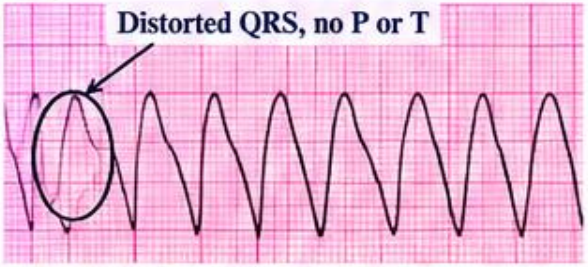


Rhythm: Regular; **HR:** >150 bpm; **P waves:** 1:1 (may be hidden)
PR: Not clear; QRS: <120 ms

Related to

- **Supraventricular tachycardia** (e.g. AVNRT, AVRT, atrial tachycardia)
- **Causes:**
  - Re-entry circuits
  - WPW syndrome
  - Digoxin toxicity
  - Stimulants (caffeine, etc.)

(e) Ventricular Tachycardia (VT)

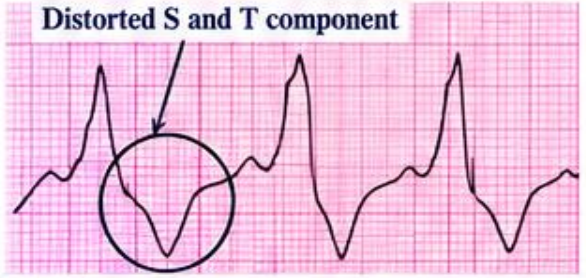


Rhythm: Regular; **HR:** >100 bpm; **P waves:** No
PR: No; QRS: >120 ms

Related to

- **Ventricular tachycardia**
- **Causes:**
  - Acute myocardial infarction
  - Cardiomyopathy
  - Scar after MI
  - Electrolyte imbalance
  - Drug toxicity

(f) Bundle Branch Block (e.g. LBBB/RBBB)

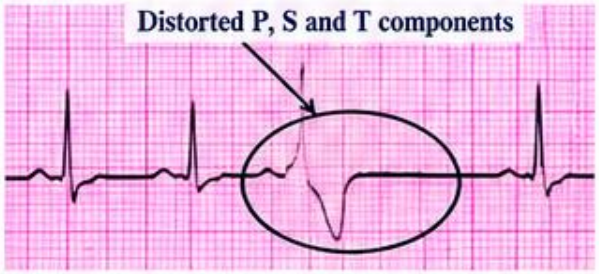


Rhythm: Regular; **HR:** Any; **P waves:** 1:1
PR: –; QRS: >120 ms (wide)

Related to

- **Bundle branch block**
- **Causes:**
  - Ischemic heart disease
  - Hypertension
  - Cardiomyopathy
  - Degenerative conduction system disease

(j) Ventricular Premature Beat (PVC)

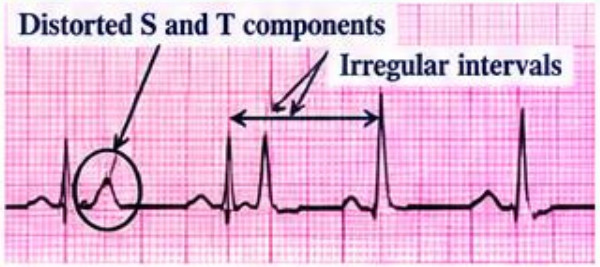


Rhythm: Regular (with premature beat); **HR:** Variable
P waves: No (for PVC); PR: –; QRS: >120 ms

Related to

- **Ventricular premature beat (PVC)**
- **Causes:**
  - Ischemic heart disease
  - Electrolyte imbalance
  - Stimulants (caffeine, nicotine)
  - Drugs (e.g. digoxin)
  - Normal in some healthy individuals

(k) Atrial Premature Beat (APC)

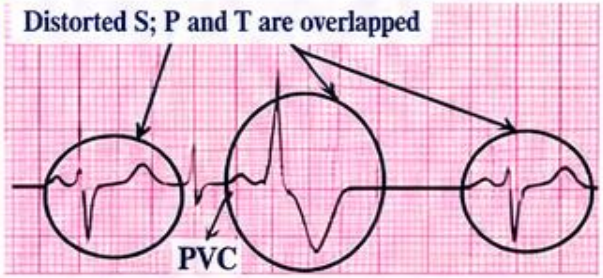


Rhythm: Regular (with premature beat); **HR:** Any
P waves: 1' (abnormal P); PR: >120 ms; QRS: <120 ms

Related to

- **Atrial premature beat**
- **Causes:**
  - Atrial irritability
  - Stress, caffeine, smoking
  - Electrolyte imbalance
  - Pulmonary disease
  - Structural heart disease

(l) Fusion Beat



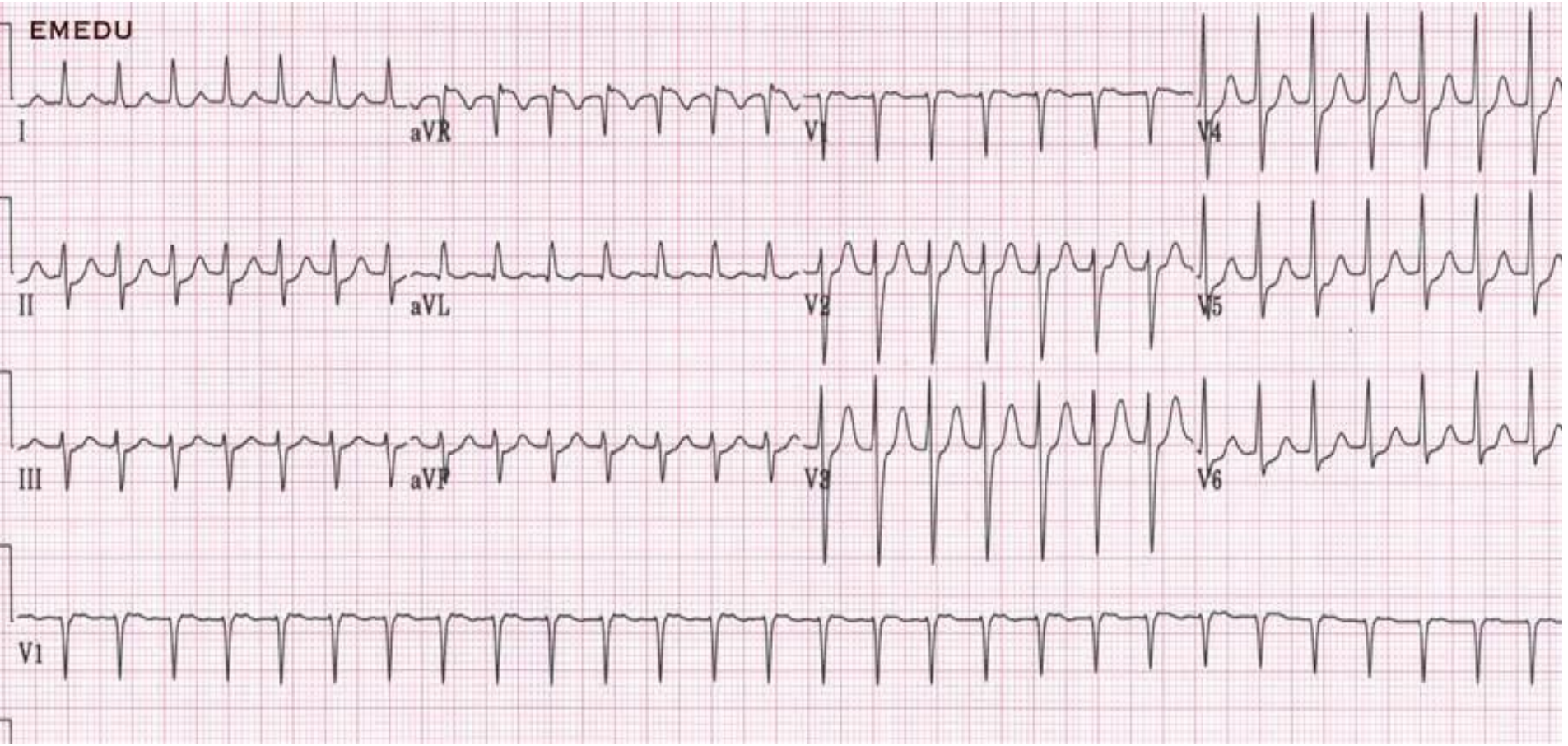
Rhythm: Regular; **HR:** Any; **P waves:** No
PR: –; QRS: >120 ms

Related to

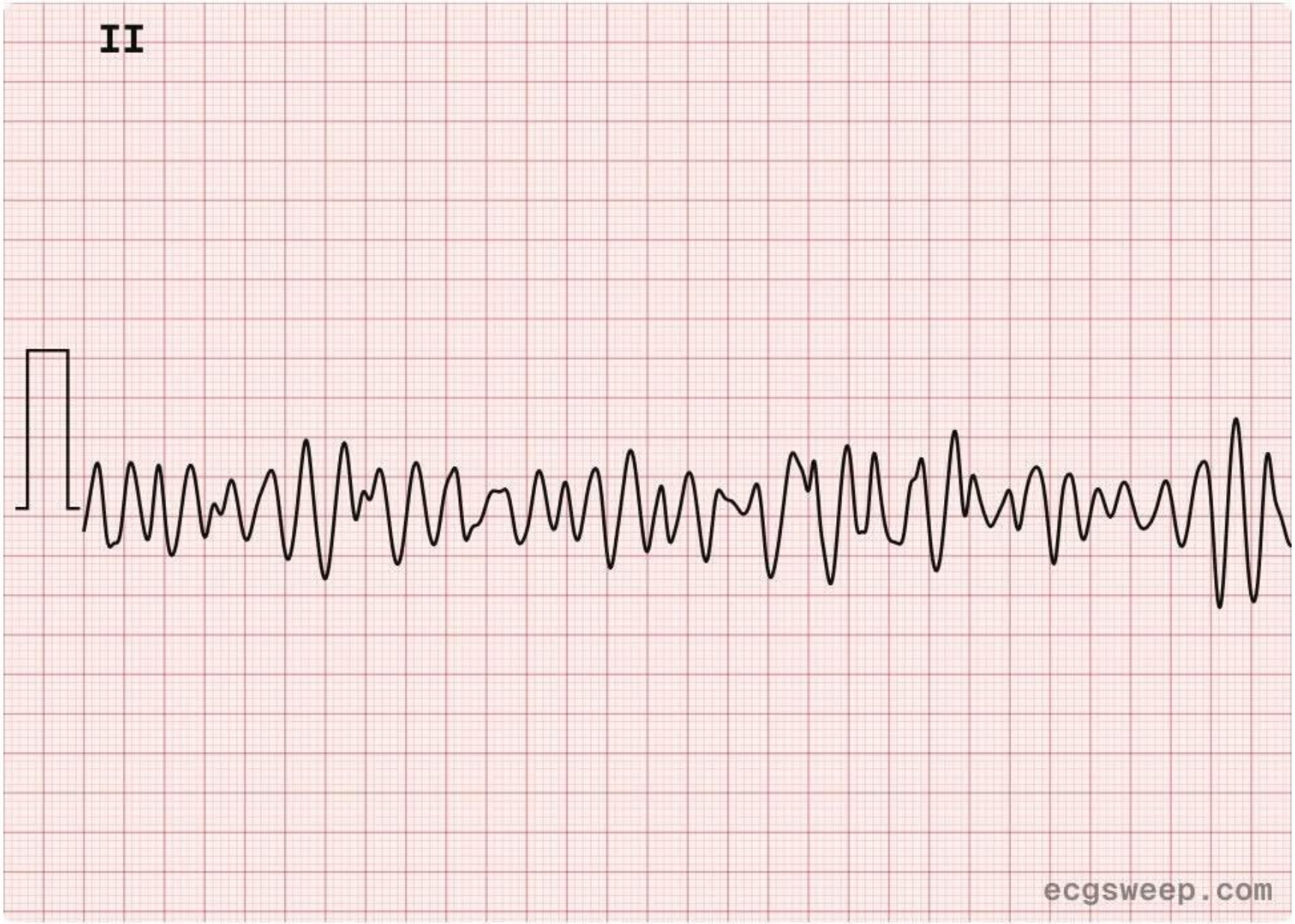
- **Fusion beat (normal + PVC)**
- **Seen in:**
  - Ventricular tachycardia
  - During intermittent capture/fusion
  - Indicates coexistence of sinus rhythm and VT

Electrocardiogram

Abnormal ECG



What's the diagnosis?



Heart Sounds

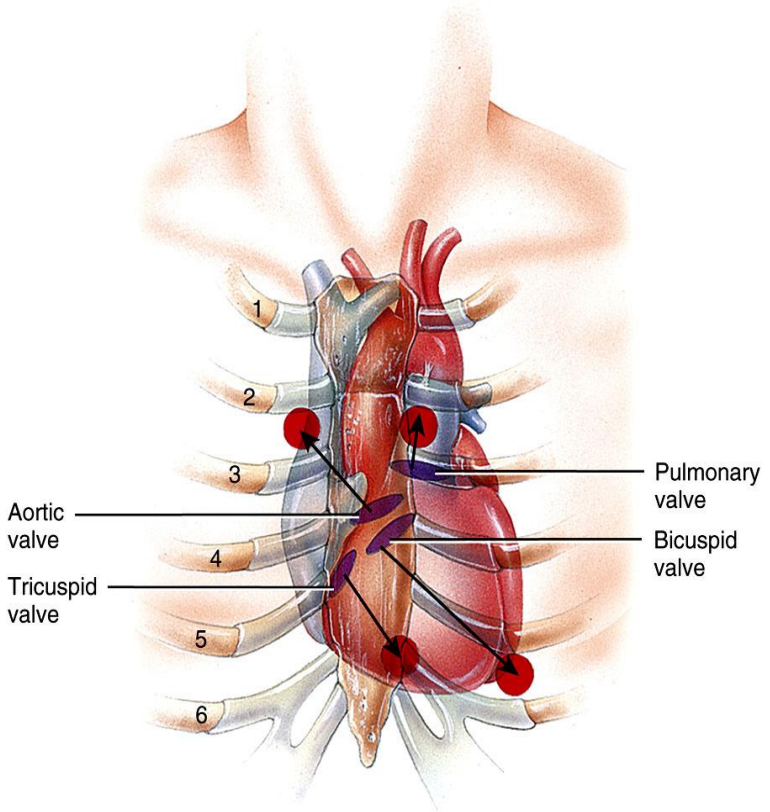


- Auscultation
- Using stethoscope
- Sound of heartbeat comes
- Primarily from blood pressure on valves
- Caused by closing of heart valves
- Sound
  - Lubb (S1)– AV valves close
  - Dupp (S2)– SL valves close

Abnormality

Heart murmurs due to:

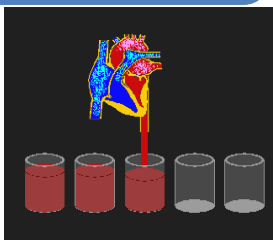
- Valves stenosis
- Valves regurgitation
- Heart holes



Anterior view of heart valve locations and

Cardiac output

- Amount of blood pumped by each ventricle of the heart in one minute



CO =(stroke volume [SV]) x (heart rate [HR])

Heart rate (HR)

- The number of cardiac cycles per 1 minute (60-100 bpm)
- Means HR = 70 bpm

Stroke volume (SV)

- Volume of blood pumped by each ventricle in one contraction
- SV= EDV-ESV= 130ml-50ml=80ml

- CO = SV X HR
  - FOR A RESTING ADULT
- CO = 80mL/beat x 70 beats/min
- = 560 mL/min = 5.6 L/min

Factors affect SV

1-Cardiac pumping power (Contractility)

- ❑ Sympathetic activity/Parasympathetic activity
- ❑ Ventricular hypertrophy/Arrhythmia
- ❑ Positive inotropic drugs (Noradrenalin/Digoxin)
- ❑ Negative inotropic drugs (Bisoprolol/Verapamil)

2-Preload

- ❖ Degree of stretch of cardiac muscle before it contracts
- ❖ Greater preload increases the stroke volume
- ❑ VC and/or hypervolemia of veins increases the stroke volume

3-Afterload

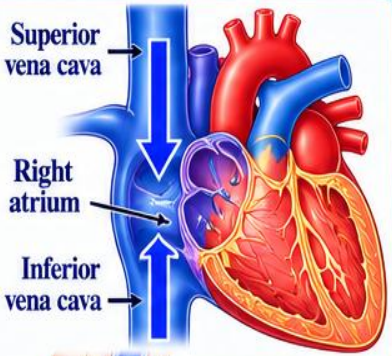
- ❖ Pressure that must be overcome before a semilunar valve can open
- ❖ Increase in afterload causes stroke volume to decrease
- ❑ Hypertension, polycythemia and atherosclerosis increase afterload

Venous Return: Determinants and Mechanisms

1 What is Venous Return?

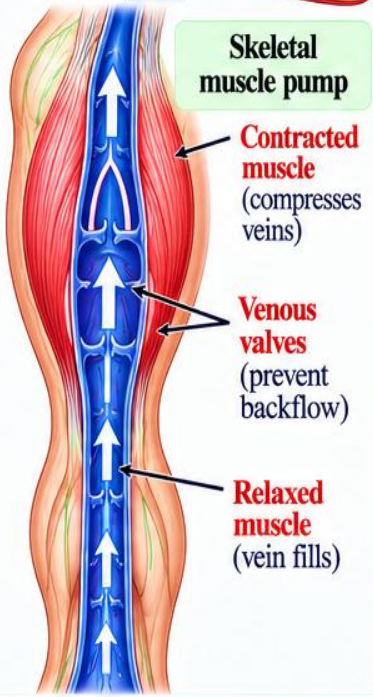
Flow of blood from systemic veins to the right atrium per minute (≈ **Cardiac output** at steady state).

Normal value ≈ 5 L/min (70 ml/kg/min)



2 Determinants of Venous Return

1. **Mean systemic filling pressure (Pms)**
   Pressure in systemic veins when there is no flow (≈ 7 mmHg).
2. **Right atrial pressure (Pra)**
   Normally low (0–2 mmHg).
3. **Venous resistance**
   Affected by venous tone (sympathetic stimulation ↓ compliance).
4. **Skeletal muscle pump**
   Contraction of skeletal muscles compresses veins (with valves).
5. **Respiratory pump**
   Inspiration ↓ intrathoracic pressure → increases venous return.
6. **Venous valves**
   Prevent backflow, ensure one-way flow towards the heart.



3 Factors That Increase or Decrease Venous Return

↑ Venous return	↓ Venous return
• ↑ Blood volume	• ↓ Blood volume (hemorrhage, dehydration)
• ↑ Venoconstriction (sympathetic)	• Venodilation
• Skeletal muscle contraction	• Standing position (venous pooling)
• Inspiration (↓ intrathoracic pressure)	• Increased right atrial pressure (e.g. right heart failure, tamponade)
• Compression of abdomen (e.g. exercise)	• Obstruction of vena cava
• Supine position (↑ venous return)	

Frank–Starling Mechanism of the Heart

1 Law of the Heart

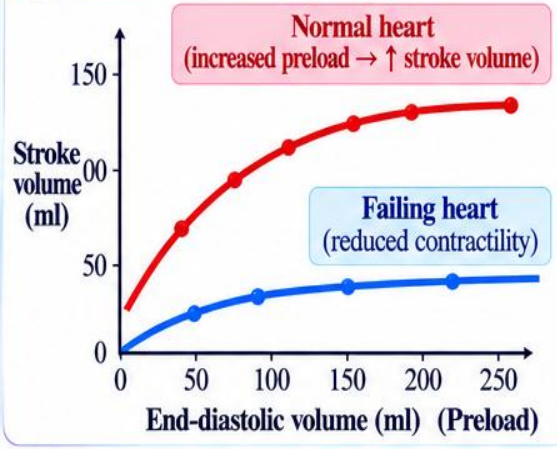
Within physiological limits, increased stretch (preload) of cardiac muscle fibers → stronger contraction → increased stroke volume.

Mechanism at the cellular level



More cross-bridge formation → higher force of contraction

2 Frank–Starling Curves



3 Effect on Cardiac Output

↑ Venous return (↑ preload, ↑ EDV)

↓

↑ Stroke volume (via Frank–Starling)

↓

↑ Cardiac output (CO = HR × SV) (at normal contractility)

4 Integration: Venous Return and Frank–Starling Mechanism

↑ Blood volume or ↑ Venous return → ↑ End-diastolic volume (preload) → ↑ Myocardial fiber stretch → ↑ Stroke volume → ↑ Cardiac output

5 Clinical Significance

Hypovolemia (hemorrhage, dehydration)



↓ Venous return
↓ Preload
↓ Stroke volume
↓ Cardiac output

Heart failure (reduced contractility)



↑ Preload (smaller effect)
Low Frank–Starling curve
↓ Stroke volume

Exercise



↑ Venous return (muscle & respiratory pumps)
↑ Preload
↑ Stroke volume
↑ Cardiac output

Postural change (standing)



↓ Venous return (venous pooling)
↓ Preload
↓ Stroke volume (temporary)

Factors affect HR

1-Factors increase heart rate

- ❑ Sympathetic nervous system
- ❑ Positive inotropic drugs (Noradrenalin/Atropine)
- ❑ Diseases: Hyperthyroidism, Pheochromocytoma
- ❑ Exercise, emotional stress, hypoxia, hypotension

❖ ↑ HR > 200 bpm →↓SV →↓CO

2-Factors decreased heart rate

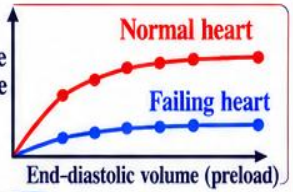
- ❑ Parasympathetic nervous system
- ❑ Negative inotropic drugs (Bisoprolol/Verapamil /Digoxin)
- ❑ Diseases: Hypothyroidism

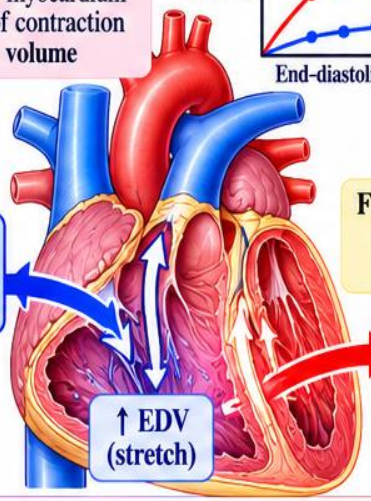
Physiology of Cardiac Regulation

Regulation of Heart Rate, Contractility and Cardiac Output

1 Intrinsic Regulation (Frank–Starling Mechanism)

↑ Venous return
(↑ End-diastolic volume)
→ stretch of myocardium
→ ↑ force of contraction
→ ↑ Stroke volume





Venous return (preload)

Frank–Starling mechanism (intrinsic)

↑ Stroke volume

↑ EDV (stretch)

2 Autonomic Nervous Regulation

Sympathetic (SNS)
Norepinephrine (heart)

- ↑ Heart rate (positive chronotropic)
- ↑ Contractility (positive inotropic)
- ↑ Conduction velocity (positive dromotropic)
- ↑ Relaxation rate (positive lusitropic)

Receptors: β_1-adrenergic (↑ cAMP)



Parasympathetic (PNS)
Acetylcholine (vagus nerve)

- ↓ Heart rate (negative chronotropic)
- ↓ Conduction velocity (negative dromotropic) (especially AV node)
- ↓ Slight ↓ in atrial contractility

Minimal effect on ventricular contractility

Receptors: M_2 muscarinic (↓ cAMP, ↑ K^+ efflux)

3 Hormonal Regulation

Adrenaline & Noradrenaline (adrenal medulla)



- ↑ HR
- ↑ Contractility
- ↑ Cardiac output

Thyroid hormones (T_3, T_4)



- ↑ HR
- ↑ β_1 receptor expression
- ↑ Contractility
- ↑ Cardiac output

Renin–Angiotensin–Aldosterone (RAAS)



Angiotensin II:
- ↑ Afterload (vasoconstriction)
- ↑ Aldosterone → ↑ blood volume → ↑ venous return → ↑ CO

Antidiuretic hormone (ADH)



- ↑ Water reabsorption → ↑ Blood volume → ↑ Venous return

4 Reflex Regulation

Baroreceptor Reflex (short-term)



↑ Arterial pressure → ↑ baroreceptor firing → ↑ vagal tone & ↓ sympathetic tone → ↓ HR, ↓ contractility

Chemoreceptor Reflex



↓ O_2, ↑ CO_2 or ↓ pH → ↑ sympathetic tone → ↑ HR and contractility

Bainbridge Reflex (atrial stretch)



↑ Venous return (atrial stretch) → ↑ HR (via sympathetic reflex)

Bezold–Jarisch Reflex



Stimulation of ventricular receptors → bradycardia, ↓ contractility, ↓ BP

Cushing Reflex



↑ Intracranial pressure → ↑ sympathetic tone → ↑ BP and reflex bradycardia

5 Other Factors Affecting Cardiac Function

Preload (EDV)



↑ Venous return → ↑ EDV → ↑ Stroke volume (Frank–Starling)

Afterload (Arterial pressure)



↑ Afterload → ↓ Stroke volume
↑ Myocardial O_2 demand

Contractility (Inotropy)



Affected by:
Sympathetic activity, circulating catecholamines, thyroid hormones, digitalis, Ca^{2+} level

Heart rate (Chronotropy)



↑ HR → ↑ CO (to a limit)
Very high HR → ↓ filling → ↓ CO

Venous return



Affected by: blood volume, venous tone, skeletal muscle pump, respiration, body position

6 Integrated Regulation of Cardiac Output

Nervous system (SNS ↑ PNS ↓)



Rapid (seconds)

Hormones (catecholamines, thyroid, RAAS, ADH)



Intermediate (minutes–hours)

Reflexes (baroreceptors, chemoreceptors, etc.)



Rapid (seconds)

→ Change in HR, Contractility, Preload and Afterload

→ Cardiac Output (CO) = HR × Stroke Volume

- Maintain arterial blood pressure
- Meet tissue metabolic demand
- Adapt to exercise, stress and disease

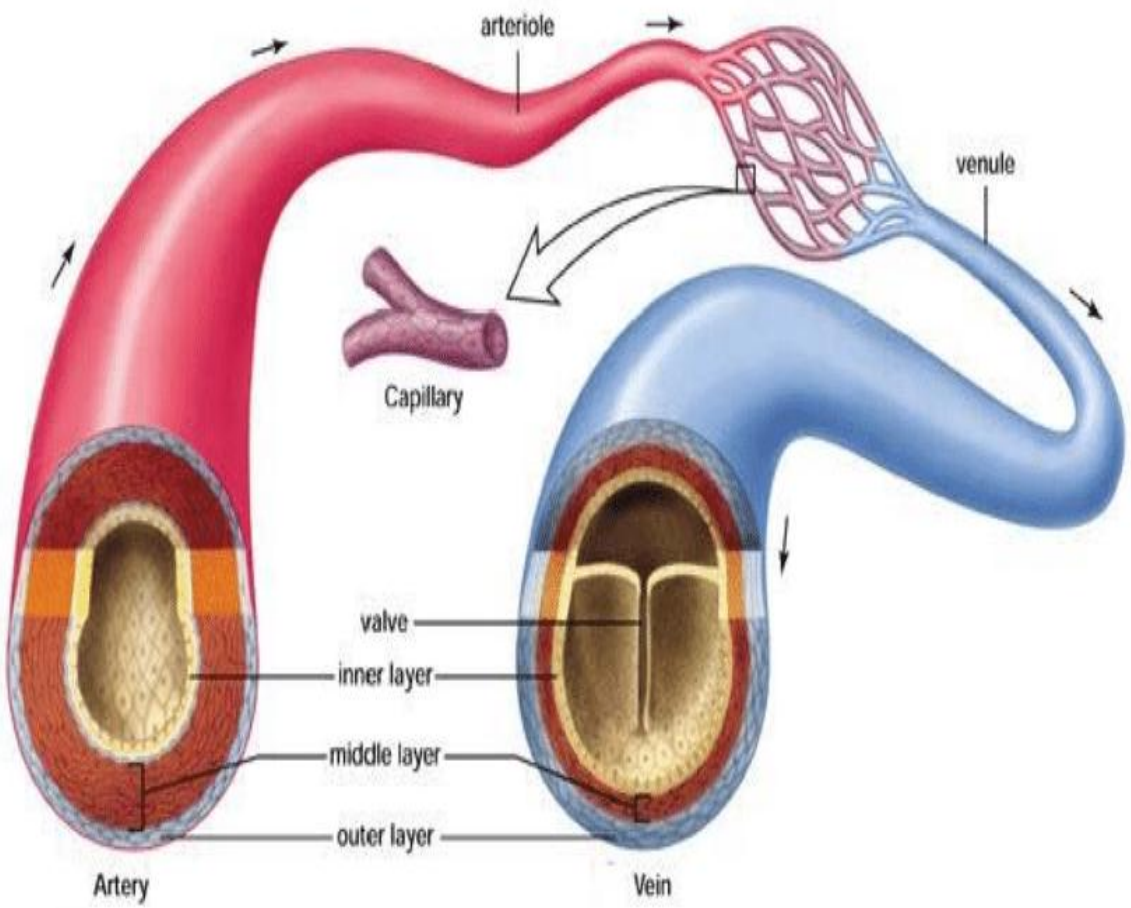
Blood vessels

Function

- Taking blood to the tissues and back

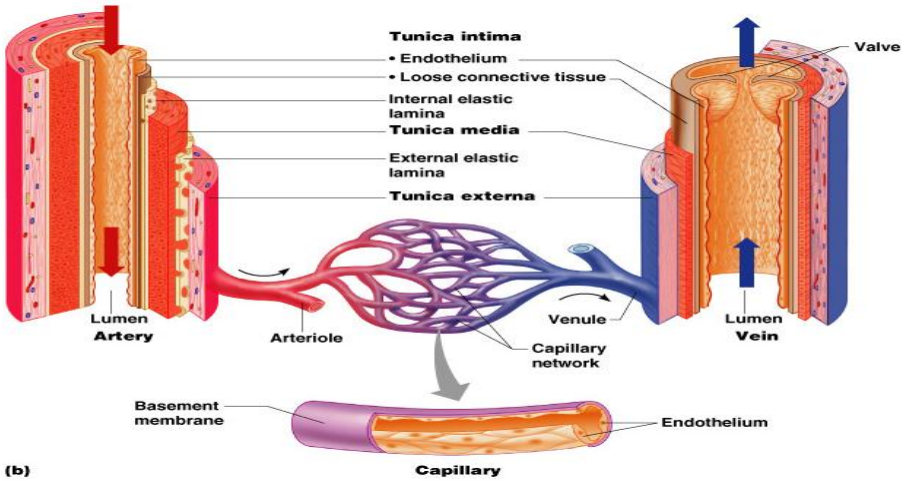
Component

- **Arteries**
  - **Large**
  - **Small**
- **Arterioles**
- **Capillaries**
- **Venules**
- **Veins**
  - **Small**
  - **Large**



Blood vessels

Three layers (tunics)



Tunica intima:

- ❖ Endothelium/Elastic fibers.
- ❖ Fenestrated
- ❑ Adjusting oscillations in blood flow (flexibility)
- ❑ Angiogenesis

Tunica media

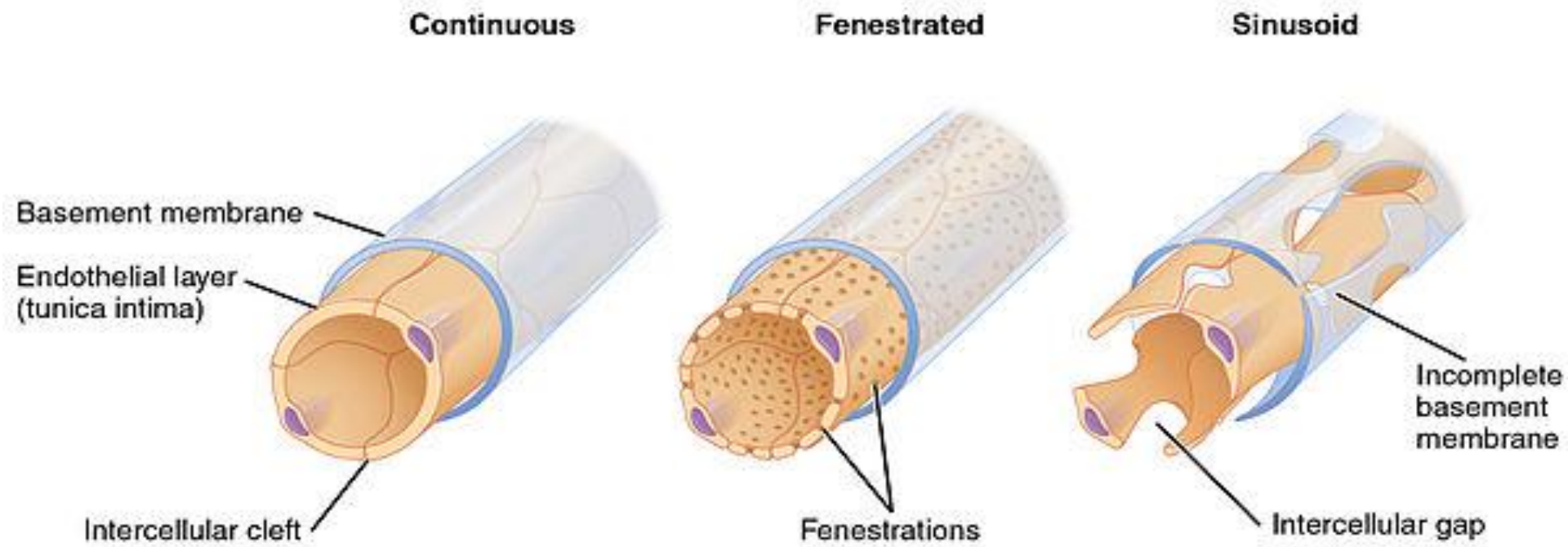
- ❖ Smooth muscle/Elastic fibers
- ❖ Controlled by sympathetic nervous system
- ❖ Regulate internal diameter of blood vessels

Tunica externa

- ❖ Mostly fibrous connective tissue
- ❖ Protect & prevent over expansion of blood vessels

TYPES OF Blood vessels

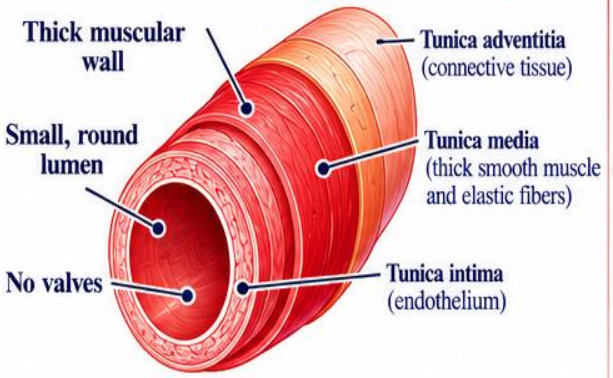
Continuous	Fenestrated	Sinusoid
Fat	Intestine	Liver
Muscles	Kidney	Spleen
Nervous system	Endocrine glands	Bone marrow



Arteries vs. Veins: Structure, Function and Clinical Significance

Arteries

Carry blood away from the heart



Main features

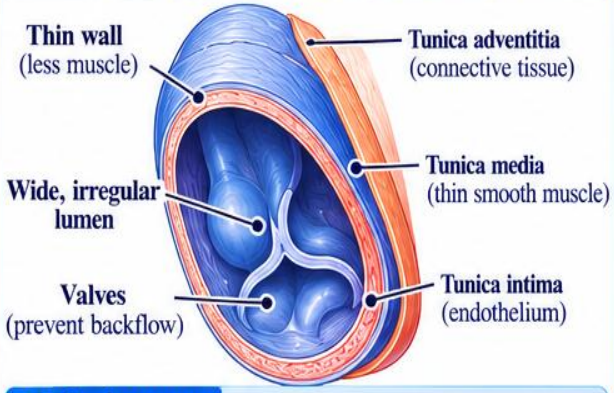
- High pressure, pulsatile flow
- Thick wall, more smooth muscle & elastic tissue
- Narrow, round lumen
- No valves (except at heart)
- Usually deep (except some e.g. radial)

Comparison

Feature	Arteries	Veins
Direction of flow	Away from heart	Toward heart
Pressure	High (pulsatile)	Low (non-pulsatile)
Wall thickness	Thick	Thin
Smooth muscle	More	Less
Elastic tissue	More (large arteries)	Less
Lumen	Narrow, round	Wide, irregular
Valves	Absent (except at heart)	Present (especially in limbs)
Location	Mostly deep	Superficial and deep
Blood volume	~ 15% of total	~ 65% of total (capacitance vessels)
Oxygen content (systemic)	Usually oxygenated (except pulmonary and umbilical arteries)	Usually deoxygenated (except pulmonary and umbilical veins)
Main function	Distribute blood under high pressure	Return blood to heart and act as blood reservoir

Veins

Carry blood toward the heart

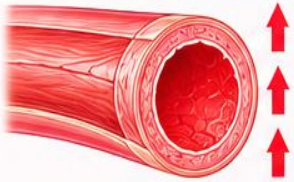


Main features

- Low pressure, non-pulsatile flow
- Thin wall, less smooth muscle & elastic tissue
- Wide, irregular lumen
- Valves (especially in limbs)
- Serve as capacitance vessels (hold most of blood volume)
- Superficial and deep veins

Clinical Significance

1. Arterial hypertension



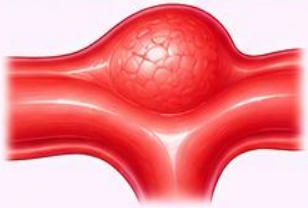
- Increased arterial resistance (vasoconstriction, structural changes, atherosclerosis)
- Leads to high blood pressure and cardiovascular risk

2. Atherosclerosis



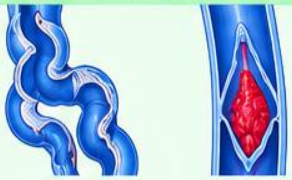
- Fatty plaque in arterial wall
- Reduces lumen → ischemia (heart, brain, limbs)
- Can cause myocardial infarction, stroke, peripheral arterial disease

3. Aneurysm



- Weakening and dilatation of arterial wall (e.g. aorta)
- Risk of rupture (life-threatening)

4. Varicose veins & DVT



Varicose veins (incompetent valves) | Deep vein thrombosis (DVT)

- Valve incompetence → venous pooling → varicose veins (legs)
- Venous stasis → thrombosis (DVT)
- Risk of pulmonary embolism (PE)

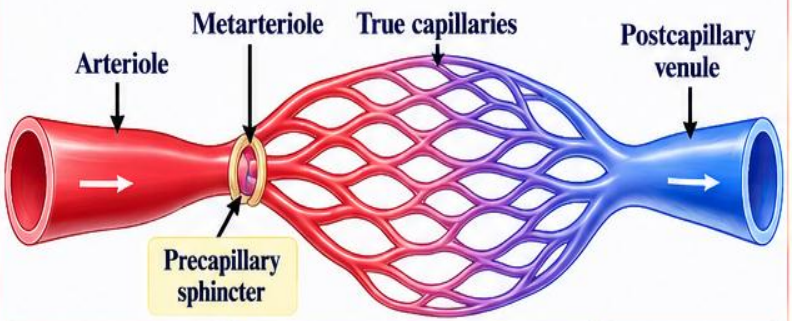
5. Clinical measurements and procedures



- Blood pressure measured in arteries (e.g. brachial artery)
- Venous access (e.g. peripheral or central venous catheter)
- Venous return affected by posture, muscle pump and intrathoracic pressure

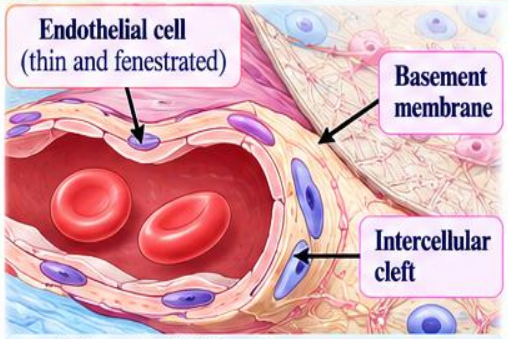
Microcirculation: Capillary Exchange & Pressure Gradients

1 Components of Microcirculation



Arteriole (high pressure)	Capillary network (site of exchange)	Venule (low pressure)

2 Structure of Capillary Wall



- Thin wall (0.5–1 μm)
- Fenestrations (except in brain)
- Allows exchange of water, solutes and some proteins

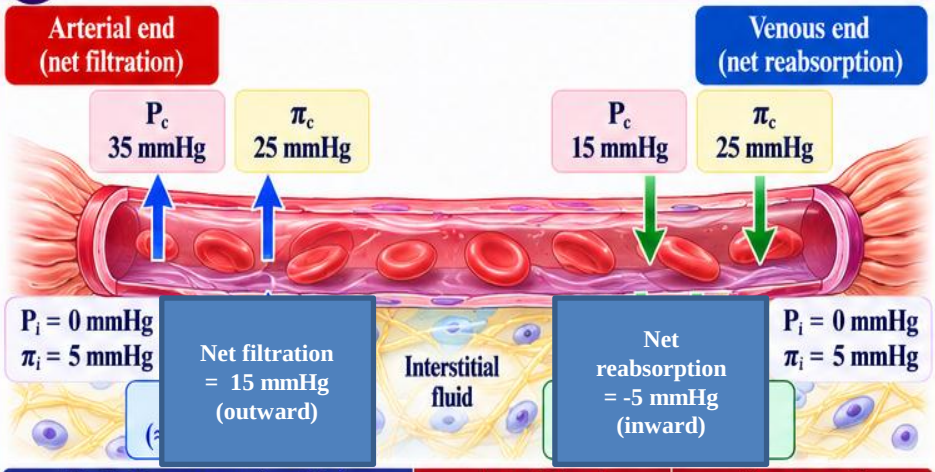
3 Capillary Exchange Forces

Starling equation:

$$J_v = K_f [(P_c - P_i) - \sigma (\pi_c - \pi_i)]$$

- J_v : Net fluid movement
- K_f : Filtration coefficient (permeability × surface area)
- P_c : Capillary hydrostatic pressure (pushes fluid out)
- P_i : Interstitial hydrostatic pressure (pushes fluid in)
- π_c : Plasma oncotic pressure (pulls fluid in)
- π_i : Interstitial oncotic pressure (pulls fluid out)
- σ : Reflection coefficient for proteins (0–1)

4 Pressure Gradients along The Capillary



Typical pressures (mmHg)	Arterial end	Venous end
Capillary hydrostatic pressure (P_c)	35	15
Plasma oncotic pressure (π_c)	25	25
Interstitial hydrostatic pressure (P_i)	0	0
Interstitial oncotic pressure (π_i)	5	5
Net effect	**Filtration (out)**	**Reabsorption (in)**

5 Factors Affecting Capillary Exchange

Increase filtration (↑ J_v)

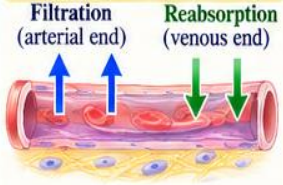
- ↑ Capillary hydrostatic pressure (e.g. heart failure, venous obstruction)
- ↓ Plasma oncotic pressure (e.g. hypoalbuminemia)
- ↑ Capillary permeability (e.g. inflammation, burns)
- ↑ Interstitial oncotic pressure (e.g. tissue injury)

Increase reabsorption (↓ J_v)

- ↓ Capillary hydrostatic pressure
- ↑ Plasma oncotic pressure (e.g. high plasma proteins)
- ↓ Capillary permeability (tight endothelium)
- ↓ Interstitial oncotic pressure

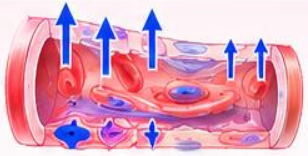
6 Clinical Examples

Normal



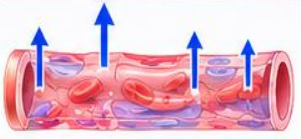
- Balanced fluid exchange
- No edema

Inflammation



- ↑ Permeability (histamine, bradykinin, etc.)
- ↑ Filtration → edema

Hypoalbuminemia



- ↓ Plasma oncotic pressure (π_c)
- ↑ Filtration → edema (e.g. liver disease, nephrotic syndrome)

Lymphatic System: Components, Pathway and Clinical Significance

1 Components and Functions

Lymphatic capillaries



- Blind-ended capillaries
- Highly permeable
- Collect excess interstitial fluid, proteins, lipids and immune cells

Function: Fluid collection from interstitial space

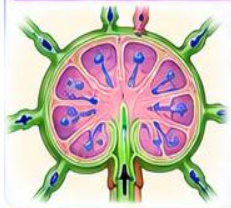
Lymphatic vessels



- Thin wall with valves
- Low pressure, one-way flow
- Propel lymph to larger vessels (helped by muscle contraction)

Function: Transport lymph toward venous system

Lymph nodes



- Filter lymph
- Remove pathogens, dead cells and foreign particles
- Site of activation of B and T lymphocytes

Function: Immune defense and filtration

Lymphatic organs



- **Spleen:** filters blood, immune response
- **Thymus:** T-cell maturation
- **Tonsils** (MALT): local immunity

Function: Immune system development and response

2 Pathway from Blood Capillaries into Vena Cava

1. Formation of lymph
- Excess interstitial fluid, proteins, lipids and waste enter lymphatic capillaries (especially in tissues).

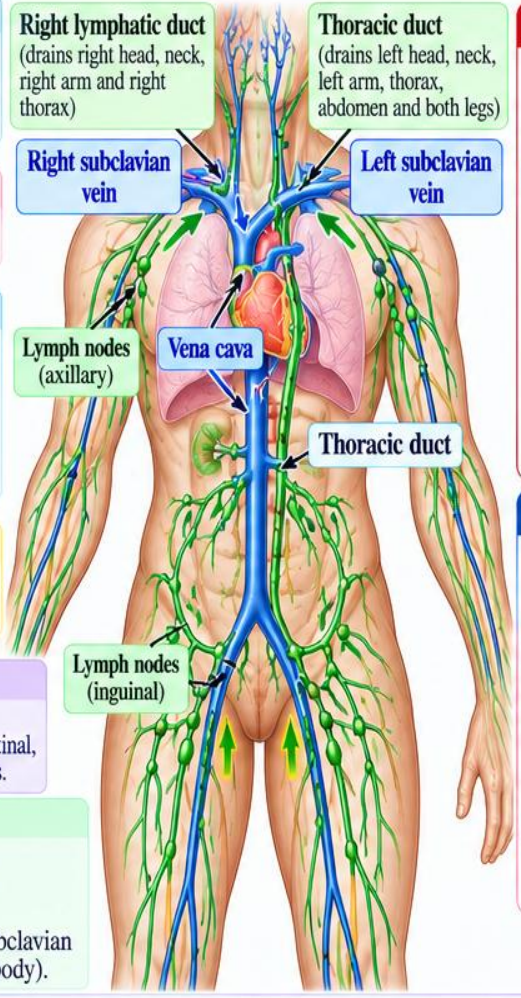
2. Lymphatic capillaries
- Blind-ended, highly permeable.

3. Collecting vessels
- Carry lymph with valves toward lymph nodes.
- Propelled by skeletal muscle contraction, respiration and arterial pulsation.

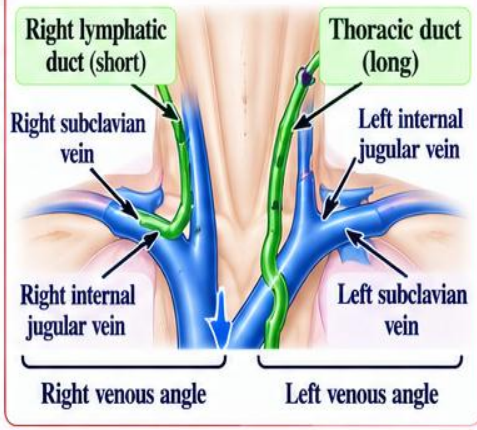
4. Lymph nodes
- Filter lymph and activate immune cells.

5. Lymphatic trunks
- Larger vessels: jugular, subclavian, bronchomediastinal, intestinal and lumbar trunks.

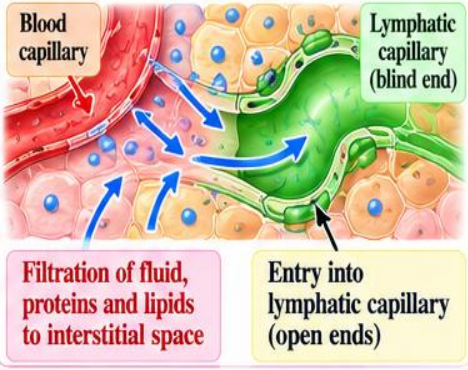
6. Lymphatic ducts
- Right lymphatic duct → right subclavian vein (right upper quadrant).
- Thoracic duct → left subclavian vein (drains rest of the body).



Details at the venous angles



Micro view: Entry into lymphatic capillary



3 Clinical Significance

A. Edema (lymphedema)



- Obstruction or removal of lymphatic vessels/nodes (e.g. cancer surgery, infection).
- Accumulation of interstitial fluid and proteins.
- Common in filariasis and after mastectomy (arm edema).

B. Infections



- Lymph nodes filter pathogens → may enlarge (lymphadenopathy).
- Lymphatic spread of infections (e.g. tuberculosis, lymphoma, metastatic cancer).
- Important in immune surveillance and vaccine response.

C. Malignancy



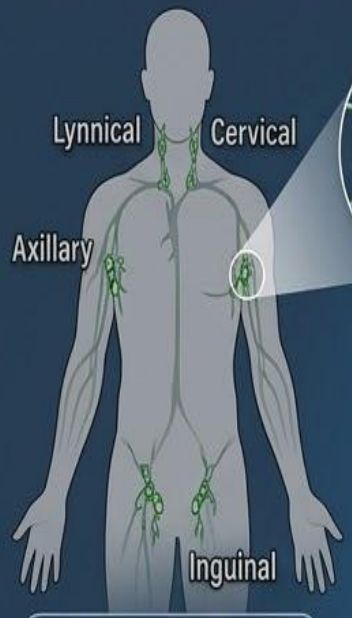
- Route for metastasis of cancers (e.g. breast, gastrointeeitial).
- Sentinel lymph node biopsy used for staging.
- Lymphoma originates from lymphoid tissue.

D. Other clinical points



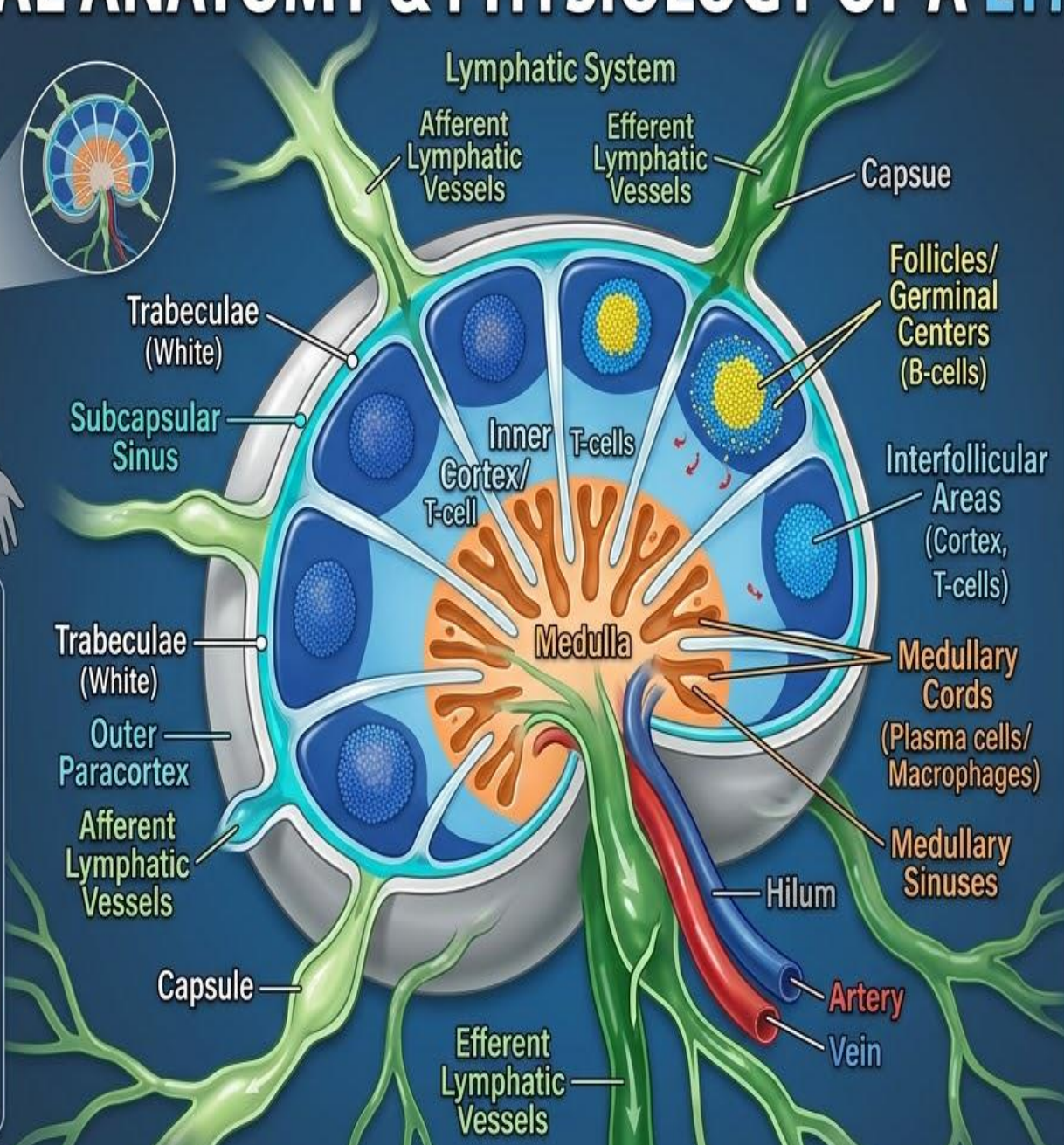
- Transport of dietary lipids (chyle) from intestinal lacteals.
- Loss of lymph → protein loss, immunodeficiency (e.g. intestinal lymphangiectasia).
- Important in fluid balance, immune function and absorption of fat-soluble vitamins (A, D, E, K).

CLASSICAL ANATOMY & PHYSIOLOGY OF A LYMPH NODE



Lymph Flow

- Afferent Vessel
- Subcapsular Sinus
- Trabecular Sinus
- Medullary Sinus
- Efferent Vessel

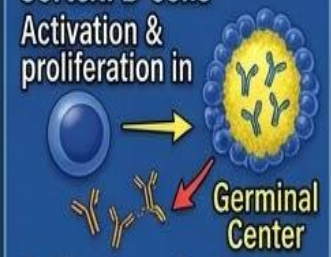


Cellular Function

Immune Response

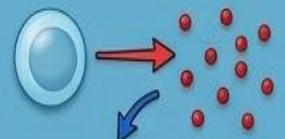
Cortex: B-cells

Activation & proliferation in



Antibody production

Paracortex: T-cells



Activation/ sampling antigens

Medulla: Macrophages



Macrophages & Plasma cells

Filter, lymph any and capture pathogens

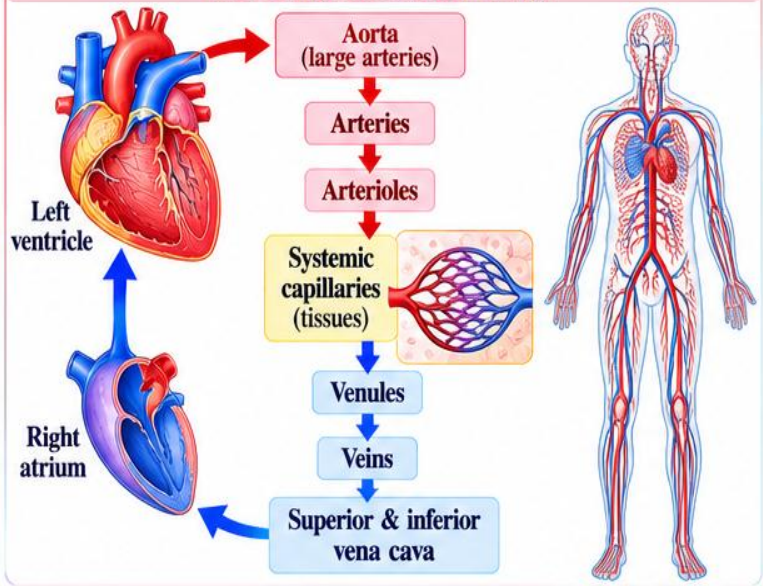
Blood Circulations

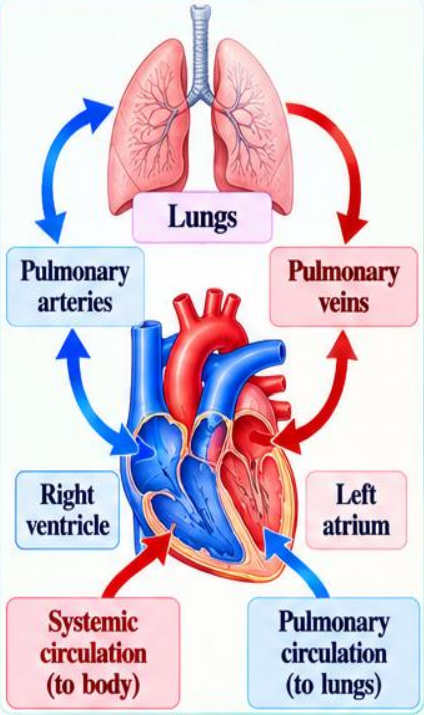
1. Systemic circulation
2. Pulmonary circulation
3. Specific circulations
   - Coronary circulation
   - Fetal circulation

Systemic vs. Pulmonary Circulation

Systemic Circulation

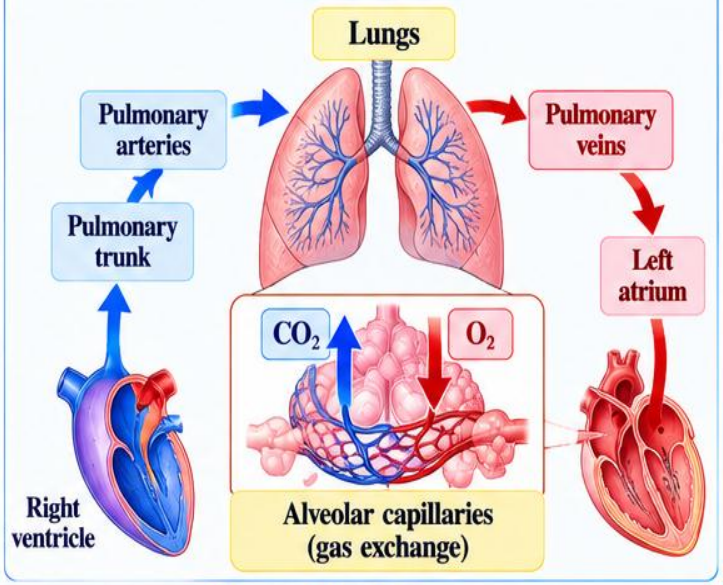
Carries oxygenated blood from the left ventricle to all body tissues (except lungs)





Pulmonary Circulation

Carries deoxygenated blood from the right ventricle to the lungs for gas exchange



Feature	Systemic Circulation	Pulmonary Circulation
Origin	Left ventricle	Right ventricle
Pathway	Aorta → arteries → arterioles → systemic capillaries → venules → veins → SVC & IVC → right atrium	Pulmonary trunk → pulmonary arteries → pulmonary capillaries → pulmonary veins → left atrium
Blood carried	Oxygenated (except in systemic veins)	Deoxygenated (except in pulmonary veins)
Main function	Supply O_2 and nutrients to body tissues; remove CO_2 and waste	Gas exchange in lungs (add O_2, remove CO_2)
Pressure	High (MAP ≈ 90–100 mmHg)	Low (mean ≈ 15 mmHg)
Vascular resistance	High (systemic vascular resistance)	Low (pulmonary vascular resistance; ~1/10 of systemic)
Vessel walls	Thicker, more muscular and elastic	Thinner, less muscular and elastic
Capillaries	Systemic tissue capillaries (continuous, fenestrated, or sinusoidal)	Pulmonary capillaries (continuous, thin for gas exchange)
Blood volume	~ 84–85% of total blood volume	~ 9–12% of total blood volume
Transit time	~ 20 seconds	~ 4–6 seconds (faster)
Oxygen content	High (SaO_2 ≈ 95–100%)	Low (SvO_2 ≈ 70–75%)
Main regulation	Systemic arterioles (sympathetic activity, local factors, hormones)	Pulmonary arterioles (hypoxia, NO, prostacyclin, endothelin)

Clinical Significance

Systemic circulation

- High resistance → high arterial pressure.
- Affected in systemic hypertension, shock, atherosclerosis.
- Supplies vital organs (brain, heart, kidneys).

Pulmonary circulation

- Low resistance → low pressure (protects lung capillaries).
- Pulmonary hypertension increases right ventricular afterload.
- Pulmonary embolism blocks flow → hypoxemia and acute right heart failure.

Key differences

- Systemic: high pressure, long distance, high resistance.
- Pulmonary: low pressure, short distance, low resistance.
- Systemic supplies tissues; pulmonary oxygenates blood.

Coronary circulation

Heart has its own network of blood vessels

Pathway

1. **Aorta →**

2. **Coronary Ostia →**

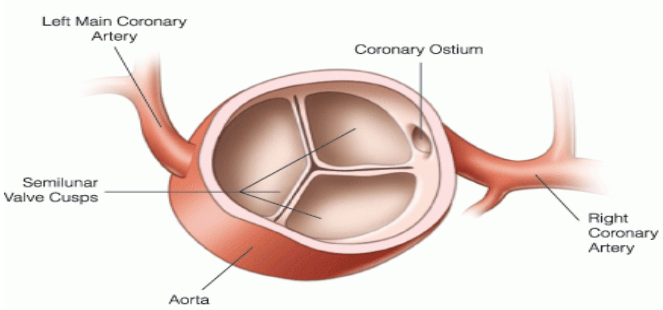
3. **Coronary arteries →**

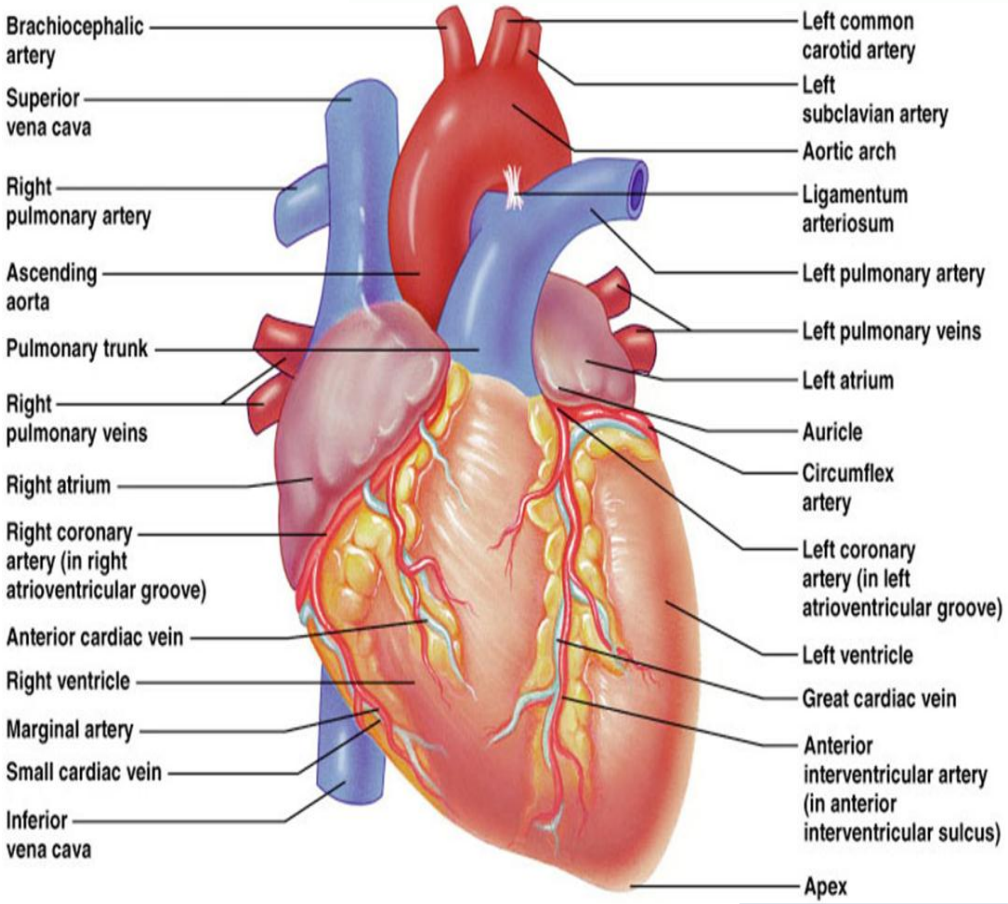
- ❑ **Right coronary artery**
  - ➢ **Right marginal artery**
  - ➢ **Posterior descending artery**
- ❑ **Left coronary artery**
  - ➢ **Left anterior descending (LAD) artery**
  - ➢ **Left circumflex artery**

1. **Coronary capillaries →**

2. **Coronary veins →**

3. **Right atrium**

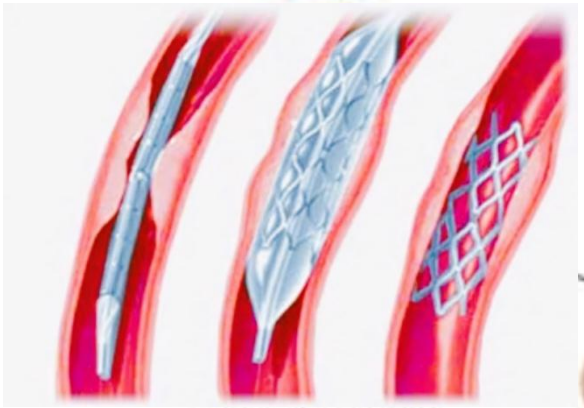


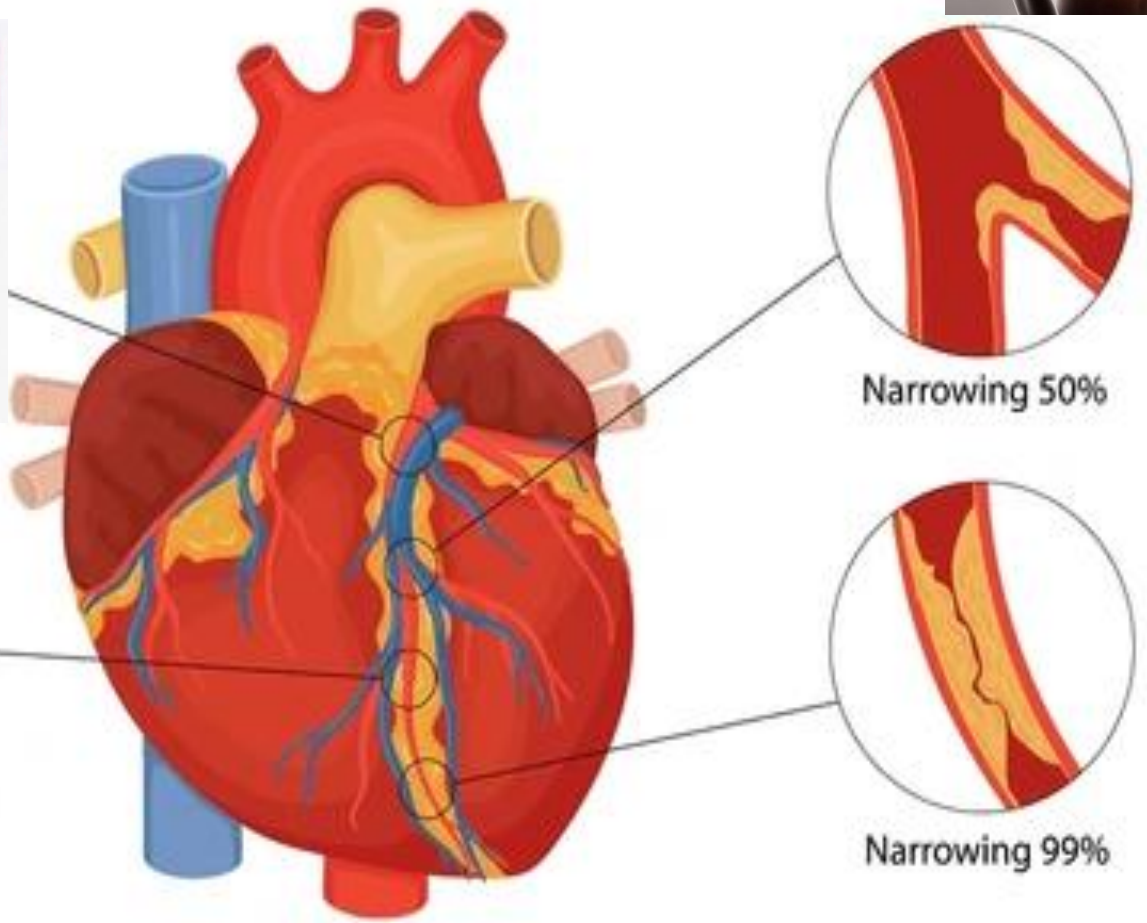


Coronary circulation

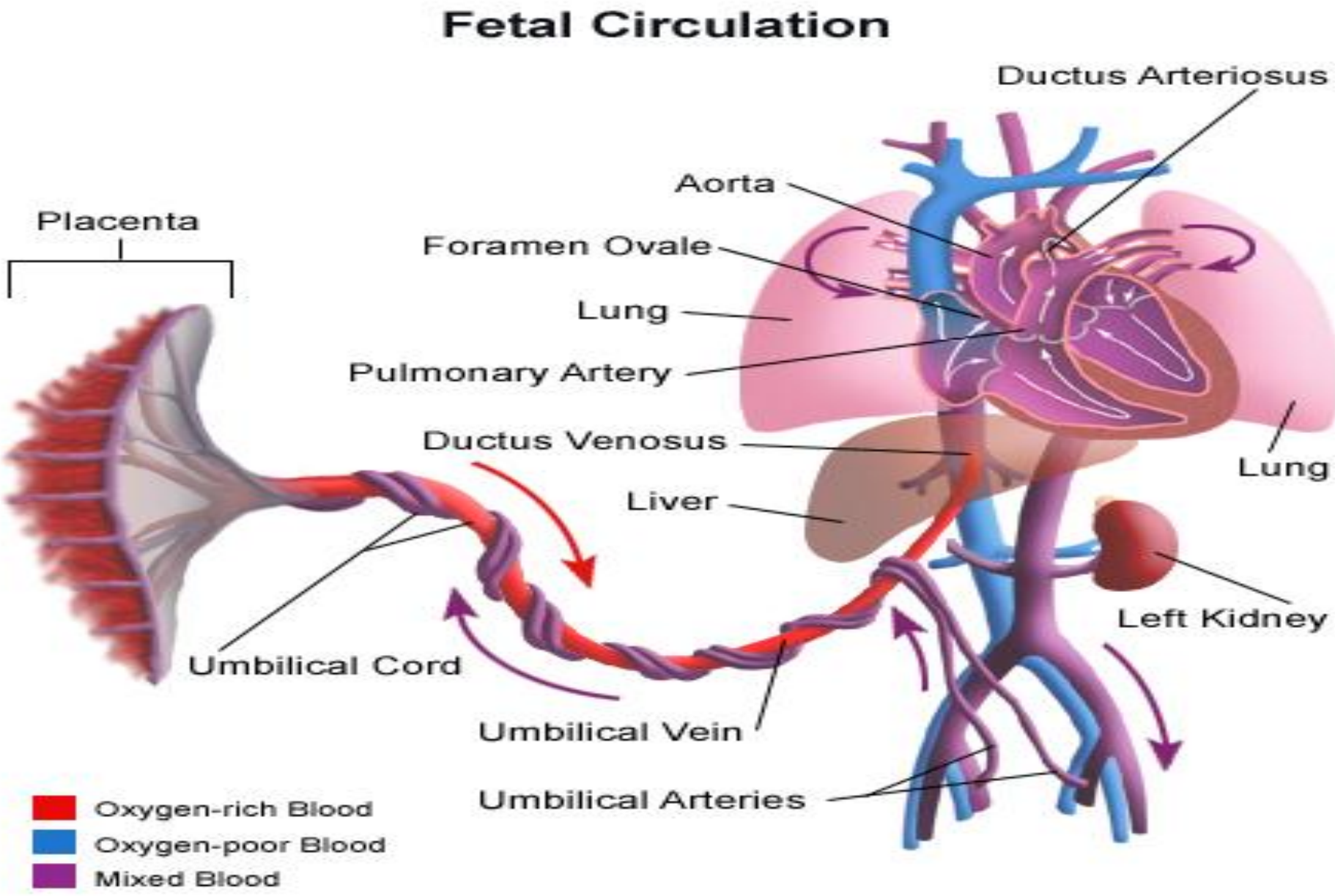
Angina pectoris



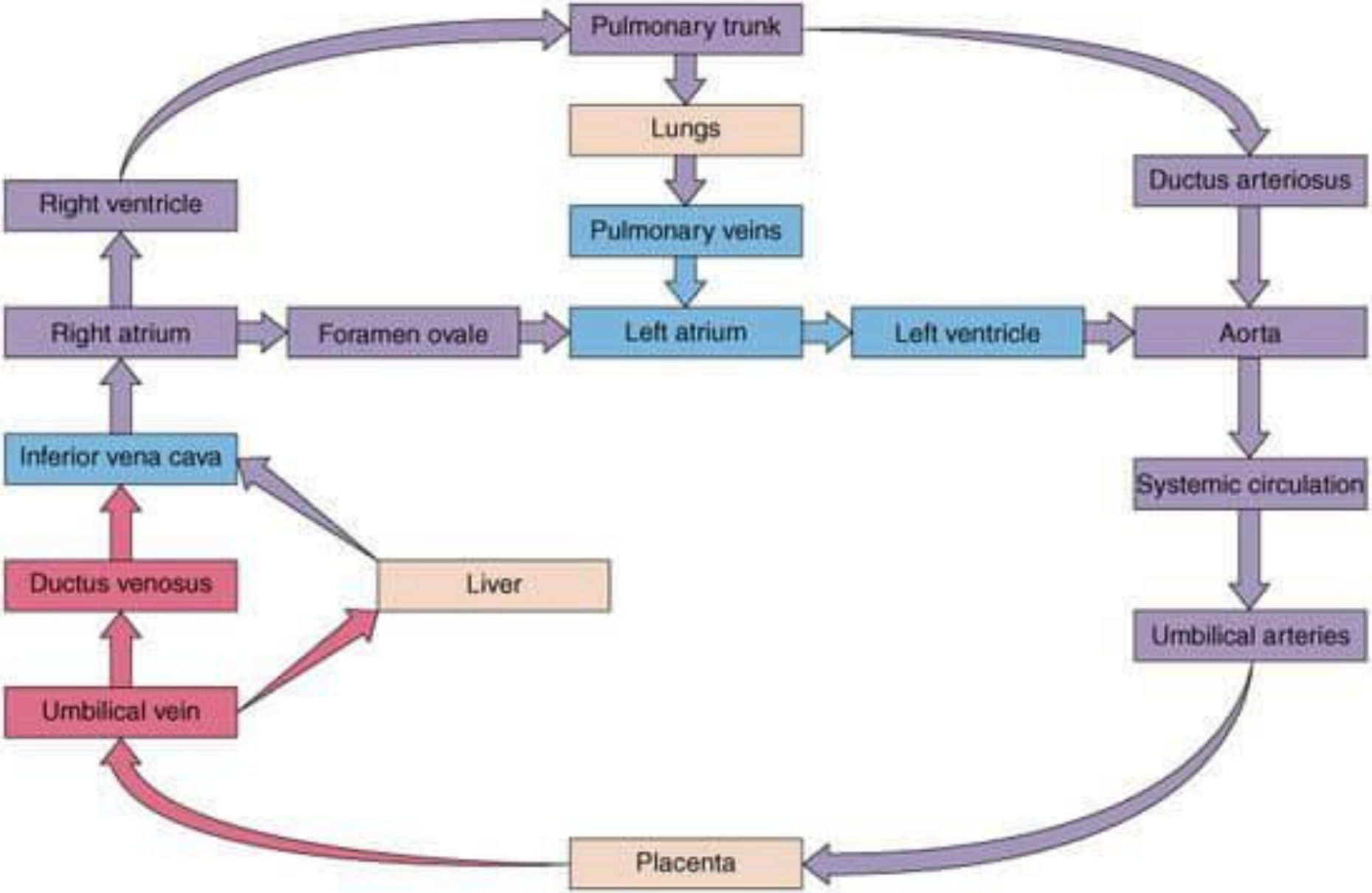




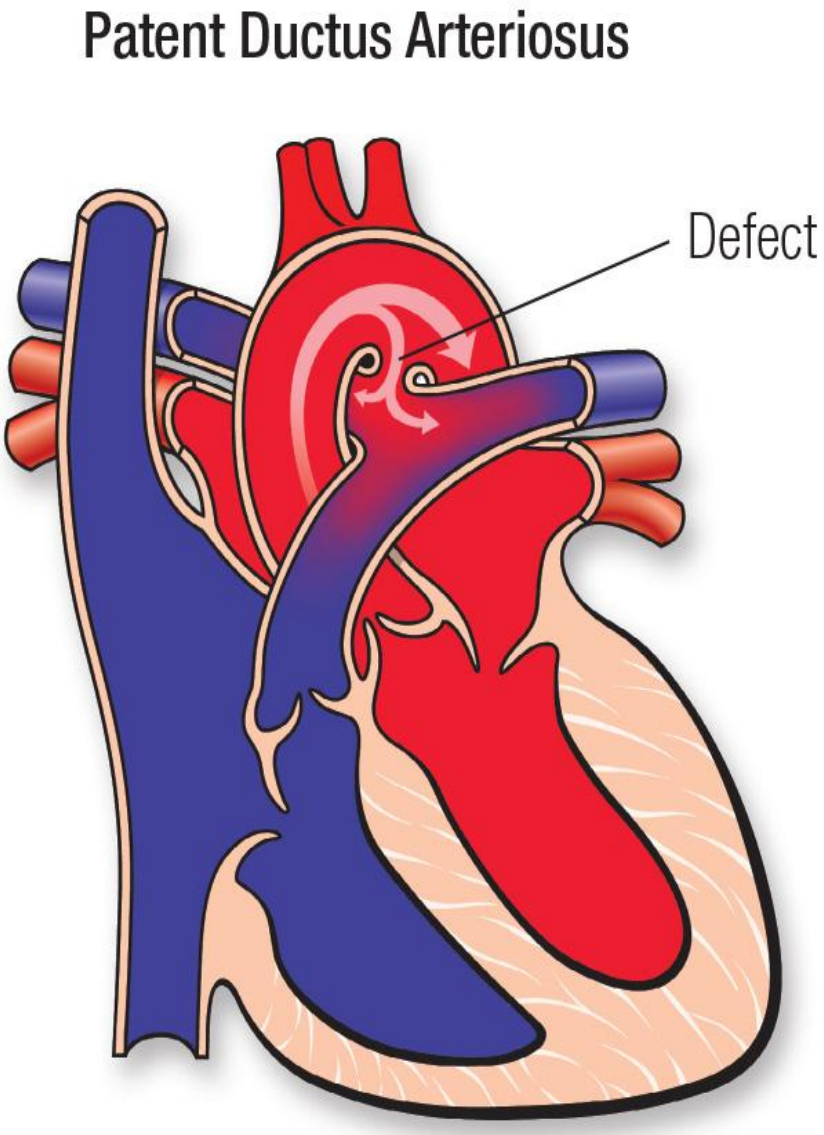
Coronary circulation

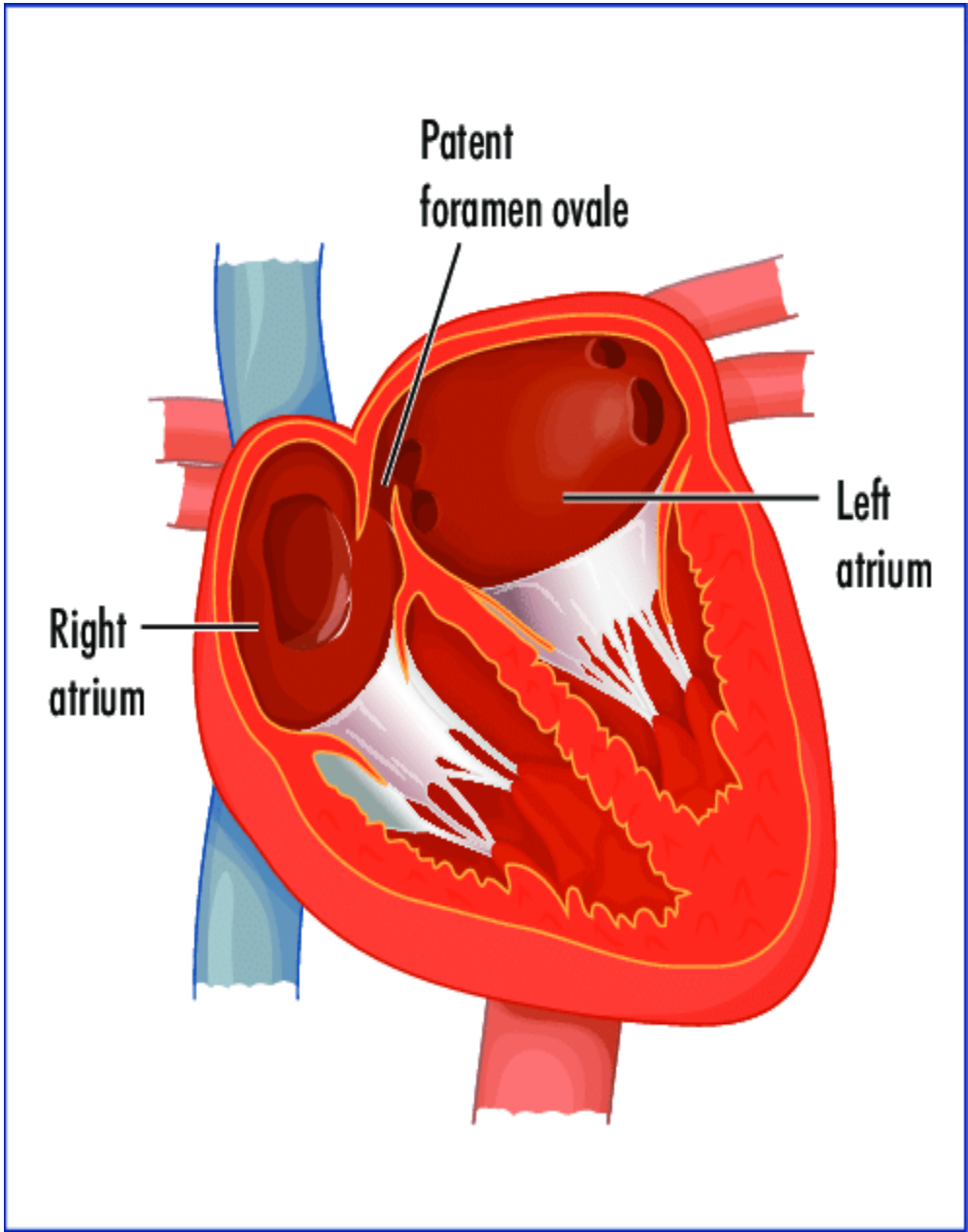


Coronary circulation



Coronary circulation





Arterial Blood pressure

- The force per unit area exerted on a vessel wall by the blood contained within it (expressed in mmHg).

Normal blood pressure

- Systolic pressure: < 120 mmHg
- Diastolic pressure: < 80 mmHg

Elevated/ Pre-hypertension:

- Systolic pressure:120-129 mmHg
- Diastolic pressure:80-89 mmHg

Hypertension

- Systolic 140 mm Hg & above
- Diastolic 90 mm Hg & above

Hypotension

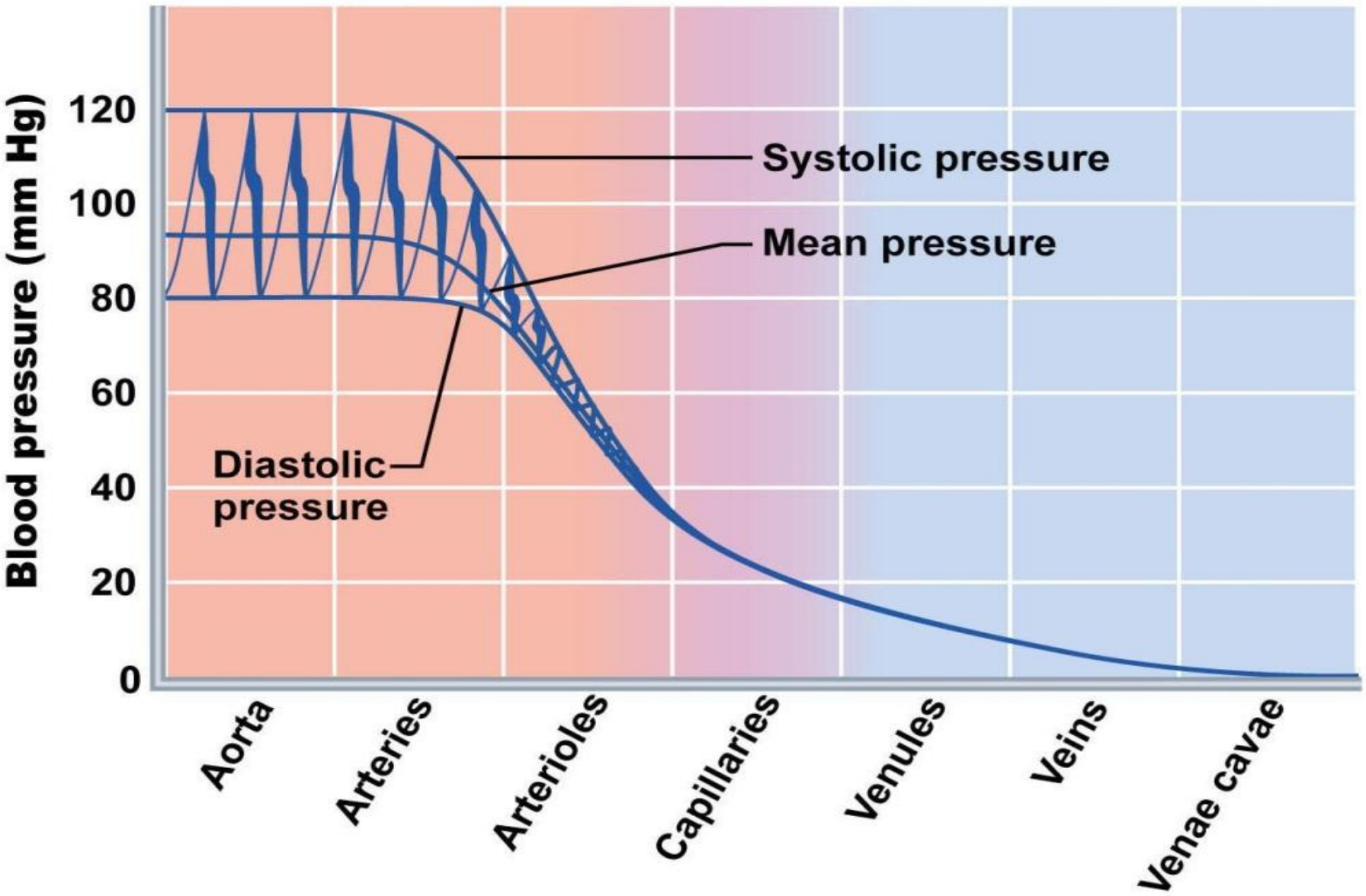
- Systolic < 90 mm Hg &/or Diastolic < 60 mmHg

- Mean arterial pressure (MAP= DP+1/3(SP-DP))
- Pulse pressure (PP= SP-DP)
  - Wide in: Valve regurgitation, Aortic stiffening, Hyperthyroidism/Narrow in: CHF, aortic stenosis, hypotension

Function of ABP:

- Maintain blood flow through different tissues & organs
- Creates hydrostatic pressure on capillaries

Blood pressure



Factors Affecting Blood Pressure

1-Cardiac Output

- Increased CO → increased BP (Exercise/ Stress)
- Decreased CO → decreased BP

2-Peripheral Resistance

- Blood viscosity (Polycythemia)
- Total blood vessel length (Obesity/Pregnancy)
- Blood vessel diameter (Atherosclerosis/Cold/Hot)

3-Blood Volume

- Increased blood volume → increased BP (↑ Salts)
- Decreased blood volume → decreased BP (Bleeding)

4-Arteries elasticity

- Decrease in elasticity → increased BP (Atherosclerosis)

Regulation of blood pressure

1-Neural regulation

- ❑ Sympathetic division (Vasomotor center)
- ❑ Parasympathetic division (Vagus center)

Pathway

- ➢ ↓Blood pressure→ stimulation of baroreceptors in carotid sinus & aortic arch → signal to VMC → VC &↑CO
- ➢ ↑Blood pressure→ stimulation of baroreceptors in carotid sinus & aortic arch → signal to Vagus center → ↓CO

2-Cardiac Regulation

- ❑ Atrial natriuretic peptide

Pathway

- ➢ ↑Blood pressure→ stimulation of volume receptors in atria →ANP secretion by atrium → circulating to kidneys→ ↑Na execration→ ↓ blood volume

3-Renal Regulation

- ❑ Renin-Angiotensin-Aldosterone System

Pathway

- ➢ ↓Renal blood pressure → stimulation of Juxtaglomerular cells in kidney→ ↑renin secretion→ Angiotensin-I to Angiotensin-II → ↑ADH & Aldosterone → ↑ blood volume and VC of blood vessels → ↑blood pressure

Physiology of Blood Pressure Regulation

1 Determinants of Arterial Blood Pressure

$$MAP = CO \times TPR$$

Mean Arterial Pressure (mmHg) | Cardiac Output | Total Peripheral Resistance



Cardiac Output (CO)

$CO = HR \times SV$

- Heart rate
- Stroke volume
- Blood volume (venous return)
- Contractility

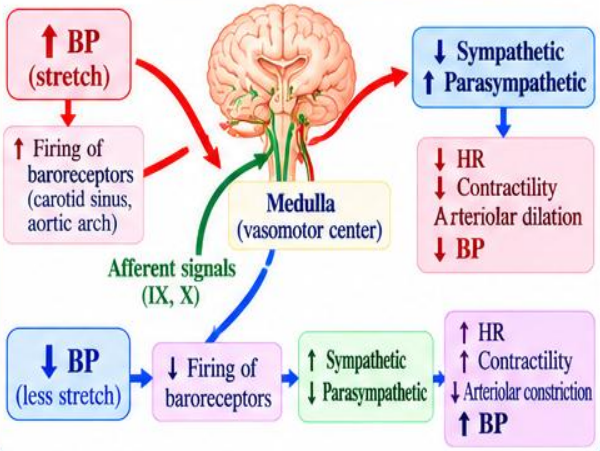


Total Peripheral Resistance (TPR)

- Arteriolar radius
- Vascular tone (sympathetic, local, hormonal)
- Blood viscosity
- Vessel length (minor)

2 Short-term Regulation (Neural)

Arterial Baroreceptor Reflex (seconds)



3 Long-term Regulation (Hormonal/Renal)

Renin–Angiotensin–Aldosterone System (RAAS)



↓ Renal perfusion
↓ NaCl at macula densa
↑ Sympathetic (β_1)
→ **↓ Renin** →
- Arteriolar constriction
- ↑ Aldosterone (Na^+ & water)
- ↑ Blood volume
- **↑ BP**

Antidiuretic Hormone (ADH, Vasopressin)



↑ Plasma osmolality
↓ Blood volume
↑ Angiotensin II
→ **ADH** (from posterior pituitary) →
- ↑ Water reabsorption (V2 receptors, kidney)
- Vasoconstriction (V1 receptors)
- **↑ BP**

Atrial Natriuretic Peptide (ANP)



↑ Atrial stretch (↑ blood volume)
→ **ANP** (from atria) →
- ↑ Na^+ and water excretion
- ↓ Vasodilation
- ↓ Renin, ↓ Aldosterone
- **↓ BP**

4 Other Neural Mechanisms

Chemoreceptor Reflex (seconds)



↓ O_2, ↑ CO_2, ↓ pH
- ↑ Sympathetic outflow
- ↑ HR
- ↑ Vasoconstriction
- ↑ BP

CNS Ischemic Response (severe ↓ BP)



- Powerful sympathetic activation (bradycardia is minimal)
- ↑ Vasoconstriction
- ↑ BP

Bainbridge Reflex (atrial stretch)



↑ Venous return → atrial stretch → ↑ HR (via vagal reflex)

5 Local (Tissue) Regulation (minutes–hours)

Metabolic Factors (Active tissue)



↓ O_2, ↑ CO_2, ↑ H^+, ↑ K^+
Adenosine, Lactate
→ Arteriolar dilation
→ ↑ Blood flow (metabolic autoregulation)

Myogenic Response



↑ Arterial pressure
→ Vascular smooth muscle stretch
→ Constriction (maintains stable flow)

Endothelial Factors



Nitric oxide (NO) → vasodilation
Prostacyclin (PGI_2) → vasodilation
Endothelin → vasoconstriction

Autoregulation



Maintains relatively constant blood flow despite changes in arterial pressure (e.g. brain, kidney, heart)

6 Integration of Mechanisms

Change in Arterial Blood Pressure

Neural (seconds)	Local (minutes)	Hormonal/Renal (hours–days)
• Baroreceptors	• Metabolic	• RAAS
• Chemoreceptors	• Myogenic	• ADH
• CNS ischemic	• Endothelial	• ANP
• Bainbridge reflex	• Autoregulation	• Fluid balance (renal excretion)

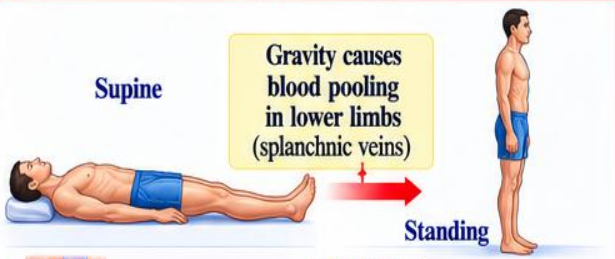
→ **Return of BP toward normal (homeostasis)**

7 Summary of Effects on Blood Pressure

Factor	Increase HR	Increase Contractility	Arteriolar Constriction (↑ TPR)	Increase Blood Volume (↑ CO)	Overall effect on BP
Sympathetic activation (NE)	↑	↑	↑	(↑ renin)	↑ BP
Parasympathetic activation (ACh)	↓	↓	–	–	↓ BP
Angiotensin II	–	↑	↑	↑ (via aldosterone)	↑ BP
ADH (vasopressin)	–	–	↑	↑ (water reabsorption)	↑ BP
ANP	–	–	↓	↓ (natriuresis/diuresis)	↓ BP
Local metabolic factors (active tissue)	–	–	↓ (local)	–	↓ local BP (↑ local flow)

Integrated Cardiovascular Responses to Posture, Exercise and Hemorrhage

1 Posture (Supine → Standing)





Venous pooling
- ↓ venous return
- ↓ stroke volume
- ↓ arterial pressure

Compensatory responses:
- ↑ sympathetic activity
- ↑ HR, ↑ contractility
- Arteriolar and venous vasoconstriction (MAP maintained)

Key Changes When Standing
- ↓ Venous return (10–30%)
- ↓ Stroke volume (10–40%)
- ↔/slight ↓ MAP (usually maintained)
- ↑ Heart rate (10–20 bpm)
- ↑ Total peripheral resistance (TPR)
- ↑ Sympathetic activity (baroreflex)

Main Mechanism
- **Baroreceptor reflex** (↓ stretch → ↑ SNS and ↓ parasympathetic)
- Venoconstriction (reduces venous pooling)
- Arteriolar constriction (maintains MAP)

2 Dynamic Exercise (Moderate to Intense)



Muscle pump
↑ venous return

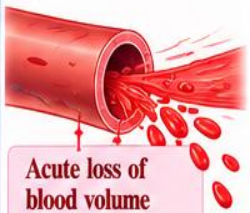
Metabolic vasodilation
in active muscles
↓ local resistance

- ↑ Sympathetic activity
- ↑ HR and contractility
- ↑ Cardiac output
(4–7 fold in intense exercise)

Redistribution of blood flow
- ↑ Muscle (up to 20×)
- ↑ Heart (2–4×)
- ↑ Skin (heat dissipation)
- ↓ Splanchnic and renal

Key Cardiovascular Changes During Exercise
- ↑ Heart rate (proportional to intensity)
- ↑ Stroke volume (to a limit)
- ↑ Cardiac output (4–7×)
- ↑ Systolic blood pressure (marked)
- ↔/slight ↓ Diastolic blood pressure (no change or slight ↓)
- ↑ Mean arterial pressure (moderate)
- ↓ Total peripheral resistance (due to vasodilation in active muscles)
- ↑ Venous return (muscle pump + sympathetic venoconstriction)

3 Hemorrhage (Acute Blood Loss)



Acute loss of blood volume
- ↓ venous return
- ↓ cardiac output
- ↓ arterial pressure

Effects
- ↓ Venous return (preload)
- ↓ Stroke volume
- ↓ Cardiac output
- ↓ Arterial blood pressure
- ↓ Organ perfusion (especially brain and kidneys)

Compensatory Responses (Integrated)



Baroreceptor reflex
- ↑ sympathetic activity
- ↓ parasympathetic activity



Hormonal responses
- ↑ Adrenaline / Noradrenaline
- ↑ Renin–Angiotensin–Aldosterone
- ↑ ADH (vasopressin)



Vasoconstriction
↑ TPR (arteriolar and venous)



↑ Heart rate, ↑ contractility
(maintain CO initially)



Fluid retention
(aldosterone & ADH)

4 Comparison of Major Variables

Variable	Supine (Rest)	Standing (Posture)	Exercise (Moderate/ Intense)	Hemorrhage (Acute)
Venous return	Normal	↓	↑	↓↓
Stroke volume	Normal	↓	↑	↓↓
Heart rate	Normal (60–80)	↑	↑↑↑	↑↑
Cardiac output	Normal (5 L/min)	↔/slight ↓	↑↑ (4–7×)	↓↓
Systolic BP	~120 mmHg	↔	↑↑	↓↓
Diastolic BP	~80 mmHg	↔	↔/slight ↓	↓↓
Mean arterial BP	~93 mmHg	Maintained	↑ (moderate)	↕↕
Total peripheral resistance	Normal	↑	↓	↑↑

5 Mechanisms Involved



Neural
- Baroreceptor reflex
- Sympathetic–parasympathetic balance
- Central command (exercise)



Hormonal
- Adrenaline / Noradrenaline (rapid)
- Renin–Angiotensin–Aldosterone (medium term)
- ADH (water retention)
- ANP (↓ during hemorrhage, may ↑ with volume expansion)



Local factors
- Metabolic vasodilation (exercise)
- Myogenic and endothelial factors (NO, prostacyclin)



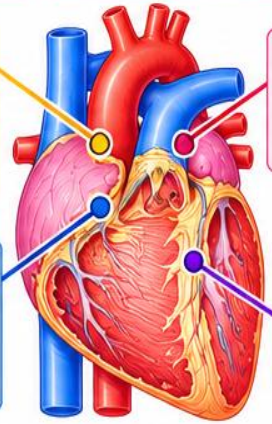
Mechanical
- Muscle pump (exercise)
- Respiratory pump
- Venous tone and blood volume

6 Key Points and Clinical Significance

1. **Posture:** Gravity ↓ venous return → baroreflex maintains blood pressure (↑ HR, ↑ TPR).
2. **Exercise:** Integrated neural, hormonal and local mechanisms → ↑ cardiac output and selective redistribution of blood flow.
3. **Hemorrhage:** Loss of blood volume → ↓ CO and BP → compensatory neural and hormonal responses to maintain perfusion of vital organs.
4. These responses are integrated to match cardiac output and vascular resistance to the body's needs and maintain arterial pressure and tissue perfusion.
5. **Clinical relevance:** impaired responses occur in autonomic failure, heart failure, shock and during anesthesia or medications (e.g. β-blockers, vasodilators).

Main Receptors in the Heart and Blood Vessels for Cardiovascular Regulation and Their Targeted Drugs

Receptors in the Heart



β_1-adrenergic receptors
SA node
AV node
Myocardium

α_1-adrenergic receptors (atria, coronary vessels)

M_2 muscarinic receptors (Parasympathetic)
SA node
AV node

AT_1 receptors (cardiac myocytes, fibroblasts)

Autonomic regulation of the heart

Sympathetic (NE, E)



Norepinephrine
Epinephrine

β1 receptors

- ↑ Heart rate
- ↑ AV conduction
- ↑ Contractility
- ↑ Renin release (kidney)

Parasympathetic (ACh)



Acetylcholine

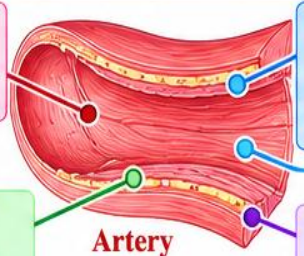
M_2 receptors

- ↓ Heart rate
- ↓ AV conduction
- ↓ Atrial contractility (slight ↓ ventricular contractility)

Receptors in Blood Vessels

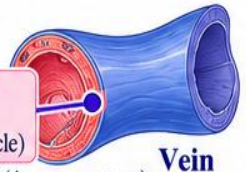
α_1-adrenergic receptors (vascular smooth muscle)
Arteriolar vasoconstriction

β_2-adrenergic receptors (vascular smooth muscle, especially in skeletal muscle)
→ Vasodilation



Endothelial receptors (M_3, B_2, etc.)
Stimulate NO, PGI_2 → Vasodilation

AT_1 receptors (vascular smooth muscle)
Vasoconstriction and vascular growth



α_1-adrenergic receptors (venous smooth muscle)
Venoconstriction (↑ venous return)

Receptor	Main Location (in heart or vessels)	Physiological Effect	Endogenous Ligand	Targeted by Drugs (examples)	Clinical Use / Notes
β_1-adrenergic	• SA node, AV node • Ventricular myocardium • Juxtaglomerular cells (kidney)	↑ Heart rate ↑ AV conduction ↑ Contractility ↑ Renin release	Norepinephrine Epinephrine	**β_1-blockers** (Metoprolol, Bisoprolol, Atenolol)	• Hypertension • Angina, post-MI • Heart failure (selected)
β_2-adrenergic	• Vascular smooth muscle (especially skeletal muscle arterioles)	**Vasodilation** (↓ peripheral resistance)	Epinephrine (> Norepinephrine)	**β_2-agonists** (e.g. Salbutamol) **Non-selective β-blockers** (e.g. Propranolol)	• β_2-agonists: acute use in vasospasm (e.g. Raynaud) • Non-selective β-blockers may cause vasoconstriction
α_1-adrenergic	• Vascular smooth muscle (arteries & veins) • Cardiac atria, coronary vessels	**Vasoconstriction** (↑ peripheral resistance and ↑ venous return)	Norepinephrine Epinephrine	**α_1-blockers** (Prazosin, Doxazosin, Tamsulosin) **α_1-agonists** (Phenylephrine)	• α_1-blockers: hypertension, BPH (tamsulosin) • α_1-agonists: acute hypotension
M_2 muscarinic	• SA node, AV node • Atrial myocardium	↓ Heart rate ↓ AV conduction ↓ Atrial contractility	Acetylcholine (vagus nerve)	**Muscarinic agonists** (Bethanechol – less used) **Muscarinic antagonists** (Atropine)	• Atropine: symptomatic bradycardia • Vagal stimulation decreases HR
M_3 muscarinic (endothelial)	• Vascular endothelium	↑ NO and prostacyclin release → Vasodilation	Acetylcholine	**Cholinergic agonists** (e.g. Bethanechol)	• Endothelium-dependent vasodilation
AT_1 (Angiotensin II)	• Vascular smooth muscle • Cardiac myocytes and fibroblasts • Adrenal cortex, kidney, brain	**Vasoconstriction** ↑ Aldosterone release Cardiac hypertrophy & fibrosis	Angiotensin II	**ACE inhibitors** (Enalapril) ARBs (Losartan, Valsartan) **Direct renin inhibitor** (Aliskiren)	• Hypertension, heart failure • Post-MI, diabetic nephropathy • Reduce cardiovascular remodeling
B_2 (Bradykinin)	• Vascular endothelium	↑ NO and prostacyclin release → Vasodilation	Bradykinin	**Enhanced by ACE inhibitors** (block breakdown of bradykinin)	• Contributes to vasodilation ACE inhibitors: ↑ bradykinin (may cause cough/angioedema)
Natriuretic peptide receptors (NPR-A)	• Vascular smooth muscle • Kidney (glomerulus, collecting duct) • Heart	**Vasodilation** ↑ Natriuresis and diuresis ↓ Blood volume and BP	ANP, BNP (from atria/ventricles)	**Recombinant BNP** (nesiritide) **Neprilysin inhibitors** (sacubitril/valsartan)	• Heart failure (↑ natriuresis, vasodilation, ↓ BP)